KU LEUVEN

FACULTY OF SCIENCE

Connectivity Bounds in Graphs and Beyond

Erdős Problem 915, from Proof to Search

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Foreword

This thesis studies how many edges or arcs a network can contain when no pair of vertices is allowed to have too many independent routes between them. The starting point is Erdős Problem 915, an extremal question about edge-disjoint paths in simple graphs. The work is told as a journey. It begins with the variants whose answers can be proved cleanly by hand, watches those proofs grow longer and more delicate as the model changes, and at the point where a single argument no longer suffices it turns to a unified program. That program holds all twelve variants behind one common interface, so a single piece of code serves them all. It measures connectivity exactly with a maximum-flow routine, the standard way to count the independent routes between two points. It runs a prover that sweeps every graph of a small fixed size and, when none beats a target, reports that sweep as a genuine upper-bound proof. And it runs a search guided by heuristics that either let hot metal cool slowly into shape or simply take the best direction on offer and never backtrack.

I am grateful to my supervisor, Prof. Stijn Cambie, for his day-to-day guidance, corrections, and mathematical feedback throughout the project. I could not ask for a more understanding and enthusiastic supervisor. I thank my promotor, Prof. Jan Goedgebeur, whose suggestion of the `nauty` generation pipeline sharply accelerated the directed enumeration. I also thank the people who discussed early versions of the arguments, checked examples, or helped clarify the presentation. Above all I thank my girlfriend Sara, whose patience and encouragement carried me through the long stretches of this work.

My use of generative-AI tools is described in the Contribution Statement. I remain responsible for every claim, proof, computation, and formulation in this thesis.

Contribution Statement

This thesis is my own work. I carried out the literature study, recast the twelve variants of the problem in one common language, and designed and wrote the single program that runs through all of them. I built every example, by hand or by computer search, checked each one with that program, wrote the proofs that appear in the thesis, and typeset the whole document. Where a result comes from the literature I say so and cite it. The theorems I use but do not claim as mine are those of Mader, Leonard, Sørensen and Thomassen, Menger, Gomory and Hu, and Baranyai. Related hypergraph-connectivity work includes Bérczi, Chandrasekaran, Király and Kulkarni [1]. The hypergraph results proved in this thesis do not depend on their splitting-off theorem.

The program has three jobs and keeps them separate: it *measures* connectivity exactly by maximum flow, it *proves* a finite optimum by sweeping a finite space of graphs, and it *discovers* dense graphs by metaheuristic search, which gives lower bounds and suggests conjectures but never an upper bound. Every general statement in the thesis rests on a proof by hand or a cited theorem. The computer searches are evidence that points the way, never a replacement for the argument.

I used Claude Code, ChatGPT, and Gemini for literature leads, exploratory proof drafts, code assistance, LaTeX editing, and proofreading. Their output was not treated as a source or as evidence: literature claims were checked against the cited publications, general mathematical claims were retained only with a proof or citation, and computational claims were checked by tests and, for critical cases, independent implementations. I reviewed and take responsibility for the final text, mathematics, and code. This statement records the role of the tools in the thesis. The same use must also be recorded in the separate KU Leuven GenAI transparency declaration submitted with the thesis.

My original results are stated fully in Chapter 3 and proved in Appendix A; the compact map is:

Variant	Contribution
Hypergraph, vertex separation	Exact value at $m = 2$; the $m = 3$ universal bound and its stated attainment range, via a new incidence-rank lemma (Theorems 1.16 and 1.17).
Directed multigraph	Exact value for all n, m, plus quadratic-branch classification for $n \geq \max(8, 2m)$ (Theorems A.31 and A.45).
Directed simple graph, arc separation	Unconditional $n^2/4+\Theta_m(n)$ bound and an $O(\sqrt{m}\, n^{3/2})$ structural distance from a one-way bipartite graph (Proposition A.23 and theorem A.25).
Multigraph, vertex separation with parallel routes distinct	Exact $m = 2$ value, counterexamples to the thickened-tree guess, matching order $\Theta(m^2 n)$, and an exact block-knapsack formulation (Theorems A.62 and A.66 and proposition A.64).
Transferred quadratic bounds	$k_m^{\mathrm{dir}}(n) = n^2/4+\Theta_m(n)$ and leading term $(m-1)n^2/(4(r-1))$ for directed hypergraphs under both separations and all orientations (Theorems A.29, A.59 and A.60).

An external AI review suggested the counting idea behind the directed stability estimate. I verified it, sharpened it, isolated the weaker route-counting hypothesis, and supplied the separate arguments that apply it to directed vertex connectivity and directed hypergraphs.

Short Summary

Erdős Problem 915, posed by Bollobás and Erdős [2], asks for the maximum number of edges in a graph on n vertices such that no two vertices are connected by m edge-disjoint paths. In modern notation this asks for

$$\ell_m(n) = \max\{|E(G)| : |V(G)| = n,\ \lambda_G^{\max} \le m-1\}.$$

Mader proved that, for simple graphs and $n \ge m \ge 2$,

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

This thesis revisits this theorem and studies the corresponding extremal problem under three independent changes: the graph model (simple graphs, multigraphs, and hypergraphs), the path-separation notion (edge-disjoint or internally vertex-disjoint), and the presence or absence of directions.

The thesis is organised as a journey from hand proofs to computation. The early variants are settled rigorously: the multigraph value $L_m(n) = (m-1)(n-1)$, the equality of the edge and vertex problems for $m \le 4$, and the first divergence at $m = 5$, where Sørensen and Thomassen prove $k_5(n) = \lfloor 8n/3 \rfloor - 4$, which pulls above the edge value from $n = 14$ onwards. The directed case forces the turn: the value $\ell_2^{\mathrm{dir}}(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ is proved by a long induction, and for $m \ge 3$ the hand proof gives way to a conjecture, after an explicit ten-vertex graph rules out the naive guess. The centrepiece is then a unified program. One multiplicity-matrix representation models almost all twelve variants, an exact maximum-flow checker measures local connectivity, a cut-counting optimisation certifies upper bounds for small sizes, and a search guided by edge sensitivity, run as both simulated annealing and tabu search, discovers extremal graphs. The program rediscovers, each in its own variant, the double star (undirected multigraphs at $m = 2$), the directed hub, the one-directional bipartite digraph, and the augmented-bipartite construction (the directed cases), independently checks the directed multigraph values $L_m^{\mathrm{dir}}(n) = 2(n-1)(m-1)$ at every $n \le 6$, which was as far as any proof then reached, and frames the remaining open problems on the directed frontier.

The interplay then pays back in the other direction. The hypergraph vertex problem is settled completely at $m = 2$. At $m = 3$ the formula $\lfloor 2(n-1)/(r-1) \rfloor$ is a universal upper bound for multihypergraphs and is exact whenever $r \ge 3$ and $2 \le \binom{n-2}{r-2}$, the upper bound following from a new rank bound on incidence graphs proved by way of Tutte's triconnected decomposition. On the directed side the multigraph problem closes completely, $L_m^{\mathrm{dir}}(n) = (m-1)\max(2(n-1), \lfloor n^2/4 \rfloor)$ for every n and every m, by deleting a sparse reachability-preserving subgraph at a time rather than a vertex at a time. For every fixed $m \ge 3$, the simple digraph problem, arc and vertex alike,

is pinned at $n^2/4 + \Theta_m(n)$ with no assumption on the shape of the extremiser, and at $m = 2$ its exact value has only an $O(1)$ correction to $n^2/4$. The directed hypergraph, which had only bounds a factor of four apart, has its leading term settled at $(m-1)n^2/\big(4(r-1)\big)$, so its bipartite construction is asymptotically optimal.

Abbreviations and Symbols

MILP	mixed-integer linear program
SA	simulated annealing
SPQR	the triconnected decomposition tree (series, parallel, rigid pieces)

$G = (V, E)$	graph with vertex set V and edge set E
$D = (V, A)$	directed graph with vertex set V and arc set A
n	number of vertices
m	forbidden-connectivity parameter, with all local connectivities at most $m - 1$
r	hyperedge size (every hyperedge spans r vertices)
$\lambda_G(u, v)$	maximum number of edge-disjoint u-v paths
$\lambda_G^{\max}$	maximum local edge-connectivity over all vertex pairs
$\kappa_G(u, v)$	maximum number of internally vertex-disjoint u-v paths
$\kappa_G^{\max}$	maximum local vertex-connectivity over all vertex pairs
$\ell_m(n)$	simple undirected edge-disjoint extremal function
$L_m(n)$	undirected multigraph edge-disjoint extremal function
$k_m(n)$	simple undirected vertex-disjoint extremal function
$\ell_m^{\mathrm{dir}}(n)$	simple directed arc-disjoint extremal function
$L_m^{\mathrm{dir}}(n)$	directed multigraph arc-disjoint extremal function
$k_m^{\mathrm{dir}}(n)$	simple directed vertex-disjoint extremal function
$\ell_m^{(r)}(n)$, $k_m^{(r)}(n)$	r-uniform hypergraph edge- and vertex-disjoint extremal functions
$K_m^{\mathrm{multi}}(n)$	multigraph vertex-disjoint extremal function with parallel copies counted as distinct routes (Section A.9)
$M(n)$	$\max\bigl(2(n-1), \lfloor n^2/4 \rfloor\bigr)$, the directed multigraph value per unit of $m - 1$
$M^*(n)$	the m-free directed multigraph optimum, $L_m^{\mathrm{dir}}(n) = (m-1)\, M^*(n)$, equal to $M(n)$ by Corollary A.38
$B_{p,q}$	one-directional complete bipartite digraph, all arcs from a p-part to a q-part
$\mu(u, v)$	multiplicity of an edge or arc between u and v
$\sigma(e)$	edge sensitivity $\lambda^{\max}(G) - \lambda^{\max}(G - e)$

Contents

Chapter 1

The Problem and Its Base Cases

A network is useful not only because it connects, but because it connects in more than one way. A single road between two towns is a liability, and a second independent road is insurance. The mathematical content of that intuition is the number of routes between two points that share no segment, and the question this thesis circles is how many connections a network can hold before some pair of points is forced to have too many such routes. This chapter builds the question from nothing, settles the variants whose proofs are short enough to read in an afternoon, and then follows the same question along three axes until the proofs stretch past breaking. By the end the case for a machine has made itself.

1.1 Graphs and routes

To state the question precisely we need only a few ideas. A *graph* G is a finite set of points, called *vertices*, together with a set of links, called *edges*, each joining two of the vertices. We write $V(G)$ for the set of vertices and $E(G)$ for the set of edges, so that $|E(G)|$ is the number of edges, the very quantity this thesis tries to make as large as possible. One pictures the vertices as dots and the edges as lines between them.

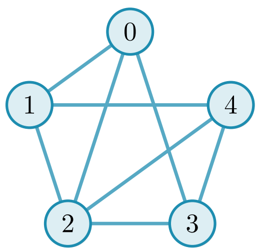


Figure 1.1. The eight-edge simple graph $K_5 - \{04, 13\}$. It returns in Figure 1.2 to illustrate independent routes.

A graph is the bare skeleton of a network: the vertices are the towns or the devices, and the edges are the roads or the links between them. We write n for the number of vertices, and the *degree* of a vertex is the number of edges meeting it. Unless we say otherwise our graphs are *simple*, meaning that at most one edge joins any given pair of vertices and its endpoints are distinct.

A *path* from a vertex u to a vertex v is a sequence of distinct vertices that starts at u, ends at v, and joins each consecutive pair by an edge. It is the formal version of a route through the network. Two paths between the same endpoints are *edge-disjoint* if no edge belongs to both, so that the loss of a single edge can break at most one of them. The largest number of pairwise edge-disjoint paths between u and v is their *local edge-connectivity* $\lambda_G(u,v)$, the count of genuinely independent routes the network offers to that pair. The pair with the most independent routes sets the ceiling for the whole graph:

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u,v),$$

the largest local edge-connectivity anywhere in G. A theorem of Menger [3], recalled in Appendix A, gives this quantity a second face: $\lambda_G(u,v)$ also equals the least number of edges one must remove to cut every route from u to v (Figure 1.2). Many independent routes between a pair is exactly the same thing as a costly cut between them.

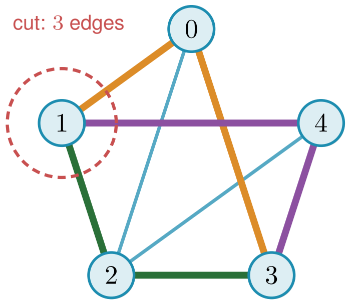


Figure 1.2. Menger's theorem on Figure 1.1: three edge-disjoint 1–3 routes (coloured) meet the three-edge cut around vertex 1, so $\lambda(1,3) = 3$.

1.2 Erdős Problem 915

The constraint at the heart of this thesis is a ceiling on connectivity. Requiring $\lambda_G^{\max} \leq m-1$ says that no pair of vertices is joined by as many as m edge-disjoint paths, or equivalently that every pair can be severed by removing fewer than m edges. The question is how many edges a graph can hold while staying under that ceiling.

Question 1.1 (Erdős Problem 915 [4]). How many edges can a graph on n vertices have if no two of its vertices are joined by m edge-disjoint paths?

Writing the answer as an extremal function,

$$\ell_m(n) = \max\{\, |E(G)| : |V(G)| = n,\ \lambda_G^{\max} \leq m-1 \,\},$$

the problem asks for $\ell_m(n)$, the most edges possible under the ceiling. Mader settled the simple-graph case completely, proving both halves at once: no graph under the ceiling has more than $\lfloor m(n-1)/2 \rfloor$ edges (the upper bound), and some graph reaches that count (the lower bound). The brackets $\lfloor \cdot \rfloor$ round their content down to the nearest whole number, and their ceiling counterpart $\lceil \cdot \rceil$ rounds up. Both recur throughout, because an edge count is always an integer while the formulas that bound it need not be.

Theorem 1.2 (Mader [5]). *For all $n \geq m \geq 2$,*

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor .$$

Mader's own proof is a direct structural induction. The proof given in full in Appendix A takes a different road, by way of the Gomory–Hu tree, together with a characterisation of the graphs that attain equality (Theorem A.10). This is the road the thesis adopts throughout, because the very same tree returns for the multigraph and hypergraph variants (Theorem 1.3, Proposition 1.15), so a single argument serves all of them rather than each needing its own. Every graph has such a tree (Theorem A.2). A Gomory–Hu tree of G is a weighted tree on the same vertex set in which $\lambda_G(u, v)$ equals the lightest edge on the tree path from u to v. We single out this one identity because the same idea reappears when the program of Chapter 2 stores connectivities as a tree (a graph without cycles). All $\binom{n}{2}$ local connectivities are therefore encoded by just $n-1$ numbers, the weights carried by the tree edges. The largest of them, $\lambda_G^{\max}$, is simply the heaviest edge of the tree. To see why, recall that every pairwise value is the lightest edge on some tree path, so no pair can exceed the heaviest edge anywhere in the tree. That value is attained by the two endpoints of that heaviest edge, since their own tree path consists of that single edge.

The whole argument (Appendix A) is one act of double counting on that tree. The word *charged* below is just accounting: to charge an edge to a tree edge is to assign it there for the count, and the proof works by charging every edge of G to tree edges and then adding the charges up. Each of the $n - 1$ tree edges stands for a minimum cut of G, and an original edge of G is charged once for every tree edge lying between its endpoints, so summing the tree-edge weights counts every edge of G at least once and, crucially, counts most of them at least twice. The only edges charged a single time are those joining tree-adjacent vertices, and a *simple* graph has at most $n - 1$ of those, one per tree edge. Every other edge is charged twice. Since each tree weight is a local connectivity and so at most $m - 1$, the two counts meet at $2|E(G)| - (n - 1) \leq (m - 1)(n - 1)$, which rearranges to the floor. The factor of two, and with it the gap between Mader's $\lfloor m(n-1)/2 \rfloor$ and the larger value $(m - 1)(n - 1)$ that parallel edges will buy in Theorem 1.3, is exactly the dividend of simplicity, the one place the argument uses that no pair carries a parallel edge.

One example fixes the scale. The complete graph K_n joins every pair, so every pair has $n-1$ routes that share no edge, $\lambda(u, v) = n - 1$, while it carries all $\binom{n}{2}$ edges. It therefore meets the ceiling $\lambda^{\max} \leq m - 1$ exactly when $m \geq n$, and from that point on the question is settled trivially: the complete graph is feasible, no simple graph on n vertices is denser, so $\ell_m(n) = \binom{n}{2}$ for every $m \geq n$ and raising m further changes nothing. For smaller m the extremal graph must be strictly sparser than complete, and pinning down exactly how much sparser is the whole problem. That is why the theorem, and the thesis, assume $m \leq n$.

1.3 Three axes of variation

Mader's theorem is the floor we build on, and everything afterwards is a variation on it. There are three independent ways to change the rules, and they generate the landscape this thesis traverses.

The first axis is the *model*, drawn in Figure 1.3. Two independent choices generate it. The first is *arity*: an edge may join exactly two vertices, or a *hyperedge* may join any $r \geq 2$ of them, a route then passing through it between any two of its members. At $r = 2$ a hyperedge is an ordinary edge, so graphs are the $r = 2$ special case and the genuinely new setting begins at $r \geq 3$. The second is *multiplicity*: a *simple* object carries at most one copy of each edge, while a *multi* object allows parallel copies, recorded by a multiplicity $\mu(u, v)$, so a single pair can already carry several independent routes. Because the two choices are independent they form a small lattice. At the top sits the most general object, the *multi-hypergraph*, and lowering one toggle at a time restricts it downward: forbidding repeats gives the simple hypergraph, dropping the arity to two gives the multigraph, and doing both gives the simple graph at the bottom. Every theorem in this thesis is a statement about one node of this lattice, and the three rows of every comparison figure are its simple-graph, multigraph, and hypergraph levels.

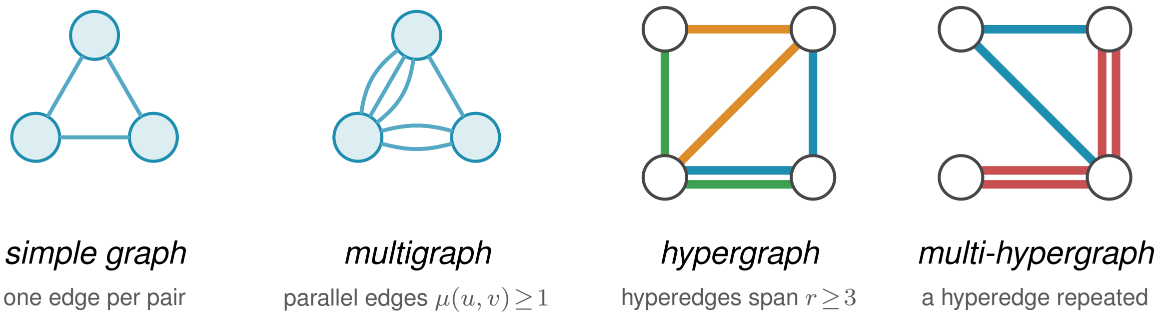


Figure 1.3. The model axis: a simple graph, a multigraph with parallel edges, a 3-uniform hypergraph drawn as metro lines, and a multi-hypergraph with a repeated hyperedge.

The second axis is the *direction*, a third toggle independent of the other two. Setting it at the top of the model lattice gives the single most general object in the thesis, the *directed multi-hypergraph*, of which every other class is a restriction. Figure 1.4 redraws the four models of Figure 1.3 with arcs in place of edges. An undirected object is the symmetric special case of a directed one, each edge standing for a matched pair of opposite arcs $u \to v$ and $v \to u$, so direction strictly enlarges the family of objects. What it does *not* do is let the directed and undirected problems merge into one. The two opposite arcs offer a round trip the single edge uv does not. This is why the extremal digraphs (directed graphs) are deliberately *lopsided*, with out-degree and in-degree pulled apart in a way no symmetric object can imitate, and why the Gomory-Hu tree that settles every undirected case has no directed analogue (Appendix A). The one-way variants therefore stay distinct in both difficulty and method.

A digraph carries three degree counts at each vertex. The *out-degree* $d^+(v)$ counts the arcs leaving v, and the vertices they reach are its *out-neighbours*. The *in-degree* $d^-(v)$ counts the arcs entering v, and the vertices they come from are its *in-neighbours*. The *total degree* $d(v) = d^+(v) + d^-(v)$ adds the two. A vertex with high out-degree fans arcs outward, one with high in-degree gathers them, and keeping the two balanced is the structural idea behind several of the directed constructions to come. The directed measure itself is easiest to meet on one concrete digraph. Figure 1.5 shows a digraph carrying three arc-disjoint routes between one chosen pair of vertices. The directed form of Menger's theorem says that route count equals the size of the smallest arc-cut separating the pair, and the figure also marks the out-degree and in-degree that bound both quantities from above, which is exactly how Menger reads in a one-way network.

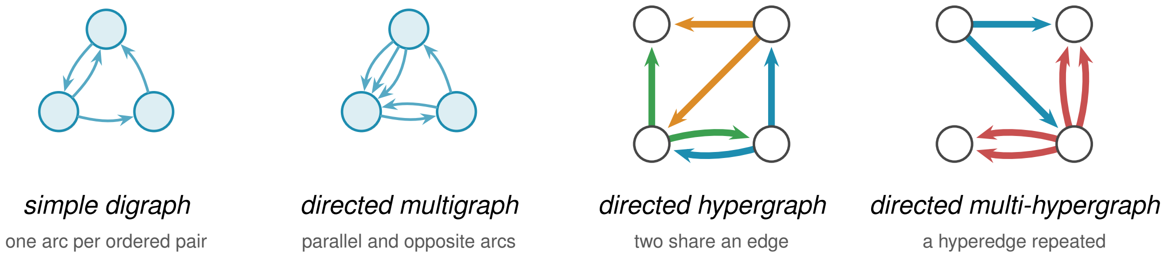


Figure 1.4. The four models of Figure 1.3 with direction added. A directed hyperedge is a star from one tail to its heads; repetition creates parallel copies of those arcs.

A directed r-edge has one tail and $r - 1$ heads. In the figures, one colour denotes one hyperedge, drawn either as a star or as a line oriented away from its tail; it is a single edge on several vertices, not a path of ordinary edges. A directed Berge route enters that edge at its tail and leaves at one of its heads.

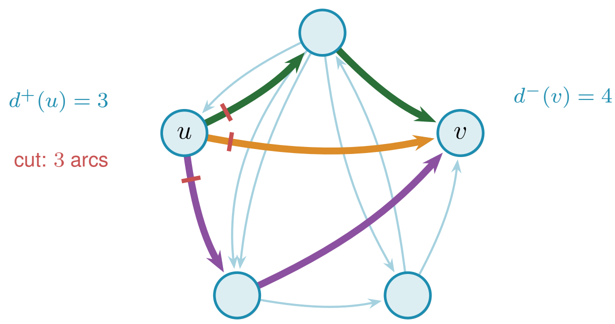


Figure 1.5. Three arc-disjoint $u \to v$ routes (coloured) and the matching three-arc directed cut (red). Thus $\lambda(u, v) = 3$, meeting the degree bound $\min(d^+(u), d^-(v)) = 3$.

The third axis is the *separation*: routes may be required to avoid sharing an edge, as above, or the stricter *internally vertex-disjoint*, sharing no intermediate vertex, which gives the local vertex-connectivity $\kappa_G(u, v)$ and its maximum $\kappa_G^{\max}$. Whitney's inequality $\kappa_G(u, v) \leq \lambda_G(u, v)$ [6] records that avoiding shared vertices is at least as demanding as avoiding shared edges, and Figure 1.6 shows the gap on a graph where two routes are edge-disjoint yet forced through one common vertex.

Three models, two directions, two separations give twelve versions of one question. This chapter proves the short cases before the machine treats the family uniformly; the final status

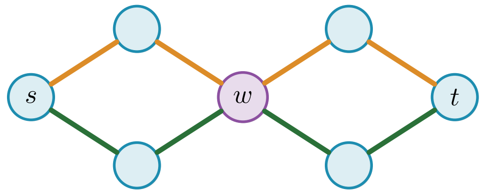


Figure 1.6. The separation axis: two edge-disjoint s–t routes both pass through w, so $\lambda(s,t) = 2$ but $\kappa(s,t) = 1$.

map appears in Figure 3.2.

1.4 Cases with short proofs

Allowing parallel edges removes the parity subtlety, the dependence on whether $m(n-1)$ is even or odd, that produces the floor in Mader's theorem, and the answer becomes exactly linear. Write $L_m(n)$ for the multigraph analogue of $\ell_m(n)$.

Theorem 1.3. *For undirected multigraphs on n vertices, $L_m(n) = (m-1)(n-1)$.*

Proof. For the lower bound, take a spanning tree of K_n, that is, a connected cycle-free subgraph that touches all n vertices and so has exactly $n-1$ edges, and give every tree edge multiplicity $m-1$, meaning $m-1$ parallel copies. The result has $(m-1)(n-1)$ edges. Any two vertices are joined by a unique path in the tree, of some length $k \geq 1$. To separate them one must cut some edge of that path, and the cheapest choice is the lightest tree edge on it, which still carries $m-1$ copies, so $\lambda(u,v) = m-1$ for every pair and $\lambda^{\max} = m-1$. The graph sits right at the ceiling.

For the upper bound, suppose $\lambda_G^{\max} \leq m-1$ and build a Gomory–Hu tree T of G (Theorem A.2), the same $(n-1)$-edge summary of all connectivities used for Mader's theorem. Each tree edge e stands for a minimum cut of G, of capacity $c(e) = \lambda_G(u,v) \leq m-1$ for the pair (u,v) it represents. Now charge each edge of G to a tree edge: an edge $\{x,y\}$ crosses the cut of the lightest tree edge on the tree path from x to y, so it is counted inside that tree edge's capacity. Summing the $n-1$ capacities therefore counts every edge of G at least once, which already gives $|E(G)| \leq \sum_{e \in T} c(e) \leq (m-1)(n-1)$. Unlike Mader's simple-graph count there is no second charge to halve, because parallel edges are now allowed and nothing forces an edge to be counted twice. The tree-of-multiplicity-$(m-1)$ construction above meets this bound exactly, which is why the answer is the clean product $(m-1)(n-1)$ and not a floor. □

The vertex-disjoint question is just as quick at the first non-trivial value. Forbidding two internally vertex-disjoint paths is a strong constraint: it forbids cycles.

Proposition 1.4. *For simple graphs, $k_2(n) = n-1$, attained exactly by trees.*

Proof. If G contains a cycle, a closed loop of distinct vertices, then any two vertices on that cycle are joined by the two paths of the cycle, which are internally vertex-disjoint, so $\kappa_G^{\max} \geq 2$. Hence

$\kappa_G^{\max} \le 1$ forces G to be acyclic, free of cycles, and an acyclic graph on n vertices has at most $n-1$ edges, with equality for trees. A tree indeed has $\kappa^{\max} = 1$: nonadjacent vertices have a unique path whose intermediate vertex separates them, while adjacent vertices are joined by their single edge alone. □

Here $k_m(n)$ denotes the vertex-disjoint extremal function, the counterpart of $\ell_m(n)$ for $\kappa^{\max}$. We have just seen $k_2(n) = n - 1 = \ell_2(n)$. The agreement of the edge and vertex problems is no accident at small m, and the moment it breaks is the first place the proofs begin to strain.

1.5 The first divergence: vertex-disjoint paths

Replacing edge-disjoint by internally vertex-disjoint routes changes nothing at first. Leonard proved that the edge and vertex problems give the same answer while m stays small.

Theorem 1.5 (Leonard [7])**.** *For simple graphs, all $n \ge m$ and $m \le 4$,*

$$k_m(n) = \ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor .$$

The hypothesis $n \ge m$ is the same one Mader's theorem carries, and it is not decoration. Below it the complete graph K_n is itself feasible, since $\kappa_{K_n}^{\max} = \lambda_{K_n}^{\max} = n - 1 \le m - 1$, so the answer is the trivial $\binom{n}{2}$ and the formula overshoots it. At $n = 3$, $m = 4$ the formula gives 4 where a graph on three vertices has only 3 edges. Every closed form in this thesis carries the same caveat, stated or inherited from its constructions, and the figures cap each curve at the trivial maximum for exactly this reason.

For $m \le 4$ the extremal values for the two problems coincide, and every edge-extremal graph is therefore also vertex-extremal: Whitney's inequality $\kappa \le \lambda$ makes every edge-feasible graph vertex-feasible (Figure 1.7, the three overlapping blue curves). Equality of the values alone does not assert the converse classification. One is tempted to expect the numerical agreement forever. It fails at the very next value, and the failure is sharp.

Theorem 1.6 (Sørensen-Thomassen [8])**.** *For simple graphs and all $n \ge 6$ with $n \ne 7$ and $n \ne 12$,*

$$k_5(n) = \left\lfloor \frac{8n}{3} \right\rfloor - 4,$$

while $k_5(7) = 15$ and $k_5(12) = 27$. The formula first exceeds the edge value at $n = 14$: for $6 \le n \le 13$ one has $k_5(n) = \left\lfloor \frac{5(n-1)}{2} \right\rfloor = \ell_5(n)$, and for every $n \ge 14$,

$$k_5(n) \; > \; \left\lfloor \frac{5(n-1)}{2} \right\rfloor = \ell_5(n).$$

Two points about how this is stated. The constant is -4 rather than the -3 printed in [8], because that paper counts the other way round, and Remark 1.7 does the conversion. The two exceptional sizes are genuine holes in the closed form rather than an artefact of the proof: at $n = 7$ it predicts 14 against the true 15, and at $n = 12$ it predicts 28 against the true 27, which is why Sørensen and Thomassen exclude both and give those two values separately. The divergence point matters for Chapter 3, which reads this theorem as the first place the edge and vertex problems part company. They agree throughout $6 \le n \le 13$, where the paper proves $k_5(n) = \lfloor 5(n-1)/2 \rfloor$ outright, and they separate from $n = 14$ onwards, where $k_5(14) = 33$ against $\ell_5(14) = 32$.

Remark 1.7 (A counting convention that shifts every quoted constant by one). The additive constant above needs a word of explanation, because two conventions are in circulation and they differ by exactly one. This thesis defines $\ell_m(n)$ and $k_m(n)$ as the *largest* number of edges a graph can carry while *no* pair is joined by m independent routes. Sørensen and Thomassen, and the Erdős Problems database [4] after them, state the same results the other way round, as the *smallest* number of edges that *forces* such a pair. Those two quantities are adjacent integers by construction: if k is the largest feasible edge count then no graph on n vertices with $k+1$ edges is feasible, so $k+1$ is exactly the forcing threshold.

The primary source settles which is which. [8] defines $f_k(n)$ as "the least integer r so that every graph with n vertices and r or more edges contains a k-rail", which is the forcing convention, and its Theorem 4 reads $f_5(n) = \lfloor 8n/3 \rfloor - 3$. Subtracting one gives the $\lfloor 8n/3 \rfloor - 4$ of Theorem 1.6. A k-rail there is a union of k paths pairwise meeting only at their two common endvertices, so it is precisely a pair at vertex-connectivity k, and the two problems are the same problem.

The shift is visible in every entry the database and this thesis both state. It gives $k_2(n) = n$ where a tree has $n - 1$ edges, $k_3(2n) = 3n - 1$ against $\lfloor 3(2n-1)/2 \rfloor = 3n - 2$ here, $k_4(n) = 2n - 1$ against $2n - 2$, $\ell_5(2n) = 5n - 2$ against $5n - 3$, $\ell_6(n) = 3n - 2$ against $3n - 3$, and it writes the Bollobás-Erdős conjecture as $1 + \binom{m}{2} n$ where the convention here gives $\binom{m}{2} n$. Every one is exactly one apart, and the values in this thesis are the ones an exhaustive search returns: $k_3(4) = 4$, $k_3(5) = 6$ and $k_4(5) = 8$ were each confirmed by walking every graph of that size.

One trap is worth naming, because an earlier draft of this thesis fell into it. The comparison $k_5 > \ell_5$ is invariant under the shift, since *both* functions move by one, so no internal check can detect a half-converted statement. What does detect it is evaluating the two sides in different conventions, which manufactures a spurious divergence one size early. Reading $\lfloor 8 \cdot 13/3 \rfloor - 3 = 31$ against this thesis's $\ell_5(13) = 30$ appears to separate the two problems at $n = 13$, but the 31 is a forcing count and the 30 is an avoiding count. Done consistently, $k_5(13) = 30 = \ell_5(13)$ and the separation begins at $n = 14$, which is also what [8] proves directly by showing $f_5(n) = \lfloor 5(n-1)/2 \rfloor + 1$ throughout $6 \le n \le 13$.

The range likewise comes from the paper rather than from the database, which records only the clean tail $n \ge 13$. Theorem 4 there covers every $n \ge 6$ apart from $n = 7$ and $n = 12$, and supplies those two values separately, which is where the $k_5(7) = 15$ and $k_5(12) = 27$ of Theorem 1.6 come from.

At $m = 5$ the vertex problem genuinely diverges from the edge problem, visible in Figure 1.7 as the shaded gap that opens between the dashed blue and solid orange curves. Leonard had already found the first crack, an explicit 57-vertex, 141-edge counterexample to the general conjecture at $m = 5$ [9], before Sørensen and Thomassen pinned the exact value down. The reason the crack opens at all is structural. Leonard's earlier method, the one that closes $m \leq 4$, leans on extremal graphs assembled from cliques. A k-*tree* starts from K_{k+1} and repeatedly joins one new vertex to every vertex of an existing k-clique. Through $m = 4$ these clique scaffolds are forced, and counting their edges gives the bound. At $m = 5$ they are no longer the unique extremal shape. Additional configurations appear, and the clean count breaks. Sørensen and Thomassen recover the exact value by a delicate structural induction, which we cite rather than reproduce. For $m \geq 6$ the exact value is unknown, and the vertex-disjoint problem has stood open since 1974. What is known there is a negative and a bound. Bollobás and Erdős had conjectured that the best graph is formed from as many copies of K_m sharing one vertex as fit, namely $\lfloor (n-1)/(m-1) \rfloor$ full copies plus a bounded remainder, which would give $k_m(n) = mn/2 + O(1)$. Mader disproved that for every $m \geq 6$, showing that the gap $k_m(n) - mn/2$ grows without bound [5]. Sørensen and Thomassen then supplied a replacement lower bound, $k_m(n) > \frac{m(m-1)-2}{2m-3}(n-m)$ for infinitely many n [8], whose rate exceeds $m/2$ by the factor $1 + (m-4)/(2m^2 - 3m)$, which is $1 + (1 + o(1))/(2m)$ for large m and only about 1.04 at $m = 6$. So the shape of the answer is known to be linear in n with a rate strictly above $m/2$, and it is the rate that is open. It is the one direction this thesis does not attempt to force, and Chapter 3 explains why.

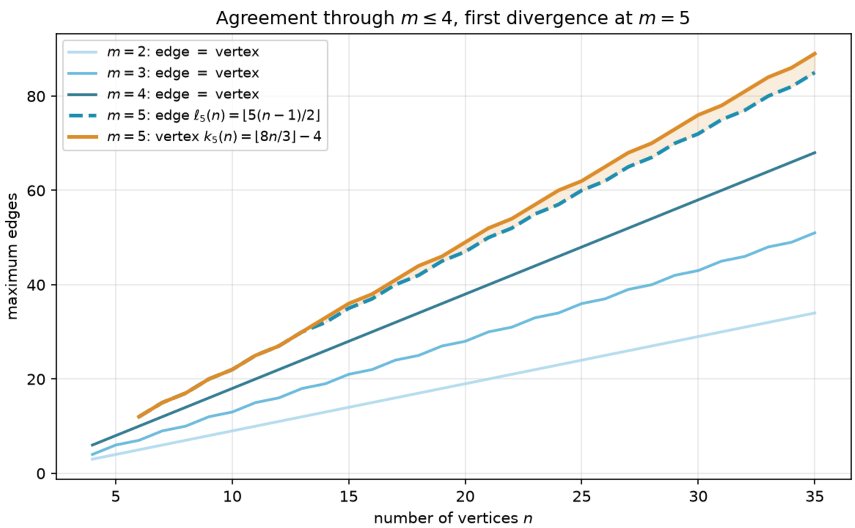


Figure 1.7. The edge and vertex extremal functions for $m = 2, 3, 4$ (blue shades, single curve per m since they agree) and $m = 5$ (dashed blue for edge, solid orange for vertex). Through $m = 4$ the two problems give identical answers, and at $m = 5$ the vertex formula $k_5(n) = \lfloor 8n/3 \rfloor - 4$ separates above the edge formula $\ell_5(n) = \lfloor 5(n-1)/2 \rfloor$, with the gap shaded. The two curves still touch at every n from 6 to 13 and part for good at $n = 14$, so the shading begins there.

1.6 The directed case and the quadratic phenomenon

Direction breaks a symmetry that undirected graphs never had. In an undirected graph the number of edges is tied to the connectivity through every vertex's degree. In a digraph a vertex can have many arcs in and none out, and a whole graph can lean entirely one way. Send every arc from one half of the vertices to the other and nothing comes back: between any two vertices there is at most one route, yet the number of arcs is quadratic.

Construction 1.8 (One-directional bipartite digraph)**.** Split V into parts A and B with $|A| = \lfloor n/2 \rfloor$ and $|B| = \lceil n/2 \rceil$, and include every arc from A to B. All arcs run from A to B, and none within A, within B, or from B back to A. Since B has no outgoing arcs, a vertex in A cannot reach any other vertex of A, and two vertices of B cannot communicate at all. For any $a \in A$ and $b \in B$ the only directed path from a to b is the single arc $a \to b$, so $\lambda(a, b) = 1$ and $\lambda^{\max} = 1$. The arc count is $|A| \cdot |B| = \lfloor n^2/4 \rfloor$.

Figure 1.8 draws the construction. Every arrowhead lands in B and none leaves it, so the picture is a wall of one-way traffic: from any a on the left there is exactly one way to reach any b on the right, the single arc between them, and there is no way at all to travel between two vertices on the same side. The arc count $|A| \cdot |B|$ is quadratic, yet no pair carries more than one route. This is the lopsidedness direction buys, and it has no undirected analogue. It competes with a second construction built around a single centre.

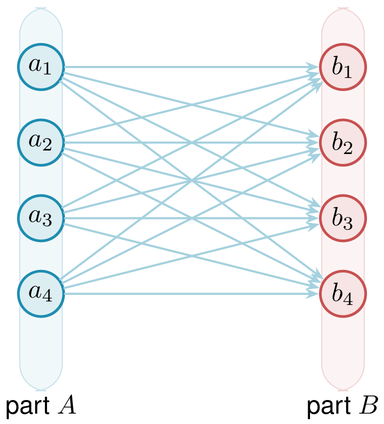


Figure 1.8. The one-directional complete bipartite digraph $B_{4,4}$. Its $16 = \lfloor n^2/4 \rfloor$ arcs all run from A to B, so every reachable pair has only its direct route and $\lambda^{\max} = 1$.

That second construction is the directed hub. It keeps each vertex's in-degree and out-degree balanced, the directed measure already met in Figure 1.5, and spends its arcs on a central vertex rather than a one-way wall.

Construction 1.9 (Directed hub)**.** For $n \geq m \geq 2$, take a hub vertex h and label the remaining $N = n - 1$ vertices $0, 1, \ldots, N - 1$. Add the $2N$ hub arcs $h \to i$ and $i \to h$ for every i, and on the non-hub vertices add the circulant arcs $i \to i + j \pmod N$ for $j = 1, \ldots, m - 2$, the target label wrapping around past N back to 0 (that ring pattern is what *circulant* means, and "$\mathrm{mod}\ N$"

is the wrap-around). Counting these gives $2N$ hub arcs (each of the N spokes joined to h in both directions) plus $(m-2)N$ circulant arcs ($m-2$ layers, one arc per spoke), so the total is $2N+(m-2)N = mN = m(n-1)$ arcs. Every non-hub vertex has $d^+ = d^- = 1+(m-2) = m-1$. Every ordered pair (u, v) has at least one non-hub member, and that member's relevant degree is exactly $m-1$, smaller than the hub's own degree $n-1$ since $n \geq m$, so it is always the binding one regardless of which of u, v is the hub. The directed degree bound (Lemma A.3, which says $\lambda(u, v) \leq \min(d^+(u), d^-(v))$, since a route out of u must use one of u's out-arcs and one of v's in-arcs) therefore gives $\lambda^{\max} \leq m-1$.

At $m = 2$ this construction is easiest to picture: the circulant layer is empty and what remains is the *bidirected star* of Figure 1.9, a hub joined to each spoke by a matched pair of opposite arcs. Every spoke can reach every other only by going in to the hub and back out, two hops, so no pair is ever joined by two independent routes, and the arc total is exactly $2(n-1)$. For $m \geq 3$ not all $m(n-1)$ arcs can touch the hub, since a simple digraph has at most $2(n-1)$ arcs incident to one vertex: the surplus lives in the circulant layers among the spokes, tuned so that no degree exceeds the budget. The two constructions compete on different scales, linear against quadratic, and which one wins depends on n. At $m = 2$ the competition resolves exactly.

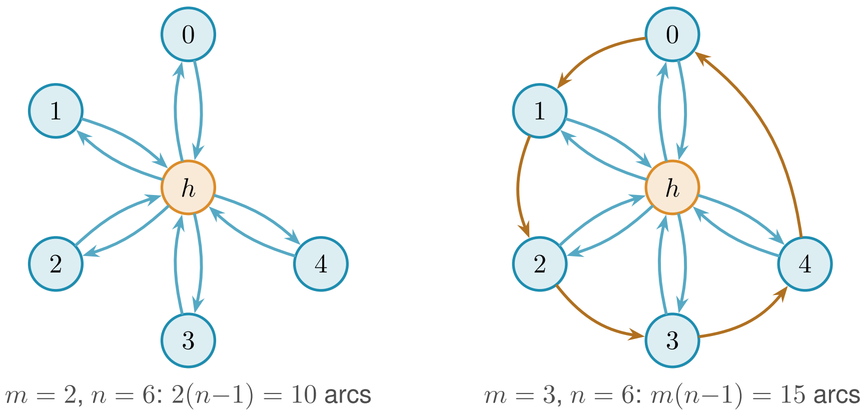


Figure 1.9. The directed hub at $n = 6$. Left: the $m = 2$ bidirected star has 10 arcs and $\lambda^{\max} = 1$. Right: adding one circulant layer gives the $m = 3$ construction with 15 arcs and $\lambda^{\max} = 2$.

Theorem 1.10 (Directed arc value at $m = 2$)**.** *For simple digraphs,*

$$\ell_2^{\mathrm{dir}}(n) = \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right).$$

The hub construction leads for $n < 7$, the one-directional bipartite construction leads for $n > 7$, and the two tie at $n = 7$, where both reach 12 arcs.

The lower bound is the better of the two constructions. The upper bound is where the trouble starts.

Idea of the upper bound at $m = 2$. The value $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ is a contest between a linear term and a quadratic one, and which leads is pure arithmetic: $2(n-1) \geq \lfloor n^2/4 \rfloor$

precisely for $n \le 7$, with equality at $n = 7$, where both give 12. So M follows the bidirected star up to the crossover and the bipartite digraph past it, and $n = 7$ is the single value where the two constructions tie. The proof that no digraph beats $M(n)$ is an induction on n (Appendix A). Write $a = |A(D)|$ for the number of arcs and delete a vertex v of *minimum* total degree. Two facts about that deletion meet. The degrees sum to twice the arc count, so the smallest of them is at most the average,

$$\deg(v) \;\le\; \frac{2a}{n},$$

and the remaining digraph on $n-1$ vertices still respects the ceiling, so once $n-1 \ge 7$ the inductive bound applies to it,

$$|A(D-v)| \;\le\; \left\lfloor \frac{(n-1)^2}{4} \right\rfloor .$$

The arc count of D itself splits as $a = |A(D-v)| + \deg(v)$, the arcs away from the deleted vertex plus the arcs at it. Feeding both facts into that split leaves a copy of a on each side, and gathering them before dividing by $1 - 2/n$ finishes the step:

$$\begin{aligned} a \;&\le\; \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + \frac{2a}{n}, \\ a\Big(1 - \tfrac{2}{n}\Big) \;&\le\; \left\lfloor \frac{(n-1)^2}{4} \right\rfloor, \\ a \;&\le\; \frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor . \end{aligned}$$

A short parity check then shows the right-hand side never exceeds $\lfloor n^2/4 \rfloor$ once $n \ge 8$: for even n it lands exactly on the bipartite value, and for odd n a sub-unit slack is swallowed by the fact that a is an integer. The averaging closes the step with no separate hub case, which is why the difficulty is not depth but bulk. Nothing in the argument is deep. It is a careful walk through two parity regimes, with the base cases $n \le 7$ settled by exhaustive computer search rather than by hand. That experience of a correct but long and almost entirely mechanical case-check is exactly what motivates the chapters that follow. One genuinely wants to hand the case-checking to a machine.

The vertex-disjoint version of the same question has the same answer at $m = 2$, but it is worth being careful about why, because Whitney's inequality is easy to read backwards here. Since $\kappa \le \lambda$ pairwise, a digraph that respects the arc ceiling automatically respects the vertex one, so the vertex-feasible family *contains* the arc-feasible one and is in general strictly larger. The free consequence therefore runs one way only: the two constructions above keep $\kappa^{\max} = 1$ as readily as $\lambda^{\max} = 1$, which makes them lower bounds for the vertex problem too. No upper bound comes with them. What settles the vertex value is that the same induction goes through unchanged, because deleting a vertex lowers $\kappa^{\max}$ for exactly the reason it lowers $\lambda^{\max}$, and because the base cases $n \le 7$ were re-run under the stricter separation and returned the same six numbers. Appendix A gives both halves, and Remark A.17 there spells out the direction trap in full.

Theorem 1.11 (Directed vertex value at $m = 2$)**.** *For simple digraphs,*

$$k_2^{\mathrm{dir}}(n) = \max\left(2(n-1),\ \left\lfloor \frac{n^2}{4} \right\rfloor\right) = \ell_2^{\mathrm{dir}}(n).$$

1.7 The failure of direct proof for $m \geq 3$

For $m \geq 3$ the induction no longer closes, and the natural guess is wrong. The natural initial conjecture was that the two constructions above continued to be the whole story, giving $\max(m(n-1), \lfloor n^2/4 \rfloor)$. A single graph on ten vertices refutes it.

Remark 1.12 (A thirty-arc counterexample). Take $A = \{a_1, \ldots, a_5\}$ and $B = \{b_1, \ldots, b_5\}$ with all 25 arcs from A to B, and add a directed five-cycle $b_1 \to b_2 \to b_3 \to b_4 \to b_5 \to b_1$ inside B. This digraph has 30 arcs. Every vertex of B has exactly one extra in-arc from inside B, so two arc-disjoint routes from a_i to b_j exist: the direct arc $a_i \to b_j$, and the detour $a_i \to b_{j-1} \to b_j$ using the 5-cycle predecessor of b_j (indices mod 5). For the matching upper bound, the only in-arcs of b_j reachable on a path from a_i are the direct arc $a_i \to b_j$ and the cycle arc $b_{j-1} \to b_j$. Removing both disconnects a_i from b_j, while neither alone does, so the minimum cut has size 2. Hence $\lambda(a_i, b_j) = 2$. Routes between two vertices of B never leave B and give $\lambda(b_i, b_j) = 1$. Hence $\lambda^{\max} = 2$, so the graph is feasible at $m = 3$, yet $30 > \max(3 \cdot 9,\ 25) = 27$.

Figure 1.10 draws this thirty-arc digraph and traces the mechanism. The twenty-five faint arcs are the complete one-directional bipartite layer from A to B, the construction we already understand. On top of them the orange five-cycle gives each vertex of B a single extra in-arc from inside B. That one extra arc is exactly what doubles the connectivity. For the marked pair a_1, b_2 the two heavy routes show how: the blue arc is the direct hop $a_1 \to b_2$, and the green path $a_1 \to b_1 \to b_2$ borrows the cycle arc $b_1 \to b_2$ for its second step. The two share no arc, so $\lambda(a_1, b_2) = 2$, and the same pair of routes exists for every a_i, b_j. Five extra arcs lift the whole digraph from one independent route per pair to two, and the thirty arcs overshoot the old prediction of twenty-seven.

The counterexample generalises: the five-cycle gave every vertex of B exactly one extra in-arc from inside B, and the budget for such arcs is $m - 2$ per vertex, not one. Figure 1.10 is the instance $m = 3$, where that budget is one and a single cycle suffices. For larger m each b_j collects $m - 2$ cyclic predecessors and the same picture acquires more concentric chords inside B.

Construction 1.13 (Augmented bipartite digraph)**.** For $m \geq 2$ and $n \geq m$, set $|B| = \lceil (n + m - 2)/2 \rceil$ and $|A| = n - |B| = \lfloor (n - m + 2)/2 \rfloor$, include every arc from A to B, and inside B give each vertex b_j the $m - 2$ in-arcs from its cyclic predecessors $b_{j-1}, \ldots, b_{j-(m-2)}$ (indices mod $|B|$). The total is $\lfloor (n + m - 2)^2/4 \rfloor$ arcs.

For the connectivity, fix $a \in A$ and $b \in B$. Two kinds of route run from a to b: the direct arc, and the two-step detours $a \to b' \to b$ that pass through an in-neighbour b' of b inside B. There are $m - 2$ detours, so $m - 1$ routes in all, and no two of them share an arc. They are also all

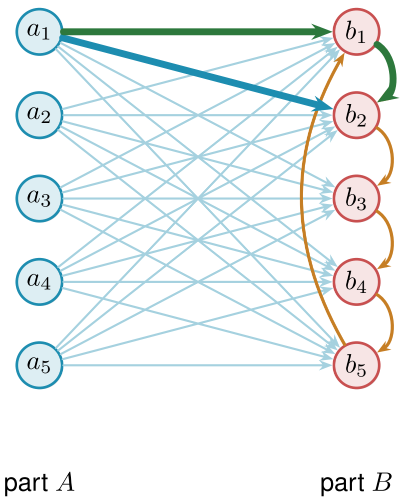


Figure 1.10. The thirty-arc counterexample at $m = 3$, $n = 10$: 25 arcs run from A to B and a directed 5-cycle lies inside B. The highlighted direct route and detour show the binding value $\lambda^{\max} = 2$.

there are, because the only in-arcs of b that a can reach are exactly those $m-1$ arcs. Deleting them therefore cuts every route from a to b. Pairs inside B are limited by the in-degree $m-2$, pairs inside A and pairs from B to A have no routes at all, so $\lambda^{\max} = m-1$.

At $m = 2$ the inside of B is empty and the construction reduces to Construction 1.8, with the balanced partition $(\lfloor n/2 \rfloor, \lceil n/2 \rceil)$. At $m = 3$ the formula gives $|B| = \lceil (n+1)/2 \rceil$, which coincides with $\lceil n/2 \rceil$ at odd n but gives $|B| = n/2 + 1$ at even n. Both the balanced and the shifted partition achieve the same arc count $\lfloor n^2/4 \rfloor + \lceil n/2 \rceil$ in this case, so the two are interchangeable at $m = 3$. The shift to a strictly larger B is structurally significant only at $m \geq 4$, where the balanced partition is suboptimal. At $m = 3$, $n = 10$ it is the thirty-arc counterexample of Remark 1.12, drawn in Figure 1.10 (using the equivalent balanced partition $|A| = |B| = 5$).

The two rival shapes invite a political picture. The hub is a single king through whom all traffic must pass, cheap to build but limited by the one throne. The bipartite family is a flat cabinet of ministers, each handling its own dossier with no central seat, and it scales quadratically. Which arrangement packs more arcs under the ceiling depends on n, and for $m \geq 3$ the cabinet overtakes the king once n is large. This is the corrected shape, and it leads to the standing conjecture for the directed arc problem.

Conjecture 1.14. *For simple digraphs and all $n \geq m \geq 2$,*

$$\ell_m^{\mathrm{dir}}(n) = \max\left(m(n-1),\ \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor \right).$$

The hypothesis $n \geq m$ is the one both constructions already carry, and it is needed for the same reason as in Theorem 1.5: below it the complete digraph is feasible and the answer is the trivial $n(n-1)$, which the formula exceeds. At $n = 3$, $m = 4$ the formula gives 8 where three vertices carry only 6 arcs. The exhaustive search confirms the value is 6 there.

For $m = 2$ this is Theorem 1.10. For $m = 3$ the formula $\lfloor (n+1)^2/4 \rfloor$ equals $\lfloor n^2/4 \rfloor + \lceil n/2 \rceil$, so the conjecture reduces to the previous formula for $m = 2$ and $m = 3$. For $m = 3$, $n = 10$ the second branch gives $\lfloor 121/4 \rfloor = 30$, matching Remark 1.12. For $m \geq 4$ the new formula and the old balanced-partition count $\lfloor n^2/4 \rfloor + (m-2) \lceil n/2 \rceil$ no longer agree. The quadratic branches already differ, and at $m = 4$ the new one is larger by exactly one at every even n. At small n, though, the hub term $m(n-1)$ dominates both branches and hides that difference. The full conjectured value therefore first overtakes the old one only at $n = 12$, for $m = 4$. The construction of Construction 1.13 with the shifted partition witnesses the difference. The lower bound holds for all m, witnessed by Construction 1.9 on the first branch and Construction 1.13 on the second. For fixed $m \geq 3$, the matching upper bound is open, though not entirely: Theorem A.25 proves unconditionally that the value is $n^2/4 + \Theta_m(n)$, so what the conjecture still claims beyond what is known is the coefficient of that linear term rather than the shape of the answer. Chapter 3 returns to it with what structural progress the machine and the hand can jointly supply.

1.8 The hypergraph variant

The third model deserves its own short treatment, not because its base case is hard but because it is the one variant whose objects are stored and manipulated differently from all the rest, and it is worth meeting that difference now rather than at the moment of computation. A hypergraph keeps no multiplicity matrix, only the list of its hyperedges, each a set of r vertices. We fix the edge size r, following the extremal literature, for a structural reason: under a connectivity ceiling a small edge is always at least as efficient per route as a large one, so a hypergraph free to mix sizes would maximise its edge count by using pairs only, and the question would collapse back to the graph case. Fixing r is what makes the hypergraph question a new question.

The route between two vertices is a *Berge path*, which alternates vertices and hyperedges so that each consecutive pair of vertices lies in the hyperedge listed between them, and two such routes are edge-disjoint when they share no hyperedge. Concretely, in a three-uniform hypergraph a route from u to w might be the single step $u, \{u, x, w\}, w$ that crosses one hyperedge, or a longer alternation $u, \{u, x, y\}, y, \{y, z, w\}, w$ that switches hyperedge at the intermediate vertex y.

With routes redefined this way the same Gomory–Hu machinery that proved the multigraph value carries over, and the base case has the same linear shape. The picture to keep in mind is a metro map: each hyperedge is a line, and a route from one vertex to another rides a line and changes to another line at a station the two share.

A single hyperedge may serve only one route, so two Berge routes that share no hyperedge are edge-disjoint, the hypergraph counterpart of two edge-disjoint paths in a graph. In the metro picture, two hyperedges that share several vertices are distinct lines calling at several of the same stations, as in Figure 1.3. Two routes are hyperedge-disjoint exactly when they never ride the same line.

The separation axis needs its own definition here, and it is worth fixing now because the natural guess is not the one this thesis uses. Two Berge routes are called *internally disjoint* when they share neither an intermediate vertex nor a hyperedge. Both halves are required. Demanding only that the routes avoid each other's intermediate stations would let two routes ride the same line, and a line that can run only one train at a time cannot serve two routes at once, so the resulting count would not correspond to anything the metro map can carry. Requiring both is also what makes the vertex measure the exact analogue of the graph one under the translation of Appendix A, where a hypergraph becomes its incidence graph, each hyperedge becomes a node, and internally disjoint routes become internally disjoint paths in the ordinary sense. It has the pleasant side effect that Whitney's inequality survives the move to hypergraphs by definition, since an internally disjoint family is in particular hyperedge-disjoint. Write $\kappa_{\mathcal{H}}(u,v)$ for the largest such family and $\kappa_{\mathcal{H}}^{\max}$ for its maximum over pairs.

Proposition 1.15 (Hypergraph edge bound)**.** *An r-uniform hypergraph on n vertices with Berge edge-connectivity at most $m-1$ has at most $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. A simple hypergraph attains the bound whenever $m-1 \le \binom{n-2}{r-2}$, and a multihypergraph attains it whenever $(r-1) \mid n-1$.*

The proof, given in Appendix A, runs the multigraph argument through the hypergraph Gomory–Hu theorem (Theorem A.47, the same tree construction of Theorem A.2 applied to the hyperedge-cut function in place of the edge-cut function), charging each hyperedge to the at least $r-1$ tree cuts it crosses. The star hypertree (Figure 1.11) attains the $m=2$ case, a hub joined to disjoint blocks of $r-1$ vertices, with Berge connectivity one everywhere and $\lfloor (n-1)/(r-1) \rfloor$ hyperedges, exactly as a tree attained the multigraph bound. One genuine surprise is worth recording here: for $r \ge 3$ the bound is attained by *simple* hypergraphs, with no repeated hyperedge, whenever $m-1 \le \binom{n-2}{r-2}$ (Theorem A.48). In the graph case simplicity is exactly what separates Mader's $\lfloor m(n-1)/2 \rfloor$ from the multigraph value $(m-1)(n-1)$. One edge size higher, it costs nothing.

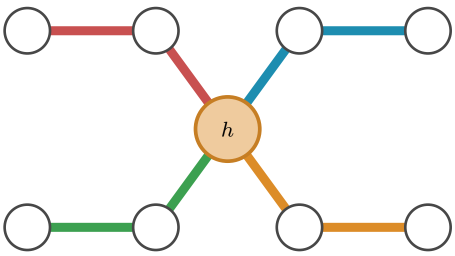


Figure 1.11. The $r=3$, $n=9$ star hypertree: four hyperedges meet only at the hub h, attaining $\lfloor (n-1)/(r-1) \rfloor = 4$ with Berge connectivity one.

The edge bound is the easy half of the hypergraph story. The vertex separation is where this thesis has something new to say, and its two statements are worth having in front of the reader before the machinery that establishes them.

Theorem 1.16 (Hypergraph vertex value at $m=2$)**.** *For $r \ge 2$ and $n \ge 1$, the maximum number* *of hyperedges of an r-uniform multihypergraph on n vertices with $\kappa^{\max} \leq 1$ is*

$$k_2^{(r)}(n) \;=\; \left\lfloor \frac{n-1}{r-1} \right\rfloor,$$

attained by the star hypertree of Figure 1.11. In particular $k_2^{(r)}(n) = \ell_2^{(r)}(n)$: the vertex and edge problems agree at $m = 2$ for every uniformity, and repeated hyperedges never help.

Theorem 1.17 (Hypergraph vertex value at $m = 3$)**.** *For $r \geq 2$, every r-uniform multihypergraph on n vertices with $\kappa^{\max} \leq 2$ has at most $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ hyperedges. For $r \geq 3$ with $2 \leq \binom{n-2}{r-2}$ the bound is attained by a simple hypergraph, so*

$$k_3^{(r)}(n) \;=\; \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor \;=\; \ell_3^{(r)}(n).$$

Both are proved in Appendix A. The first falls out once Berge routes are read as ordinary paths in the incidence graph, where the ceiling $\kappa^{\max} \leq 1$ says exactly that the incidence graph is a forest. The second is the harder one and carries the genuinely new tool: a rank bound for graphs in which one side of a partition must stay below local 3-connectivity, proved by way of Tutte's triconnected decomposition, which the incidence translation then turns back into the hypergraph statement. At $m \geq 4$ that route stops for want of a 4-connectivity analogue of the decomposition, and the value is open.

The proposition counts hyperedges, but it does not say how to *measure* the Berge connectivity of a hypergraph we are handed, and that measurement is the genuinely new part. The metro picture shows why. An ordinary edge is a one-stop link between a single pair, so policing edge-disjointness means watching each pair on its own. A hyperedge is a whole line: it stops at all r of its vertices and so touches $\binom{r}{2}$ pairs at once, yet like a line that can run only one train at a time it may serve only one route. When several routes want to board the same line at different stations, the checker must wave exactly one through and turn the rest away, and it must do this for every line simultaneously. That bookkeeping is the one new piece of machinery the next chapter builds, and it is why the hypergraph rejoins the family only once the checker is in place.

1.9 Where this work sits

It is worth pausing, before the machine, to place the question and the method against the two traditions they descend from: the extremal graph theory that posed the problem and the computational mathematics that will help settle it.

The mathematical lineage is the older of the two. The problem belongs to the family of extremal connectivity questions opened by Bollobás and Erdős, who asked how dense a graph can be while every pair of vertices stays below a fixed connectivity [2], logged today as Erdős Problem 915 [4]. Mader's theorem [5] closed the simple undirected edge case with the clean floor $\lfloor m(n-1)/2 \rfloor$ recalled in Theorem 1.2, and it is the floor every variant in this thesis is measured against. The vertex-disjoint branch has a longer and rougher history. Leonard [7] showed that the edge and vertex problems coincide through $m \leq 4$, and Sørensen and Thomassen [8] found the first divergence at $m = 5$, where $k_5(n) = \lfloor 8n/3 \rfloor - 4$. Past that point the exact values $k_m(n)$ remain unknown. What is known there is one negative and one bound: the original conjecture of Bollobás and Erdős is false for every $m \geq 6$ [5], and a general lower bound is available [8]. That gap between a settled edge case and a stalled vertex case is exactly the terrain Chapter 1 has been mapping, and the directed and hypergraph axes extend it further. To this particular lineage, the undirected vertex problem, the thesis adds no new closed form. It does add exact formulas and asymptotic results on the directed and hypergraph axes.

The computational tradition this thesis's pipeline belongs to is the practice of settling combinatorial questions by machine in a way a human can audit, and it splits into a few schools worth contrasting with the certify-then-prove loop built next. The oldest is *exhaustive generation*: enumerate every object of a given size up to isomorphism and check each one. McKay and Piperno's `nauty`/`geng` toolkit [10] is the standard engine for this, and its flagship results include exact small Ramsey numbers such as $R(4,5) = 25$ [11]. The enumeration of Chapter 2 is the same idea, run on the twelve variants of Problem 915. The second school is *SAT-assisted proof*, in which a question is encoded in propositional logic and a solver emits a machine-checkable certificate. Two results are the canonical examples of certified solver output at scale: Heule, Kullmann and Marek's resolution of the Boolean Pythagorean triples problem [12], whose verified proof runs to some two hundred terabytes, and Heule's determination of Schur number five [13]. The third is *flag algebras*, Razborov's framework [14] that converts asymptotic extremal bounds into semidefinite-programming certificates, the dominant tool for density problems in the large-n limit. Closest of all to the method here is the fourth, *certification by linear and integer programming* (LP/ILP). A feasibility or optimality argument over the rationals or integers is written as a linear program, meaning an optimisation whose constraints are all linear. Such a solve can in principle be accompanied by primal and dual data that a separate checker verifies. The cut-counting optimisation of Chapter 2 records the solver's status and an integral primal witness, but does not emit that independently replayable certificate. Section 2.13 states the evidential standard used here.

Against that backdrop, what is distinctive here is consolidation rather than a new solver. A single program measures connectivity exactly across all twelve problem variants through one common interface, using a matrix representation for graphs and a separate compact representation for hypergraphs, so search and certification share one codebase rather than living in separate tools. Its connectivity outputs and witnesses are exact integral objects. An optimiser's `OPTIMAL` status is recorded as machine evidence and is not presented as a formal certificate by itself. The general statements that use those finite values also have the independent hand proofs in Appendix A. Because the same program both hunts for dense graphs and checks the bounds they suggest, the discover and finite-verification steps are two modes of one loop. What makes the consolidation honest is that it never blurs three kinds of knowledge: a value can be *proved* (by hand or exhaustive enumeration), *solver-verified* at a finite size, or only *conjectured* (witnessed by a dense graph with no matching upper bound). This trichotomy, named in Section 2.13 and obeyed by the summary of Chapter 3, is the through-line of everything that follows.

Computer note

We are now stuck in three different ways: a fifty-year-old open problem at $m \geq 6$ for vertices, a long proof we would rather not check by hand at $m = 2$ for arcs, and at $m \geq 3$ a conjecture of which only the leading behaviour can be proved. The base cases of the long induction were verified by exhaustive search, and the thirty-arc counterexample was found the same way. The next chapter builds a single program that models all of these cases at once, measures their connectivity exactly, and proves the small-case upper bounds outright.

Chapter 2

Certifying and Discovering Bounds by Machine

The last chapter ended with several variants stuck for different reasons, but all ask one machine to do the same three jobs. It must represent every model, measure connectivity exactly, and then use that measurement either to *prove* a finite optimum by exhaustive certification or to *discover* a dense feasible graph. This chapter builds that single pipeline in that order. The epistemic distinction matters more than the shared implementation: measurement is exact, completed enumeration proves, and heuristic search supplies only a lower bound. Figure 2.1 is the map. The small cards below name each routine and its interface; the function bodies stay in the archive because the argument needs the contract, not the code listing.

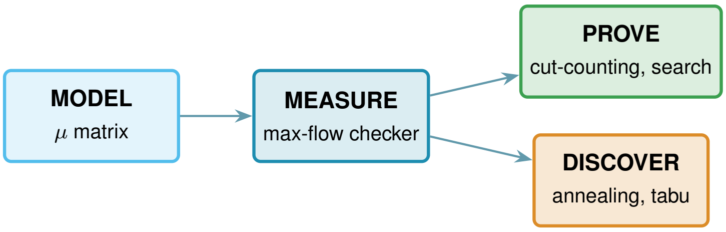


Figure 2.1. The architecture as one pipeline. A single model feeds the exact max-flow checker; certification turns that measurement into finite upper bounds, while simulated annealing or tabu search uses it to discover lower bounds. The badge colours on the code cards match the boxes here.

2.1 The solver

Everything in this chapter is a single function, `solve`, and it helps to hold its most basic version in mind from the start, because every later idea is a refinement of it and nothing is ever a separate program for a separate case. Given n and m, the basic strategy is as plain as it sounds. Walk through every graph on n vertices, measure the largest local connectivity $\lambda^{\max}$ of each with a max-flow computation, and keep the densest graph that stays at or below the ceiling $m - 1$. Because it has looked at every graph, the count it returns is the exact answer, not an estimate.

That one loop already settles the smallest cases, and the rest of the chapter is built on top of it. We first make the measurement step trustworthy and quick, via the connectivity checker and a Gomory–Hu shortcut for undirected graphs. We then make the enumeration step skip whole families of hopeless graphs, via a pruned search. Finally, for the directed multigraph case, we describe the cut-counting optimisation used for the historical finite checks. The public `solve` interface now returns the general hand theorem directly for that variant, while the optimisation remains available as an independent experiment.

`solve(n, m, directed, simple, separation, exhaustive, max_seconds)` **DRIVER**

in the variant booleans `directed` and `simple` (or `hypergraph` with uniformity `r`), the `separation` (edge or vertex), n, m, the `method`, and a time budget `max_seconds`

out a `SolveResult` carrying a value with an honest label: exact, lower bound, or upper bound

how picks the method the case allows: brute-force enumeration, the pruned search, or the cut-counting certifier when proving, and the temperature search or tabu search when discovering. It always respects the budget, and it is the one function the rest of the thesis ever calls.

2.2 A single representation for all variants

All graph variants use one multiplicity matrix μ: it is symmetric when undirected and binary when simple. Hypergraphs are stored as r-sets, with a tail distinguished in the directed model. Consequently the same checker, certifier, and search routines serve every variant; only the capacity reduction and separation test change.

2.3 The connectivity checker built on maximum flow

The whole question reduces to a flow computation. A *maximum flow* asks how much can be pushed from one point to another through a network whose links each carry a limited capacity. Menger's theorem says the most edge-disjoint u–v paths a graph offers equals its smallest u–v cut, the fewest links whose removal severs every route between the two. The max-flow / min-cut theorem of Ford and Fulkerson [15] says that smallest cut is exactly the value of a maximum flow, once each multiplicity $\mu(u, v)$ is read as the capacity of the link $u \to v$. So the route count we are after is just a maximum-flow value: measure the flow and you have measured the connectivity. The routine that runs this flow and reports the route count is the connectivity *checker*. It is the single most important routine in the program: it is the only place graph theory meets computation, and every later claim about connectivity passes through it. Because the temperature search below calls it by the million on tiny graphs, the program ships an optional compiled C helper that runs this same flow (named later in this chapter's reproducibility section). It is a speed-up only: it returns the identical value the pure-Python checker does, verified against it, so no number in the thesis depends on whether the helper is built.

`max_edge_connectivity(G)` $\to \mathbb{N}$ **MEASURE**

in a graph G

out $\lambda^{\max}(G)$, the largest local edge-connectivity over all pairs

how one max-flow per pair (ordered if directed), take the largest. This is the quantity the edge problem caps at $m-1$.

The vertex-disjoint version follows the same idea after one construction: split each vertex v into an in-copy v^{in} and an out-copy v^{out} joined by a single unit-capacity arc (Figure 2.3a). Every arc $u \to v$ of the original graph becomes $u^{\text{out}} \to v^{\text{in}}$ in the split network. The construction is not special to digraphs: for an undirected graph each edge $\{u, v\}$ is first read as the two opposite arcs $u \to v$ and $v \to u$, and then the same split applies, so one routine serves the undirected and directed vertex problems alike. Routing through v now consumes the one unit of capacity on its internal arc, so two paths sharing v as an intermediate vertex compete for that unit and cannot coexist in any maximum flow. A max-flow from u^{out} to v^{in} in the split network therefore equals the number of internally vertex-disjoint u–v paths in the original graph.

That sweep is `max_vertex_connectivity`, identical to the edge checker except that each flow runs on the split capacity matrix, and it gives every adjacency capacity one, so parallel edges never raise κ. That last convention deserves to be said in prose, because it does not merely change how the checker computes, it changes what the multigraph vertex question asks. In vertex mode every adjacency carries capacity one, so a bundle of parallel edges counts as a single direct route. The rationale is that vertex-disjointness is about intermediate vertices, and parallel copies offer no new intermediate vertices to be disjoint over.

Remark 2.1 (What parallel edges mean in vertex mode). That choice has a consequence which has to be stated rather than left implicit. If parallel copies never raise κ, they never break feasibility either, so one could pile arbitrarily many copies onto any feasible multigraph and stay under the ceiling forever. Counted with multiplicity, the maximum number of edges would simply be infinite, and the extremal question would be empty. The two multigraph vertex variants are therefore posed here with the objective counting *adjacencies*, that is, pairs joined by at least one edge, rather than edges with multiplicity. Under that reading the question is exactly the simple question on the underlying graph, which is why those two rows of Table 3.1 reduce to their simple counterparts and why every figure in this thesis reports the reduced problem for them.

If parallel copies are instead counted as distinct direct routes, one obtains a different extremal problem, outside the twelve variants studied here. A single edge is a path with no interior, so q parallel copies between one pair are q internally disjoint routes on their own, and feasibility then caps every multiplicity at $m-1$ by itself. That question is genuinely different rather than a relabelling of a solved one, and Section A.9 settles what can be settled about it. The incidence graphs used in Lemma A.53 are simple, so that alternative never affects the arguments below.

The checker has an exact mode for reported values and a capped yes-or-no mode for search, which stops after finding one route too many. For undirected edge connectivity, one Gomory–Hu tree reduces the pairwise sweep from $\binom{n}{2}$ flows to $n-1$ and serves as an independent cross-check.

A hyperedge is handled by adding one helper point, split by a unit-capacity arc, and joining it to every member (Figure 2.3b). Every route using that hyperedge must traverse the bottleneck, so two such routes cannot coexist. The hypergraph Menger theorem (Theorem A.46) proves that the resulting maximum flow is exactly the Berge connectivity, and at $r=2$ the construction reduces to the ordinary graph checker. Figure 2.2 shows four gate-disjoint routes in a whole hypergraph.

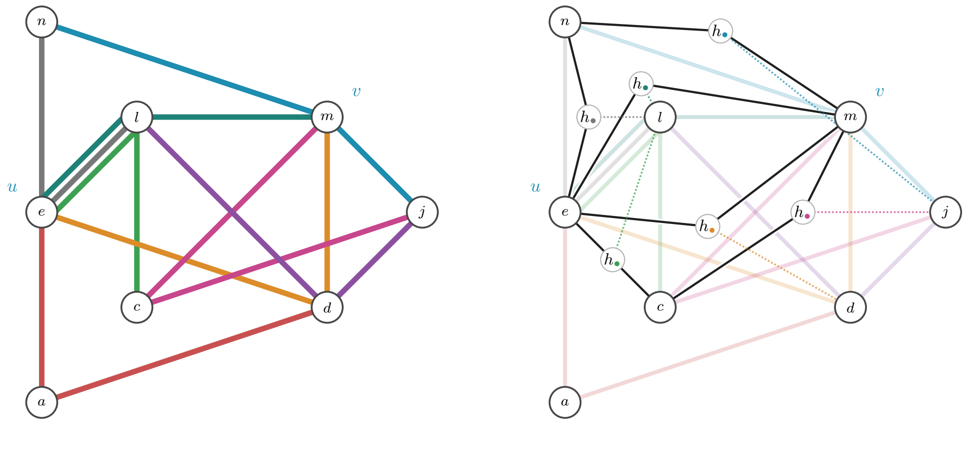
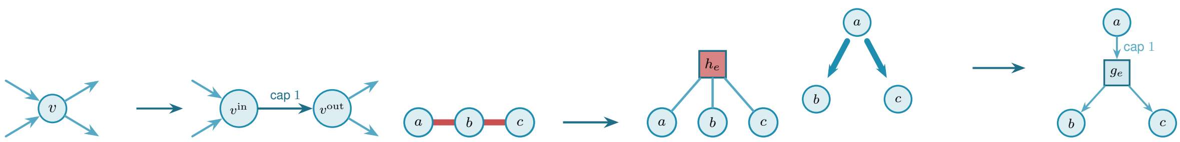


Figure 2.2. An undirected 3-uniform hypergraph and its gate network for u, v. The four black routes use distinct capacity-one gates, so the checker reads $\lambda(u,v)=4$; a fifth would reuse a gate.

Self-tests recover $n-1$ on K_n and 3 on $K_4^{(3)}$. Splitting the ordinary vertices as in Figure 2.3a gives the vertex-disjoint hypergraph checker; only direction still has to be added.

2.3.1 The directed hypergraph checker

A *directed* hyperedge has the one-tail, $r-1$-heads convention fixed in Chapter 1, and a directed Berge route may enter it only at the tail and leave at a head. The flow gadget makes the helper point directional: for $e=(a;b,c,\dots)$ it adds a gate g_e, a capacity-one arc $a \to g_e$, and arcs $g_e \to b$, $g_e \to c$, and so on (Figure 2.3c). Thus a route reaches the gate only through the tail, and at most one route uses the hyperedge. Directed and undirected hyperedges therefore use the same capacity-one gate; only the incident wiring changes.



(a) Split an ordinary vertex. **(b)** Gate an undirected hyperedge. **(c)** Orient the hyperedge gate.

Figure 2.3. The three capacity-one reductions used by the checker. Each forces competing routes through one unit-capacity bottleneck; direction changes only how the gate is wired.

Adding ordinary vertex splitting to the directed gate measures the vertex variant as well, so the same reduction covers all four hypergraph cases. The gate is indifferent to how a hyperedge's vertices are split between tails and heads. It draws one capacity-one arc in from every tail and one out to every head, so the single forward tail it was built for is only the simplest case: give it a hyperedge with several tails and one head, or several of each, and it still counts the routes that enter at a tail and leave at a head. The same checker therefore measures every *orientation model* of a directed Berge edge, not just the one-tail one, and Section 3.3.1 takes up which of those models are worth distinguishing and what the program finds for them.

With the checker in place the proposed value of Proposition 1.15 can be tested on any small hypergraph the way every other variant is, which is the subject of the next section.

2.4 Checking every graph of a fixed size

The checker measures one graph. To bound *every* graph of a given size, the most direct method is the one the base cases of Chapter 1 already needed: enumerate them all. A simple graph on n vertices is a choice of $\mu(u, v) \in \{0, 1\}$ on each off-diagonal cell, a multigraph a choice in $\{0, \ldots, m-1\}$, and a digraph varies every ordered pair rather than only the upper triangle. Walking that product, measuring each candidate with the checker, and keeping the densest feasible one settles the value exactly, and because it has looked at every graph it is a genuine upper bound, not evidence.

This is the first thing `solve` does when it is asked to prove a small case. With its exhaustive flag set it walks that product, measures each candidate with the checker, and keeps the densest feasible one. The value it returns is exact and exhausted, a true upper bound for that n, but only while the count of graphs stays small. No new routine is involved, only `solve` choosing to enumerate.

On the small cases this exhaustive mode independently reproduces every value proved by hand in Chapter 1; those values are not repeated here.

The honesty of the method is also its limit. The number of graphs it must visit grows as $2^{\binom{n}{2}}$ for simple undirected graphs and $2^{n(n-1)}$ for digraphs, and capping multiplicities at $m-1$, so that each cell ranges over the m values $\{0, \ldots, m-1\}$, raises the base from two to m, giving $m^{\binom{n}{2}}$ undirected multigraphs and $m^{n(n-1)}$ directed ones. The enumeration is exact wherever it finishes, and it finishes only for the smallest sizes. Section 2.8 measures exactly how fast this wall arrives. For now the point is that the wall arrives precisely where Chapter 1 needed help, at the directed base cases $n = 6, 7$, so the next sections reach a little further than blind enumeration can.

2.5 Reaching further: a pruned search for the directed base cases

The base cases $n \leq 7$ of the directed $m = 2$ theorem (Theorem 1.10) are the finite checks the induction of Appendix A depends on, and they sit just past where blind enumeration of digraphs is tractable. They can be reached by enumerating more cleverly. Feasibility is monotone: adding an arc can never lower $\lambda^{\max}$, so once a partial digraph is infeasible every completion of it is infeasible too and the whole branch is abandoned at once. A counting bound prunes further. At any partial digraph, count the arcs already placed and add the most the still-undecided cells could ever contribute. That sum is a ceiling on every completion of the branch, so if even the ceiling cannot beat the best feasible digraph found so far, the branch is dropped unexplored (Figure 2.4). What remains is a branch and bound over simple digraphs with the connectivity checker as the feasibility test, exact and complete at these sizes where the blind walk is not.

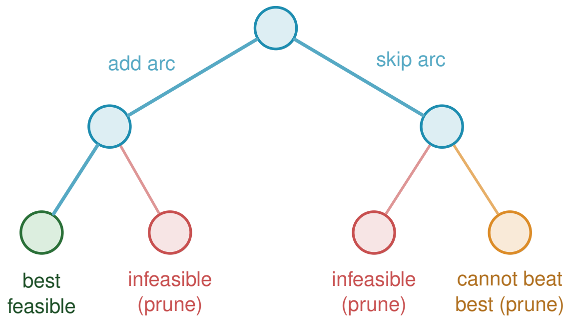


Figure 2.4. Branch and bound over digraphs. Each level fixes one adjacency; infeasible partial graphs (red) and branches unable to beat the incumbent (orange) are discarded, allowing the exact search to reach $n \leq 7$.

Run on the directed $m = 2$ problem, with `solve` switching to this branch and bound behind an unchanged interface and the booleans alone deciding which engine it uses, the sweep returns the complete sequence $2, 4, 6, 8, 10, 12$ for $n = 2, \dots, 7$, each proved complete at its size. The densest digraph it finds at each n is exactly the extremal shape Chapter 1 named, the bidirected star at every $n < 7$ and, at the crossover $n = 7$ itself, a graph of the tying value 12 that both the hub and the one-directional bipartite digraph attain. The sweep stops there, so the bipartite side of the crossover is witnessed by the construction and the checker rather than by this search (Section 2.12 records that the cooled walk in fact stalls on the hub at $n = 8$). These are exactly the base cases the induction requires, and with them the long proof of Theorem 1.10 rests on a single auditable search rather than on anything done by hand. Table A.1 tabulates that sweep beside its vertex counterpart.

2.6 Generating the directed cases faster

Theorem A.31 (Appendix A) now proves the directed multigraph value $L_m^{\mathrm{dir}}(n) = (m-1)M(n)$ by hand, for every n and m at once, so nothing in this section is load-bearing for that theorem any more. It remains useful for a narrower and genuinely separate question: which $n = 7$ extremisers satisfy the maximum-total-degree cap 8 used in the extension analysis (Remark A.42). This is not the unrestricted classification, since every doubled bidirected tree is extremal at the tie size, and the program lists only the capped family up to renaming, in two independent ways: when two different methods return the same list, that agreement is itself a check.

The first way is a search written from scratch: a depth-first walk over multiplicity matrices that abandons a branch the moment it turns infeasible or can no longer beat the best graph found, keeping one representative of each isomorphism class by reducing every finished graph to its canonical relabelling, the smallest matrix in dictionary order. The canonical form keeps the memory small, but the time is spent walking labelled partial graphs, and time is what runs out: the full classification at $n = 5$ costs this in-house search about 456 seconds, and it climbs steeply from there.

The second way, suggested by Prof. Jan Goedgebeur, does not search at all, it generates. It hands the hard part, listing one graph per isomorphism class, to the `geng` program of the `nauty` toolkit of McKay and Piperno [10], invoked through a subprocess and decoded from its `graph6` output. `geng` lists the non-isomorphic *undirected* graphs on n vertices, and the program decorates each one with the directed multiplicities the problem needs, so the isomorphism reduction is inherited for free from the support graph. The decoration walks the support's edges in a fixed order and gives each edge $\{u, v\}$ an ordered pair of multiplicities $\mu(u, v), \mu(v, u)$, each between 0 and $m - 1$ but not both zero, since an edge of the support must carry at least one arc. The checker measures the partly decorated multigraph as it goes, and the moment some pair exceeds the connectivity ceiling the whole branch of assignments is abandoned, since adding further arcs can never lower a connectivity, the same monotonicity prune as Figure 2.4. A decoration surviving to the end with the target arc count is kept and reduced to canonical form, so each isomorphism class is reported once, and different supports share nothing, so they are processed independently across the processor cores. The companion orientation tools `directg` and `watercluster2` are not used, because each gives only one orientation per edge and so cannot express the bidirected arcs and repeated multiplicities the extremal graphs are built from. To be precise about credit: Prof. Goedgebeur is not an author of `nauty`, which is the work of McKay and Piperno cited above. He suggested the generation route, and it is his suggestion that the speed-up rests on.

The two ways agree to the last graph. The generated pipeline reproduces the in-house classification exactly at every settled size, two non-isomorphic extremal families at $n = 4$ and three at $n = 5$, and it runs substantially faster, since the isomorphism work it never repeats is precisely what `geng` has already done. One difference decides which to trust further out: the in-house search leans on a conjectured counting bound once $n \geq 7$, while the generated enumeration prunes only with proved bounds, so it is sound and complete for the parameter range and any explicitly imposed degree cap. That soundness has now paid off. Pointed at the degree-bounded classification at $n = 7$, the generation enumerator completed the sweep in about seven hours across ten processor cores and returned exactly three extremal families, $2 \cdot B_{3,4}$, $2 \cdot B_{4,3}$, and the doubled bidirected path P_7 (Remark A.42).

2.7 Proving every m at once

The pruned search proves one (n, m) at a time. Historically, the directed multigraph checker instead scaled out m and tested every threshold at a fixed n in one pass. The general theorem of Theorem A.31 has superseded this route in `solve`, but the formulation remains an independent finite check.

Divide each multiplicity of a feasible multigraph by $m-1$. This produces weights $w(u,v) \in [0,1]$, pairwise maximum flow at most one, and total weight $|A|/(m-1)$, as Figure 2.5 illustrates. Thus, with

$$M^*(n) = \max\Big\{\textstyle\sum_{u\neq v} w(u,v) : w(u,v) \in [0,1],\ \operatorname{maxflow}(s,t;w) \le 1 \text{ for all } s,t\Big\}$$

for the largest total weight of such a matrix, scaling gives

$$L_m^{\mathrm{dir}}(n) \ \le\ (m-1)\, M^*(n) \qquad \text{for every } m \ge 2.$$

The reverse inequality requires an optimum whose entries scale back to integers. The historical optima for $n \le 6$ were $\{0,1\}$ double stars, so there

$$L_m^{\mathrm{dir}}(n) = (m-1)\, M^*(n) \qquad \text{for every } m \ge 2,$$

for every m. The hand proof now establishes this identity for every n (Corollary A.38); the finite optimisation is retained to record the independent route by which the cases $n \le 6$ were first checked.

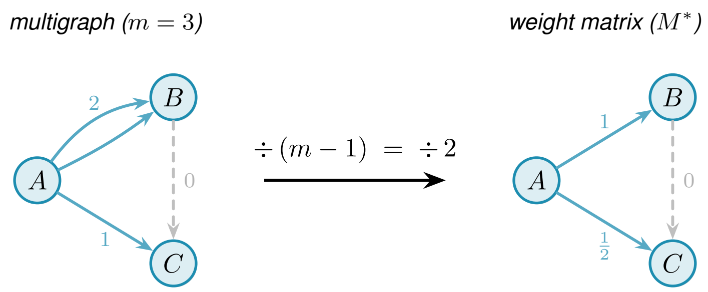


Figure 2.5. Scaling at $m = 3$: dividing multiplicities by $m - 1 = 2$ turns the integer multigraph into a weight matrix with entries in $[0, 1]$ and pairwise maximum flow at most one. Hence one m-free optimisation bounds every threshold.

By max-flow/min-cut, $\operatorname{maxflow}(s,t) \le 1$ exactly when some s–t cut has capacity at most one. The optimisation therefore chooses one cut per ordered pair. A binary label x_u^{st} records its side, and p_{uv}^{st} counts $w(u,v)$ only when $u \to v$ crosses from source to sink (Figure 2.6). This describes the exact feasible region. The archived wrapper retained an `OPTIMAL` status and primal witness but no independently replayable matching bound, so the runs are finite solver checks rather than formal certificates.

```
prove_directed_multigraph(n) → ProofResult                                PROVE
```

in the vertex count n (and tightening flags such as `use_deletion_cuts`)

out $M^*(n)$ with the solver's termination status and a primal weight matrix. A replayable upper-bound certificate is not emitted

how builds the cut-counting MILP in `pulp`, runs CBC or Gurobi, and records the status and primal matrix; `prove_integral_arc_bound` supplies a separate integer cross-check.

With scaled weights w, binary side labels x, and non-negative crossing weights p, the model is

$$\begin{aligned}
&\text{maximise} && \sum_{u\neq v} w(u,v) && \\
&\text{subject to} && p_{uv}^{st} \geq w(u,v) + x_u^{st} - x_v^{st} - 1 && \text{for all } (s,t),\ (u,v), \\
& && \sum_{u\neq v} p_{uv}^{st} \leq 1 && \text{for all } (s,t), \\
& && w(s,t) + \sum_x \min\bigl(w(s,x), w(x,t)\bigr) \leq 1 && \text{for all } (s,t), \\
& && d(v_0) \leq d(v_1) \leq \cdots \leq d(v_{n-1}), && \text{with } d(v) = \textstyle\sum_{u\neq v}\bigl(w(u,v) + w(v,u)\bigr), \\
& && w \in [0,1], \quad x \in \{0,1\}. &&
\end{aligned}$$

Because $w \leq 1$, the first constraint forces $p_{uv}^{st} \geq w(u,v)$ only in the crossing case $(x_u^{st}, x_v^{st}) = (1,0)$; in the other three cases its right-hand side is non-positive. The second constraint then caps the chosen cut at one. The remaining two lines are redundant accelerators: the first counts the direct arc and all two-step detours, and the second removes equivalent vertex relabellings.

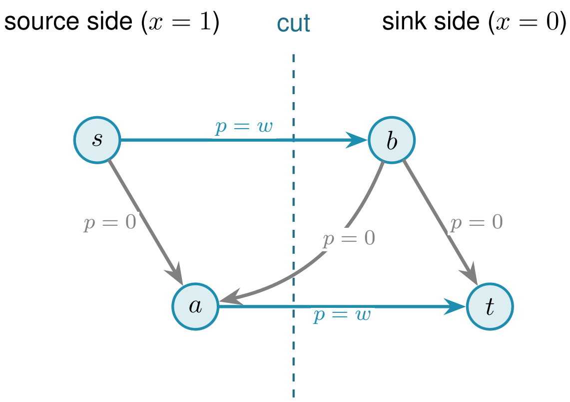


Figure 2.6. One chosen s–t cut. Only the two thick source-to-sink arcs contribute crossing weight $p = w$; same-side and backward arcs have $p = 0$.

Theorem 2.2 (Directed multigraph value, small n)**.** *For $n \leq 6$, $M^*(n) = 2(n-1)$, and the optimum is attained by a $\{0,1\}$ weight matrix. Hence for all $m \geq 2$, $L_m^{\mathrm{dir}}(n) = 2(n-1)(m-1)$, attained by the double star.*

Historical runs gave $M^*(n) = 2(n-1)$ with status OPTIMAL for $3 \le n \le 6$; $n = 7$ did not close. Their archived output lacks a separately checkable bound or gap, and no theorem now depends on it: Appendix A proves the value for all n, m by hand and Section A.10 retains the solver record.

Certification is exact only while it can cover every candidate. The second half of this chapter measures where that ends and replaces the exhaustive walk by a selective one: simulated annealing or tabu search explores a few graphs well, using the same checker to keep the best feasible construction. Its output is a concrete graph and therefore a lower bound, never an upper bound.

2.8 From certification to discovery: the enumeration wall

The exhaustive methods above visit every graph of a given size. The trouble is how many that is. A simple undirected graph on n vertices is a yes-or-no choice on each of the $\binom{n}{2}$ possible edges, so there are $2^{\binom{n}{2}}$ of them. A digraph chooses on each of the $n(n-1)$ ordered pairs, giving $2^{n(n-1)}$, and allowing multiplicities up to a cap c raises the base from two to $c+1$. These are not numbers that merely grow. They detonate. Figure 2.7 draws the growth on a logarithmic scale, where each step up the axis multiplies the count tenfold, and the message is the same in every panel: the count passes any reachable budget while n is still in the single digits.

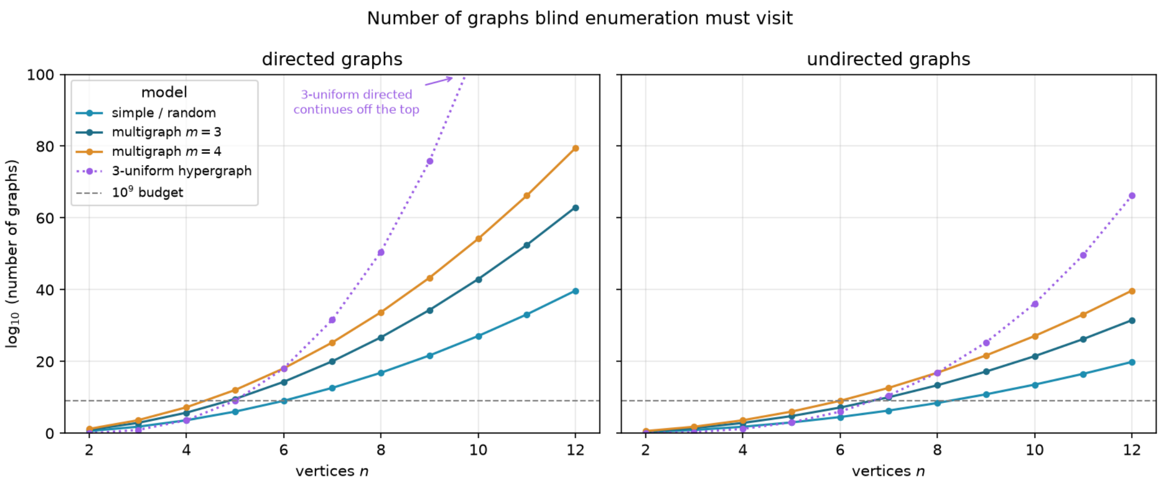


Figure 2.7. Candidate counts for blind enumeration on a logarithmic scale. Every model crosses the generous 10^9 budget in single-digit order; directed objects grow fastest because they have more independent cells to fill.

Three readings of the figure matter for what follows. Adding vertices is the sharpest cost: each new vertex multiplies the count by an exponential in n, so a single extra vertex can move a problem from seconds to centuries. Raising the connectivity budget is the next cost, lifting the base of the exponent rather than its height, which is why the multigraph curves sit above the simple one in every panel. Direction is the third: a digraph has twice as many cells to fill as the undirected graph on the same vertices (and a directed 3-uniform hyperedge, a tail with two heads, has three times as many), so the directed panel climbs fastest of all, and those are precisely the variants whose answers Chapter 1 could not prove. The separation, by contrast, costs nothing here, because edge and vertex disjointness search the same graphs and differ only in what the checker measures. The cut-counting certifier reaches a few vertices further than blind enumeration, but solving its encoding is itself hard and the gain is measured in single vertices, not orders of magnitude. Past that, no exhaustive method reaches the interesting sizes.

2.9 The temperature search

We want, for fixed n and m, a graph with as many edges as possible subject to $\lambda^{\max} \leq m-1$ (or $\kappa^{\max} \leq m-1$ for the vertex variants). The space of graphs on n vertices is enormous and its feasible region has no helpful shape: adding an edge can be harmless until, together with edges added long before, it suddenly creates a forbidden pair. A greedy walk that only ever accepts an improving move stalls at the first such collision, in a graph that is locally maximal but globally far from extremal.

The classical method for this kind of landscape is *simulated annealing* (SA), named after the metallurgical process of cooling a heated material slowly so that its atoms settle into a low-energy crystal rather than a disordered glass. Applied to graphs, annealing escapes local optima by sometimes accepting a worsening move, and by accepting fewer of them as the run proceeds. Each graph G is a state with energy

$$E(G) = -|E(G)| + \Lambda \cdot \max\big(0,\ \lambda^{\max}(G) - (m-1)\big),$$

which rewards edges and penalises any excess connectivity. The penalty is a soft constraint rather than a hard one, so the walk may pass briefly through infeasible graphs rather than being walled inside the feasible region, and we return below to why that freedom is worth having. A move toggles one adjacency. The Metropolis rule accepts an improving move always and a worsening move only with probability $e^{-\Delta E/T}$, a chance that shrinks both as the move gets worse and as the temperature drops, and the temperature T falls geometrically, multiplied by a fixed factor just below one at every step.

This is what `solve` runs in its discovery mode. With the exhaustive flag off it anneals instead of enumerating, using geometric cooling and removals biased toward low-sensitivity edges, and returns the densest feasible graph it encountered together with the full temperature trace. What it returns is a witness, hence a lower bound, never a proof.

`search_for_dense_graph(n, m, variant, separation)` $\rightarrow$ `SearchResult` **DISCOVER**

in the size n, the ceiling m, the `Variant`, the separation, and a step budget

out a `SearchResult`: the densest feasible graph found, with the full step-by-step trace

how one Metropolis cooling run. The driver `best_of_searches` dispatches several independent runs across processor cores and keeps the densest, so a single local optimum never decides the answer.

It is fair to ask why the walk is allowed above the ceiling at all. Reaching every feasible graph does not require it. Removing an edge can only lower $\lambda^{\max}$ and $\kappa^{\max}$, never raise them, so every subgraph of a feasible graph is again feasible, and the empty graph is feasible trivially. From any feasible target we can therefore delete edges one at a time down to the empty graph without ever crossing the ceiling, and reading that path backward builds the target up the same way. The feasible region is connected through the empty graph by single-edge moves that stay inside it, so a walk that never went infeasible could in principle visit every feasible graph there is.

We let it go infeasible anyway, because reachability and efficiency are different things. A feasible graph can be locally maximal, meaning every edge we might add creates a forbidden pair, and yet still sit far below the densest feasible graph. To improve such a graph the walk has to add an edge and then remove a different one, a swap whose first half is infeasible. A walk confined to feasible graphs could only find that better graph by retreating most of the way back to the empty graph and climbing a different slope, a detour so long that the cooling schedule usually ends first. Allowing a brief excursion above the ceiling lets the walk tunnel straight across instead, trading one load-bearing edge for a better one in two moves rather than dozens. The soft penalty also turns the search into a single smooth landscape with no hard walls for the Metropolis rule to bounce off, and the same machinery then carries over unchanged to variants whose feasible region is not so conveniently connected. Going above the bound is never the goal, then. It is the shortcut the annealer takes on its way back down to it.

2.10 Temperature and edge sensitivity

The removal bias is where the temperature acquires its meaning, and it is the heart of the method. For an edge e define its *sensitivity*

$$\sigma(e) = \lambda^{\max}(G) - \lambda^{\max}(G - e),$$

the amount by which deleting e relaxes the binding connectivity. A load-bearing edge sits on a tightest cut and has $\sigma(e) > 0$. Removing it would drop $\lambda^{\max}$ and so loosen the very constraint that limits how many edges the graph may hold. A free edge has $\sigma(e) = 0$: it can be removed without easing the binding pair, which usually means it can be put to better use elsewhere. The program measures this by deleting e and letting the checker remeasure (`edge_sensitivity`, with a whole-graph version `sensitivity_map` that reuses one shared baseline). This single number is the dial the cooling turns.

When the search proposes a removal it draws the edge with probability proportional to $e^{-\sigma(e)/T}$. The dial is now explicit. At high temperature these weights flatten and the walk toggles edges regardless of how load-bearing they are, exploring widely and willing to dismantle essential structure. As the temperature falls the weights concentrate on $\sigma = 0$, so the walk almost never disturbs a load-bearing edge and instead rearranges only the free ones, until the graph crystallises onto an extremal shape. Thus the temperature tracks how strongly removing an edge changes the bound.

The same sensitivities also explain a finished graph. Colouring a graph's arcs by σ shows at a glance which arcs are structural and which are slack. Figure 2.8 illustrates all four sensitivity levels on one directed multigraph: the darkest arcs carry the most flow and lower $\lambda^{\max}$ the most when removed, the cool arc contributes nothing. The picture is not decoration: it makes the tightness visible, showing at a glance which arcs are load-bearing, and it tells the search exactly which arcs to leave alone as the temperature falls.

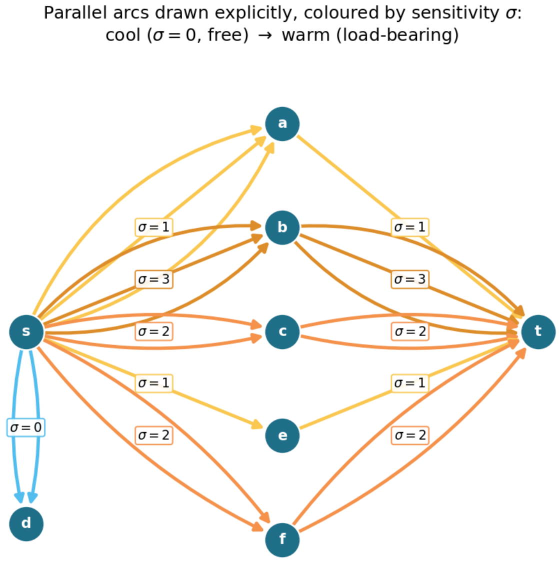


Figure 2.8. Arc sensitivity in a directed multigraph with $\lambda^{\max} = 9$. Equal multiplicities can have different σ: the downstream bottleneck makes $s \to a$ cooler than $s \to b$, while arcs into the dead end d have $\sigma = 0$.

A single cooling schedule can still settle into a local optimum, so in discovery mode `solve` runs several independent schedules from different starting seeds and keeps the densest result across all of them. Because the restarts share no state and none depends on the outcome of another, they run in parallel. The program dispatches them across the available processor cores, so the wall-clock cost of k restarts is a small multiple of a single run rather than k times it, bounded below by the slowest restart and by the cost of starting the worker processes. A handful of parallel restarts is enough to reach the known optima at the sizes we examine. This is a reliability measure, not a change of method, and the parallelism is invisible to every number the thesis reports: the same best graph would be found by running the schedules sequentially, just more slowly.

2.11 Keeping the checker cheap inside the search

The energy above reads the binding connectivity $\lambda^{\max}(G)$, and a literal implementation would call the checker afresh on the whole graph at every step. That checker is the most expensive routine in the program, one maximum flow per ordered pair, and the walk runs for thousands of steps with several restarts, so such a search would spend nearly all of its time remeasuring graphs that barely changed from the step before. Two elementary facts remove almost all of that work without altering a single number the search reports.

Proposition 2.3 (Monotonicity of the binding connectivity)**.** *Let G be a graph or directed multigraph, and let G' be obtained from G by changing one adjacency by a single unit of multiplicity. Then*

(1) (monotonicity) *removing the unit cannot raise $\lambda^{\max}$, and adding it cannot lower $\lambda^{\max}$, and*

(2) (unit step) *adding the unit raises $\lambda^{\max}$ by at most one.*

Both statements hold verbatim for the vertex measure $\kappa^{\max}$.

Proof. Fix an ordered pair (s, t). By the max-flow / min-cut theorem $\lambda_G(s, t)$ is the smallest capacity of any cut separating s from t, where the capacity of a cut is the total multiplicity of the arcs that cross it from the source side to the sink side.

For part (1), reducing one adjacency by a unit lowers the capacity of every cut that the changed arc crosses and leaves all other cut capacities untouched, so no cut capacity rises. The smallest cut capacity therefore cannot rise either, which gives $\lambda_{G'}(s, t) \leq \lambda_G(s, t)$. Taking the maximum over all pairs yields $\lambda^{\max}(G') \leq \lambda^{\max}(G)$. Adding a unit is the same statement read backwards.

For part (2), write G'' for G with one unit added to a single adjacency. That extra unit of capacity crosses any fixed cut at most once, so every cut has capacity at most one greater in G'' than in G. The smallest cut capacity thus grows by at most one, giving $\lambda_{G''}(s, t) \leq \lambda_G(s, t) + 1$, and together with part (1) we have $\lambda_G(s, t) \leq \lambda_{G''}(s, t) \leq \lambda_G(s, t) + 1$. The maximum over pairs gives $\lambda^{\max}(G'') \leq \lambda^{\max}(G) + 1$.

For $\kappa^{\max}$ the argument is identical on the vertex-split network of Figure 2.3a, where changing an adjacency changes the capacity of its single arc $u^{\text{out}} \to v^{\text{in}}$ in the same way. Parallel copies leave that capacity at one and so change $\kappa^{\max}$ not at all, which only strengthens both parts. □

These two facts settle almost every step without a flow. The energy depends on $\lambda^{\max}$ only through the excess $\max(0,\ \lambda^{\max} - (m - 1))$, which is zero exactly when the graph is feasible. So a removal applied to a feasible graph leaves it feasible by part (1), with excess still zero, and needs no measurement at all. An addition made while the graph sits strictly below the bound, at $\lambda^{\max} \leq m - 2$, also stays feasible. Part (2) caps the rise from a single added unit at one, so the graph can climb at most to $\lambda^{\max} = m - 1$, which is still inside the feasible region. The excess is zero again, and again no measurement is needed. Only an addition made right at the boundary $\lambda^{\max} = m - 1$ can tip the graph over, and only there does the search run a single flow, capped so that it stops the instant it finds one route too many. The exact connectivity value is recomputed only on the rare step whose proposal is genuinely infeasible, where the penalty term needs it.

To know which case applies, the search carries a running upper bound on $\lambda^{\max}$, raised by one whenever it accepts an addition (part (2)) and left untouched when it accepts a removal (part (1)), and resynchronised to the exact value at the cadence where it already pays for flows to refresh the sensitivity map. The bound can only ever sit at or above the truth, so any step it clears is certainly feasible.

The key point is that the energy returned on every step is, by Proposition 2.3, exactly the value the naive implementation would have computed. Every accept-or-reject decision, every random draw, and the graph the walk finally returns are therefore unchanged, and the speed-up is invisible to every number the thesis reports. This is verified rather than asserted. A regression test runs both implementations, the fast one and a reference that measures exactly at every step, across all four graph variants and both separations, and checks that the two agree step for step in energy, in every accept-or-reject decision, and in the final graph, and that the returned graph is feasible under the exact checker. Nothing about the optimisation hides inside the bound: the witness the search reports is always certified by an exact measurement, exactly as in the slower search.

It is worth noting where this method earns its keep. For undirected graphs one could resynchronise the exact value cheaply from the Gomory–Hu shortcut used above, which encodes every pairwise edge-connectivity in $n-1$ flows. The directed and vertex variants admit no such tree, so there the monotone bound and the capped predicate are the only inexpensive handle on feasibility, and they are what let one annealer serve every variant at speed.

2.12 Rediscovering the known extremisers

A search method is only worth trusting if it recovers what is already known before it is pointed at what is not. Run from the empty graph on the cases settled in Chapter 1 or by the exact methods above, and told only to pack in edges without breaching the connectivity ceiling, the cooled search arrives unprompted at the named constructions and matches the established value in every tested case. Table 2.1 reports the densest graph it found against the value the theory predicts.

Construction	n	m	Search	Theory
spanning tree (simple undirected, m=2)	6	2	5	5
multiplicity-$(m-1)$ star (multi undirected, m=3)	6	3	10	10
bidirected star (simple directed, m=2)	4	2	6	6
bidirected star (simple directed, m=2)	6	2	10	10
hub–bipartite tie (simple directed, m=2)	7	2	12	12
double star (multi directed, m=3)	4	3	12	12
double star (multi directed, m=3)	5	3	16	16

Table 2.1. Validation by rediscovery. For each variant the temperature search, run from scratch with a handful of restarts, recovers exactly the extremal value proved or certified earlier. The "Search" column is an honest lower bound found by annealing, and the "Theory" column is the proved or certified value.

What the table hides is that the graphs themselves are the named constructions, not merely graphs of the right size. The simple undirected run at $m = 2$ settles on a spanning tree, the only shape with $n-1$ edges and no cycle. The undirected multigraph run settles on a star with every edge at multiplicity $m-1$, an instance of the multi-tree extremiser behind Theorem 1.3. The directed multigraph runs settle on the double star of Construction 1.9, both directions of every spoke at full multiplicity. The simple directed runs find the bidirected star at $n=4$ and, at $n=7$, land on twelve arcs, the value at which the hub and the one-directional wall tie, exactly the crossover Theorem 1.10 predicts. The cooled search does not know these names. It rediscovers the structures because, under the connectivity ceiling, there is nowhere denser to go.

Some extremisers are past the size at which a quick search lands reliably, but the exact checker built above settles them immediately by measuring their connectivity. The one-directional bipartite digraph on eight vertices has 16 arcs at $\lambda^{\max}=1$, and 16 already exceeds the linear hub value $2(n-1)=14$: the smallest size at which the quadratic phenomenon strictly wins, made concrete. It is also a measured illustration of the search's limit: repeated cooling runs at $n=8$, $m=2$ stall at the hub value of 14, because dismantling a finished hub and rebuilding a one-directional wall is far more than the single-arc moves of the walk can tunnel through. The known answer there comes from the construction and the checker, not from the search, which is exactly the division of labour this chapter argues for. The augmented bipartite digraph on ten vertices at $m=3$ has 30 arcs at $\lambda^{\max}=2$, the thirty-arc counterexample of Remark 1.12, confirmed pair by pair against the naive bound of 27 it refutes. The directed double star on seven vertices at $m=4$ has 36 arcs at $\lambda^{\max}=3$, matching $2(n-1)(m-1)$. None of these is a proof of optimality. Each is an exactly measured feasible graph, a lower bound the checker stamps as genuine.

The discovery driver defaults to tabu search, using the same energy and single-adjacency neighbourhood but taking best-improving moves with a short tabu tenure and occasional perturbations. Simulated annealing remains an independent cross-check. Both return checked witnesses and therefore lower bounds, never proofs.

2.13 Results and evidential status

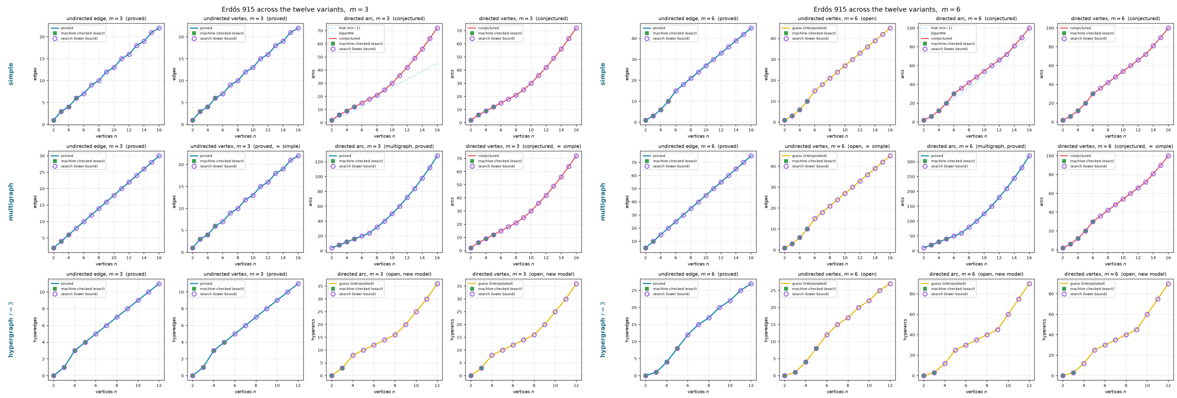
The checker *measures* a supplied graph exactly; completed exhaustion *proves* a finite optimum; the historical cut-counting optimisation is an independent finite check unless accompanied by a replayable bound; and heuristic search supplies only a witness and lower bound. A value is settled only when its lower and upper bounds meet. All computational claims rerun from `program/erdos915_unified.py`; critical values are cross-checked by independent implementations, while Section A.10 records the environment, commands, and audit evidence. No general theorem rests on an unauditable solver status.

The directed simple arc problem at $m\geq 3$ shows the boundary cleanly. Exhaustion proves the values $6,9,12,15$ for $n=3,4,5,6$ at $m=3$ under both separations (Remark A.18), but does not finish at $n=7$. Beyond that point the best reported lower bound is the better of the search and a named construction. The conjectured formula

$$\ell_m^{\mathrm{dir}}(n)=\max\left(m(n-1),\ \left\lfloor\frac{(n+m-2)^2}{4}\right\rfloor\right)\qquad(n\geq m)$$

passes through every exhausted value and through the construction-backed lower bounds beyond them. Its linear-to-quadratic crossover occurs at $n = 9$ when $m = 3$, just past the exhausted range, and at $n = 10$ it gives the thirty-arc counterexample of Remark 1.12.

The paired grids apply that reading to all twelve variants. Blue curves are theorems, red curves conjectures, and yellow curves interpolate lower bounds; filled squares are exact finite values and open circles are witnesses. Directed vertex values were checked separately because Whitney's inequality transfers witnesses but not upper bounds.



(a) $m = 3$ **(b)** $m = 6$

Figure 2.9. All twelve variants at $m = 3$ and $m = 6$. Rows select the model and columns the direction and separation; colours encode theorem, conjecture, or interpolation, while filled squares are exact finite values and open circles are witnesses.

Where an upper bound exists, the search reaches it at every plotted size except the recorded directed $m = 2$, $n = 8$ stall. On the open directed sizes the named constructions, not the cooled walk, carry the best bounds once the extremal shape changes from a hub to a bipartite wall. The grid therefore validates the search where proof can check it and marks the remaining values honestly as evidence rather than conclusions.

Chapter 3

Synthesis, Results, and Open Problems

The journey set out from one clean theorem and followed it across twelve variants, proving what proved cleanly, building a machine where the proofs ran out, certifying the small cases the machine could close, and using the search to discover the dense graphs the proofs had singled out and to point at the ones still open. This closing chapter draws the threads together. It maps the directed frontier, where proof and certificate both still fall short of the full answer, names the two genuine surprises the landscape contains, gathers the twelve answers into one picture, and says plainly what the program changed and where it stops.

3.1 The directed frontier and the missing decomposition

The directed arc problem is a race between a hub with $m(n-1)$ arcs and an augmented bipartite construction with $\lfloor (n+m-2)^2/4 \rfloor$ arcs (Constructions 1.9 and 1.13). They cross near $n = 9$ at $m = 3$, and Conjecture 1.14 asserts that their maximum is exact. Exhaustion proves the small $m = 3$ cases, but the general upper bound is structural: it would follow if every non-hub extremiser admitted the complete $A \to B$ decomposition of Conjecture A.22. The difficult clause forbids arcs returning from B to A, since one such arc lets routes escape through A and defeats the internal $m-2$ budget. Figure 3.1 places the numerical crossover and that obstruction together.

The conjecture is nevertheless correct to first order and to the order of its error. The two-step route count of Proposition A.23 and theorem A.25 gives, without assuming any partition,

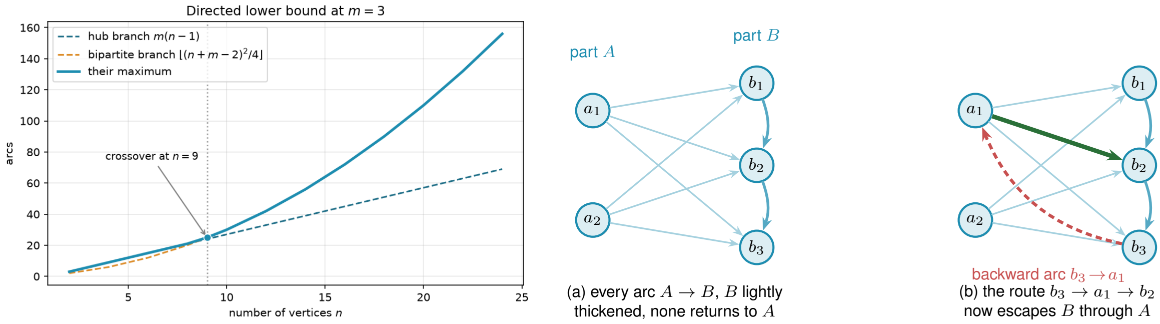
$$|A(D)| \;\le\; \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-1), \qquad \text{hence} \qquad \ell_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} + \Theta_m(n) \qquad (m \ge 3).$$

Thus only the coefficient of the linear term remains, between the conjectured $(m-2)/2$ and the proved $4(m-1)$. The same route family is internally vertex-disjoint, so Theorem A.29 independently gives $k_m^{\mathrm{dir}}(n) = n^2/4 + \Theta_m(n)$; Whitney's inequality alone would not justify that transfer.

The directed multigraph problem, by contrast, is closed exactly (Theorem A.31):

$$L_m^{\mathrm{dir}}(n) \;=\; (m-1)\max\!\Big(2(n-1),\; \left\lfloor \frac{n^2}{4} \right\rfloor\Big),$$

with doubled trees on the linear branch and the balanced one-way bipartite multigraph on the



(a) The two lower-bound branches at $m = 3$.

(b) The desired $A \to B$ form and one returning arc that breaks it.

Figure 3.1. The directed frontier in two views: the hub-to-wall crossover and the backward-arc obstruction to proving the wall optimal.

quadratic branch. Its proof peels off $m - 1$ sparse reachability-preserving subgraphs instead of deleting vertices, eliminating the former parity obstruction. Equality also classifies the quadratic extremisers for $n \geq \max(8, 2m)$ (Theorem A.45); only the smaller range remains open.

3.2 Two principal phenomena

Across all the variation, two phenomena stand out as the ones a first guess would not have predicted.

The first is the divergence at $m = 5$. The edge and vertex problems agree for $m \leq 4$, and it would be natural to expect them to agree forever. Instead they part company at the very next value, with $k_5(n) = \lfloor 8n/3 \rfloor - 4$ pulling strictly above $\ell_5(n)$ from $n = 14$ onwards (Theorem 1.6, and Remark 1.7 for the counting convention that fixes the constant). The separation is genuinely a fact about m rather than about n: at $m = 5$ the two values still agree at every size up to $n = 13$ and only then come apart, and beyond $m = 5$ the vertex problem becomes genuinely hard and has stayed open since 1974.

The second is the quadratic bipartite phenomenon. In an undirected graph every edge contributes to the degree of both its endpoints, and Mader's theorem turns that symmetry into a hard ceiling: $\lfloor m(n-1)/2 \rfloor$ edges, linear in n. A digraph breaks the symmetry. Place all n vertices into two equal parts A and B and draw every arc from A to B, one direction only. The result has $\lfloor n^2/4 \rfloor$ arcs, quadratic in n, yet $\lambda^{\max} = 1$: every pair has at most one arc-disjoint route, because every route must cross the A-to-B wall exactly once and no arc crosses the other way to offer a second path. A graph whose arc count grows like n^2 can sit at connectivity one, something no undirected graph can do past a handful of vertices.

The initial conjecture for the directed case was linear, matching the undirected intuition. The one-directional bipartite construction refutes it outright, already at $m = 2$: for $n = 8$ the wall holds 16 arcs while the linear hub holds only 14. At $m = 3$ the refutation is the thirty-arc augmented bipartite graph of Remark 1.12, which packs 30 arcs at $\lambda^{\max} = 2$ against the naive prediction of 27. The construction generalises to every m via Construction 1.13, which thickens the larger part

B slightly so that each vertex of B absorbs $m-2$ extra in-arcs from inside B. Those extra arcs add exactly $m-2$ short detours from A to each $b \in B$. That lifts $\lambda^{\max}$ from 1 to $m-1$, while the arc count climbs to $\lfloor (n+m-2)^2/4 \rfloor$.

The quadratic term is what makes the directed cases the deepest in the thesis. The upper bound argument must explain why no configuration beats the bipartite wall, which requires producing the partition in the first place, ruling out backward arcs, and controlling the internal structure of the large part B. That structural task is exactly what Conjecture A.22 has not yet resolved for $m \geq 3$. What Proposition A.23 and Theorem A.25 show is that the wall is at least the right shape, and the two say it in different currencies, which is worth keeping apart. In *size*, every feasible digraph is within $O_m(n)$ arcs of the wall, so the difficulty is a constant inside the second-order term rather than the phenomenon. In *shape*, every feasible digraph becomes a wall after deleting $O(\sqrt{m}\, n^{3/2})$ arcs, which is a weaker error term because it comes from the cruder of the two counts, the one that does not get to induct on a low-degree vertex. The undirected problem has no analogous obstacle, because Mader's degree-counting argument closes it in a few lines. The quadratic phenomenon is therefore not just a curiosity: it is the geometric reason the directed and undirected problems live on different levels of difficulty.

3.3 Summary of the twelve variants

Table 3.1 records the status of all twelve structural variants. Each entry is marked by how it is known: proved by hand here or in Appendix A, cited to the literature, conjectured with a proved lower bound, or, for the two rows where only the leading term is settled, proved asymptotically. No row now rests on a machine certificate alone. That category was live in earlier drafts, when the directed multigraph value was known only for the sizes the cut-counting optimisation could close, and Theorem A.31 retired it. The distinction is the honesty contract of Chapter 2 carried through to the summary. Figure 3.2 shows the same distinction as a picture before the table spells out every case.

Here $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$. Simple-graph formulas assume $n \geq m$; below that range the complete object gives the trivial maximum. The edge-hypergraph bound is attained when $(r-1) \mid (n-1)$ for multihypergraphs, and for simple hypergraphs when $r \geq 3$ and $m-1 \leq \binom{n-2}{r-2}$. The $m=3$ vertex-hypergraph bound is attained when $r \geq 3$ and $2 \leq \binom{n-2}{r-2}$. The two multigraph vertex rows use the counting convention of Remark 2.1. Under the other convention, where parallel copies count as distinct routes, the multigraph vertex question becomes a separate extremal problem, studied in Section A.9 and listed among the open problems below. Full statements and exceptions are in Theorems 1.2, 1.5, 1.6, 1.16, 1.17, A.25, A.29, A.31, A.48 and A.59 and proposition 1.15.

3.3.1 Orientation models for the directed hypergraph

The table uses the *forward* convention: one tail and $r-1$ heads. More generally a hyperedge may split into non-empty tail and head sets. Reversal makes the one-head *backward* model exactly

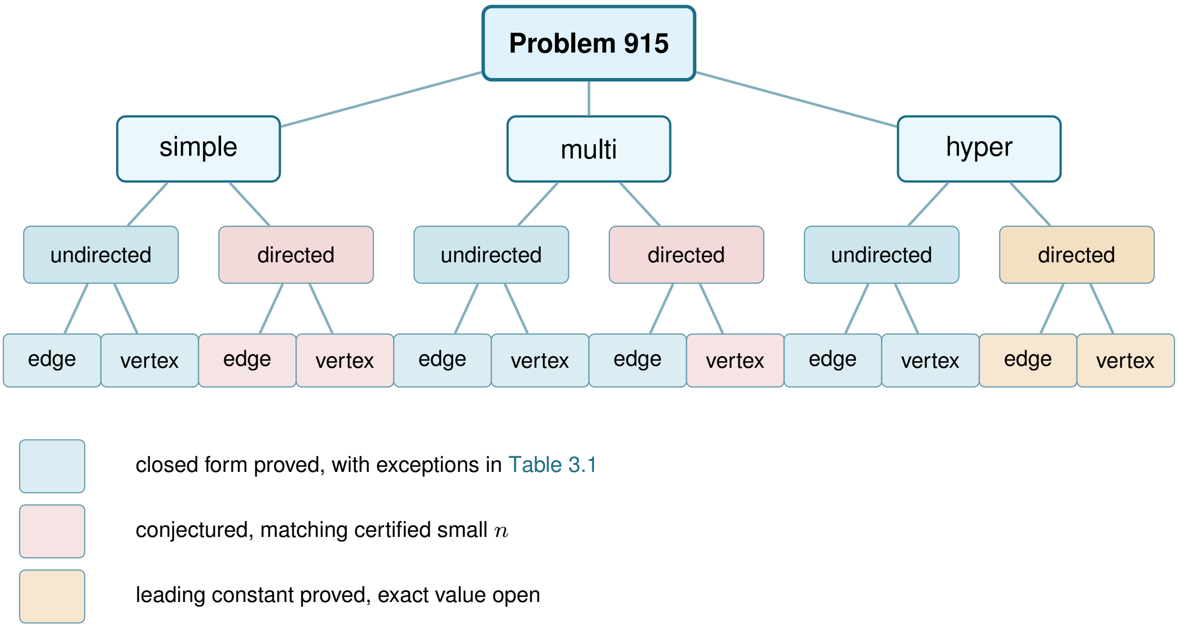


Figure 3.2. Status of the twelve variants. Blue marks a proved closed form in the ranges stated in Table 3.1, red a conjectured exact formula with proved lower bound, and amber a proved leading term with unknown exact value. The table is authoritative for exceptions and conventions.

equivalent to forward (Lemma A.56), while Theorem A.60 proves that arbitrary splits have the same leading term

$$(1 + o(1))\frac{(m-1)n^2}{4(r-1)}$$

under both separations. Thus the orientation choice does not add another asymptotic row to Table 3.1.

Finite values may differ when vertices are scarce. Whether forward and general agree exactly once $n > r$ remains open; the thesis uses the forward convention throughout its twelve-variant summary.

3.4 Contributions and limitations of the program

The program turned a per-case craft into a uniform pipeline. Before it, each variant came with its own one-off computation and its own hand argument. After it, one representation holds every variant, one exact checker measures connectivity, one certifier proves small-case upper bounds, and one cooled search discovers extremisers and the lower bounds they witness. The rediscovery of Section 2.12 is the evidence that the pipeline is faithful: pointed at solved cases, it returns the solved answers.

Its limits are as clear as its reach, and stating them is part of the honesty the thesis insists on. The search proves no upper bounds. It only ever exhibits graphs. The certifier proves upper bounds, but only for a fixed size, and the directed multigraph encoding closes unconditionally only up to $n = 6$. Past there the value is settled instead by the hand proof of Appendix A, and

Model	Edge/arc-disjoint	Vertex-disjoint
undirected simple	$\lfloor m(n-1)/2 \rfloor$; exact	edge value for $m \le 4$; $\lfloor 8n/3 \rfloor - 4$ for $m = 5$; open for $m \ge 6$
undirected multi	$(m-1)(n-1)$; exact	simple value; adjacency-count convention
undirected r-hyper	$\le \lfloor (m-1)(n-1)/(r-1) \rfloor$; equality in ranges below	$\lfloor (n-1)/(r-1) \rfloor$ for $m = 2$; $\le \lfloor 2(n-1)/(r-1) \rfloor$ for $m = 3$; equality ranges below
directed simple	$M(n)$ for $m = 2$; $n^2/4 + \Theta_m(n)$ for $m \ge 3$; exact then conjectured	same status and values
directed multi	$(m-1)M(n)$; exact	simple-digraph value; adjacency-count convention
directed r-hyper	$(1+o(1))(m-1)n^2/[4(r-1)]$; leading term exact	same leading term

Table 3.1. The twelve variants and the strongest result known for each.

the certifier contributed nothing at $n = 7$, where it exhausted its time limit without a verdict. The generation enumerator's degree-capped classification at $n = 7$ (Remark A.42) stands only as a machine-verified fact about the extremisers of maximum total degree at most 8, and it is not an unrestricted classification. No amount of either replaces the genuinely structural gaps the thesis leaves open, and the next section lists them in full. Three are worth naming here as the ones no computation, past or future, was ever going to close by itself. The decomposition of Conjecture A.22 would turn Conjecture 1.14 into a theorem, and its backward-arc clause is the hardest part of it but not the whole. The directed vertex problem at $m \ge 3$ is one Whitney's inequality pointedly does not hand over with the arc case. And the undirected vertex-disjoint problem at $m \ge 6$ has resisted for half a century, which the program can probe but not crack.

3.5 Open problems

The map is filled in where honest mathematics could fill it, and where it could not, the blanks are marked precisely, with a machine standing ready at each of them. That, finally, is the use of building the microscope: not that it answers every question, but that it shows exactly which questions are left, and hands the next reader a clean place to start.

Problem	Known here	Remaining target
Directed simple arc, $m \geq 3$	$\ell_m^{\mathrm{dir}}(n) = n^2/4 + \Theta_m(n)$ (Theorem A.25)	Prove the decomposition of Conjecture A.22, or otherwise determine the linear coefficient.
Directed simple vertex, $m \geq 3$	$k_m^{\mathrm{dir}}(n) = n^2/4 + \Theta_m(n)$ (Theorem A.29)	Decide whether its exact value equals the arc value; an arc upper bound does not transfer.
Directed multigraph classification	Exact value for all n, m and quadratic-branch uniqueness for $n \geq \max(8, 2m)$	Classify the range $n < 2m$ and the remaining linear-branch extremisers.
Undirected simple vertex, $m \geq 6$	Block formulation and $m/2 < c_m \leq 2(m-1)$	Determine c_m; the likely next structure is triconnected rather than merely blockwise.
Multigraph vertex, parallel copies distinct	Block-knapsack formulation and order $\Theta(m^2 n)$, with the per-vertex constant between $3 - 2\sqrt{2}$ and 2 (Theorems A.66 and A.67 and proposition A.64)	Determine the asymptotic constant and the optimal 2-connected blocks.
Hypergraph vertex, $m \geq 4$	Exact at $m = 2$ and, in the stated range, $m = 3$ (Theorems 1.16 and 1.17)	Locate the first failure for $r \geq 3$ and replace the rank-two decomposition used at $m = 3$.
Directed hypergraphs	Leading term $(m-1)n^2/(4(r-1))$ for all orientations and both separations	Determine finite exact values and whether forward and general agree once $n > r$.

Table 3.2. The remaining problems, separated from the results already proved in this thesis.

Appendix A

Full Proofs and Computational Audit

This appendix contains the standard machinery and long proofs deferred from the body, with figures beside the arguments that need them.

How to read this appendix. Theorems A.1 and A.2 and Mader's theorem are classical and may be skipped by readers who trust the cited results. The remaining sections contain the load-bearing directed and hypergraph arguments; Section A.10 records the source, commands, tests, and finite-output logs.

A.1 Menger, Gomory–Hu, and the degree bound

We use two classical results in their capacitated form. Unit capacities represent simple graphs, integer capacities represent multiplicities, and cut capacity is the total capacity crossing the cut.

Theorem A.1 (Menger, as max-flow min-cut [3])**.** *In any capacitated digraph, and for distinct vertices u, v, the maximum flow from u to v equals the minimum capacity of a cut separating u from v. At unit capacities this is the combinatorial statement we cite throughout: the maximum number of pairwise edge-disjoint u–v paths equals the minimum number of edges whose removal separates them,*

$$\lambda_G(u, v) = \min\{\, |C| : C \subseteq E(G),\ G - C \text{ has no } u\text{–}v \text{ path}\,\}.$$

Undirected graphs become symmetric networks, multigraph multiplicities become capacities, hypergraphs use the gate network of Figure 2.3b, and vertex-disjoint routes use the vertex splitting of Figure 2.3a. Thus one maximum-flow theorem supports every checker variant.

Theorem A.2 (Gomory–Hu [16])**.** *Every* undirected *capacitated graph G has a weighted tree T on the same vertex set, its* Gomory–Hu tree*, such that for all u, v the value $\lambda_G(u, v)$ equals the minimum edge weight on the u–v path in T, and each tree edge e corresponds to a minimum cut of G whose capacity is the weight of e. All $\binom{n}{2}$ pairwise values are thereby stored in the $n - 1$ weights of a single tree.*

The theorem applies to undirected simple graphs, multigraphs, and hyperedge cuts (Theorem A.47), but not to asymmetric directed cuts or vertex connectivity. Its standard submodular-cut construction is proved in [16]; here we use only the displayed path-minimum identity.

We also use the immediate endpoint bound.

Lemma A.3 (Degree bound)**.** *For every pair of vertices of a graph G, $\lambda_G(u,v) \leq \min(\deg u, \deg v)$. In a digraph D, $\lambda_D(u,v) \leq \min(d^+(u), d^-(v))$.*

Proof. Every edge-disjoint u–v path uses a distinct edge at u, so there are at most $\deg u$ of them, and likewise at most $\deg v$. In the directed case each arc-disjoint path leaves u by a distinct out-arc and enters v by a distinct in-arc. □

A.2 Mader's theorem in full

Let T be a Gomory–Hu tree of G and let $\mathrm{dist}_T(e)$ count the tree edges on the path between the endpoints of $e \in E(G)$. The upper bound is the following double count; simplicity enters only because at most $n-1$ graph edges can join adjacent vertices of T.

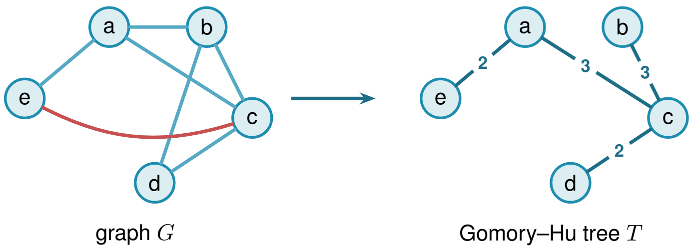


Figure A.1. A graph G and a Gomory–Hu tree T on the same vertices. Each graph edge crosses $\mathrm{dist}_T(e)$ tree cuts; at most $n-1$ edges of a simple graph have tree distance one.

Lemma A.4 (Distance-one edges)**.** *At most $n-1$ edges of a simple graph G satisfy $\mathrm{dist}_T(e) = 1$.*

Proof. An edge $e \in E(G)$ has $\mathrm{dist}_T(e) = 1$ if and only if its endpoints are adjacent in T. The tree T has exactly $n-1$ edges, and since G is simple each pair of adjacent tree vertices supports at most one edge of G. □

Lemma A.5 (Sum of tree distances)**.** $\displaystyle\sum_{e \in E(G)} \mathrm{dist}_T(e) \;\geq\; 2|E(G)| - (n-1).$

Proof. Partition $E(G)$ into $E_1 = \{e : \mathrm{dist}_T(e) = 1\}$ and $E_{\geq 2} = \{e : \mathrm{dist}_T(e) \geq 2\}$. Then

$$\sum_{e \in E(G)} \mathrm{dist}_T(e) \;\geq\; |E_1| + 2\,|E_{\geq 2}| \;=\; 2|E(G)| - |E_1| \;\geq\; 2|E(G)| - (n-1),$$

using Lemma A.4 in the last step. □

Lemma A.6 (Double-counting identity)**.** *Let G carry* unit capacities*, so that each weight $c(t)$ of a Gomory–Hu tree T is the plain* number *of edges of G crossing the cut that t induces. Then*

$$\sum_{t \in E(T)} c(t) \;=\; \sum_{e \in E(G)} \mathrm{dist}_T(e).$$

This is the form used for Mader's theorem, where G is a simple graph and so unit-capacitated by default.

Proof. By Theorem A.2 each tree edge t induces a cut of G of capacity $c(t)$, so $c(t) = \sum_{e\in E(G)} \mathbf{1}_{e \text{ crosses } t}$. Exchanging the order of summation,

$$\sum_{t\in E(T)} c(t) = \sum_{e\in E(G)} \sum_{t\in E(T)} \mathbf{1}_{e \text{ crosses } t} = \sum_{e\in E(G)} \operatorname{dist}_T(e),$$

where the last equality holds because an edge $\{u, v\}$ crosses exactly those tree cuts induced by the tree edges on the unique u–v path in T. $\square$

Proof of Theorem 1.2, upper bound. Let G be a simple graph on n vertices with $\lambda_G^{\max} \le m-1$ and let T be a Gomory–Hu tree of G. Each tree edge $t = \{x, y\}$ has weight $c(t) = \lambda_G(x, y) \le m-1$, so summing over the $n-1$ tree edges, $\sum_t c(t) \le (m-1)(n-1)$. Combining with Lemmas A.5 and A.6,

$$2|E(G)| - (n-1) \;\le\; \sum_{e\in E(G)} \operatorname{dist}_T(e) \;=\; \sum_{t\in E(T)} c(t) \;\le\; (m-1)(n-1).$$

Rearranging gives $2|E(G)| \le m(n-1)$, and since $|E(G)|$ is an integer, $|E(G)| \le \left\lfloor \frac{m(n-1)}{2} \right\rfloor$. $\square$

The factor two comes from charging all but at most $n-1$ edges to at least two tree cuts. Parallel edges destroy that refinement, leaving the multigraph bound $(m-1)(n-1)$ of Theorem 1.3.

Construction A.7 (Hub-and-spokes graph)**.** For $n \ge m \ge 2$, let $H_{m,n}$ have vertex set $\{h, v_1, \dots, v_{n-1}\}$ with (i) all $n-1$ *hub edges* $\{h, v_i\}$, and (ii) a *spoke graph* on $\{v_1, \dots, v_{n-1}\}$ with $\left\lfloor \frac{(m-2)(n-1)}{2} \right\rfloor$ edges in which every v_i has spoke degree at most $m-2$. Such a spoke graph exists by Lemma A.8. This is the general extremal construction. If $(m-1) \mid (n-1)$, take the spoke graph to be disjoint copies of K_{m-1}; adjoining h turns them into copies of K_m with one shared vertex, the shared-clique construction.

Lemma A.8 (Near-regular graphs exist)**.** *For $N \ge 1$ and $0 \le r \le N-1$ there is a graph on N vertices with $\left\lfloor \frac{rN}{2} \right\rfloor$ edges and maximum degree at most r.*

Proof. If rN is even we build an r-regular circulant graph on vertex set $\{0, \dots, N-1\}$: for even r connect each i to $i \pm j \pmod N$ for $j = 1, \dots, r/2$, and for odd r (then N is even) use $j = 1, \dots, (r-1)/2$ together with the diameter distance $N/2$, which contributes degree one. Both are r-regular with $rN/2$ edges. If rN is odd, then r and N are both odd: take the $(r-1)$-regular circulant with distances $1, \dots, (r-1)/2$ and add the near-perfect matching that pairs vertex i with $i + (N+1)/2 \pmod N$ for $i = 0, \dots, (N-3)/2$, which covers all vertices but one. The total is $\frac{(r-1)N}{2} + \frac{N-1}{2} = \frac{rN-1}{2} = \left\lfloor \frac{rN}{2} \right\rfloor$ edges, with one vertex of degree $r-1$ and all others of degree r. $\square$

Theorem A.9 (Extremal graphs exist)**.** *For all $n \geq m \geq 2$ the graph $H_{m,n}$ has $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ edges and $\lambda^{\max} \leq m-1$.*

Proof. The edge count is $(n-1)+\left\lfloor \frac{(m-2)(n-1)}{2} \right\rfloor = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, since $(m-2)(n-1)$ and $m(n-1)$ have the same parity. For the connectivity, any two distinct vertices include at least one spoke vertex v_i, whose total degree is at most $1+(m-2) = m-1$. Lemma A.3 then gives $\lambda(u,v) \leq m-1$. □

Together the upper bound and Theorem A.9 prove Theorem 1.2. The proof also reveals which graphs are extremal: equality forces both inequalities in the chain to be tight.

Theorem A.10 (Characterisation of equality)**.** *Let G be simple with $\lambda_G^{\max} \leq m-1$ and $|E(G)| = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, and let T be a Gomory–Hu tree of G. If $m(n-1)$ is even, then every tree edge has weight exactly $m-1$, every tree edge is also an edge of G, and every edge of G joins vertices at tree distance at most 2. If $m(n-1)$ is odd, there is exactly one unit of slack, located in exactly one of three places: one tree edge has weight $m-2$, or one tree edge is not an edge of G, or one edge of G joins vertices at tree distance 3. Everything else is tight.*

Proof. Write the chain of the upper-bound proof as three non-negative integer slacks, one for each inequality it uses:

$$S_1 = (m-1)(n-1) - \sum_t c(t), \qquad S_2 = \sum_{e:\, \mathrm{dist}_T(e)\geq 2} \big(\mathrm{dist}_T(e)-2\big), \qquad S_3 = (n-1)-|E_1|.$$

In words, S_1 counts tree weights below $m-1$, S_2 counts edges beyond tree distance two, and S_3 counts tree edges that support no edge of G. Retracing the proof gives two expressions for the same quantity,

$$\sum_e \mathrm{dist}_T(e) \;=\; 2|E|-|E_1|+S_2 \qquad \text{and} \qquad \sum_e \mathrm{dist}_T(e) \;=\; \sum_t c(t) \;=\; (m-1)(n-1)-S_1.$$

Substituting $|E_1| = (n-1)-S_3$ into the first and equating it with the second,

$$\begin{aligned} 2|E|-(n-1)+S_2+S_3 \;&=\; (m-1)(n-1)-S_1, \\ \text{so} \qquad 2|E| \;&=\; m(n-1)-(S_1+S_2+S_3). \end{aligned}$$

The whole characterisation reads off that last identity. If $m(n-1)$ is even, equality $|E| = m(n-1)/2$ forces $S_1 = S_2 = S_3 = 0$. If $m(n-1)$ is odd the floor absorbs exactly one unit, so $S_1+S_2+S_3 = 1$ and exactly one of the three slacks equals one. □

For $m = 2$ the characterisation says the extremal graphs are exactly the spanning trees: every tree edge must carry weight one and lie in G, so G contains its own Gomory–Hu tree and has only $n-1$ edges to spend. For $m = 4$ and $n \equiv 1 \pmod 3$ it is consistent with the trees of

K_4 blocks glued at shared cut vertices, the case of Construction A.7 that the search of Chapter 2 rediscovers.

A.2.1 The vertex problem is a block problem

The *vertex* problem $k_m(n)$ is the one open since 1974, and this short section records its block structure. Nothing here is deep. What it buys is a change of the object being searched: from graphs on n vertices, with n growing, to 2-connected graphs alone.

Theorem A.11 (The vertex problem is a knapsack over blocks)**.** *For $b \geq 2$ let*

$$h_m(b) \ := \ \max\{\, |E(B)| \ : \ B \text{ 2-connected or a single edge},\ |V(B)| = b,\ \kappa_B^{\max} \leq m-1 \,\}.$$

Then for all $m \geq 2$ and $n \geq 2$,

$$k_m(n) \ = \ \max\Big\{ \sum_i h_m(b_i) \ : \ b_i \geq 2,\ \sum_i (b_i - 1) \leq n-1 \Big\}.$$

Consequently $k_m(n) \geq k_m(n_1) + k_m(n_2)$ whenever $n = n_1 + n_2 - 1$, and

$$c_m \ := \ \lim_{n\to\infty} \frac{k_m(n)}{n-1} \ = \ \sup_{b\geq 2} \frac{h_m(b)}{b-1}$$

exists, the limit being approached from below.

Proof. Let G be feasible. Adjacent vertices u, v lie in a common block B, and no u-v path leaves B: leaving means crossing a cut vertex, and returning means crossing that same cut vertex a second time, which no path does. So $\kappa_G(u,v) = \kappa_B(u,v)$, every block inherits feasibility, and since every edge lies in exactly one block, $|E(G)| = \sum_B |E(B)| \leq \sum_B h_m(|V(B)|)$. Within one component the block sizes satisfy $\sum_B(|V(B)| - 1) =$ (order of the component) $- 1$, and summing over components loses one further vertex apiece, so $\sum_B(|V(B)| - 1) \leq n - 1$. That is $\leq$.

For $\geq$, take blocks $B_1, \ldots, B_t$ with $\sum(b_i - 1) \leq n-1$, pick one vertex, and attach all the B_i to it, pairwise disjoint otherwise. The result has $1 + \sum(b_i - 1) \leq n$ vertices, its blocks are exactly the B_i, so pairs inside a block keep their κ and pairs split between two blocks are separated by the shared vertex and have $\kappa \leq 1 \leq m-1$. It is feasible with $\sum_i |E(B_i)|$ edges.

Superadditivity is the special case of gluing two extremal graphs at a single vertex. Writing $a_n := k_m(n)$, the sequence (a_{n+1}) is superadditive in $n-1$, so Fekete's lemma gives $a_n/(n-1) \to \sup_n a_n/(n-1)$, and $a_n/(n-1) \geq h_m(b)/(b-1)$ whenever $(b-1) \mid (n-1)$, while $a_n \leq \max_b h_m(b)/(b-1) \cdot (n-1)$ from the display. So the limit is $\sup_b h_m(b)/(b-1)$. □

Three things follow at once, and they are a good test of whether the reformulation is saying anything.

At $m = 3$ it *proves* Theorem 1.5's value. A block with $\kappa^{\max} \le 2$ is an edge or a cycle, since a block that is neither contains a theta subgraph and a theta gives three internally disjoint paths. So $h_3(2) = 1$ and $h_3(b) = b$ for $b \ge 3$, the rate $b/(b-1)$ is largest at $b = 3$, and the knapsack packs triangles: $k_3(n) = 3\lfloor (n-1)/2 \rfloor + ((n-1) \bmod 2) = \left\lfloor \frac{3(n-1)}{2} \right\rfloor$.

At $m = 4$ the picture is $h_4(b) = 2(b-1)$ for every $b \ge 4$, so the rate is exactly 2 at every block size and $k_4(n) = 2(n-1)$. This one is a repackaging rather than a second proof, and it is worth separating the two halves. The upper bound $h_4(b) \le k_4(b) = 2(b-1)$ is Theorem 1.5 itself read on a block, so it cannot turn round and reprove it, and nothing as elementary as the theta argument above is on offer for the blocks with $\kappa^{\max} \le 3$. What the reformulation does add is the extremal block, and it is one the appendix has already built: the hub-and-spokes graph $H_{4,b}$ of Construction A.7, a hub joined to every other vertex with a cycle on the rest. Unlike the shared-clique graph it is a single block, and it meets $2(b-1)$ at every size. An exhaustive sweep of the 2-connected graphs confirms that it is optimal, and the unique optimum, at every b from 4 to 8.

At every m it identifies the Bollobás–Erdős construction, as many copies of K_m sharing one vertex as fit, as the case $b = m$: $h_m(m) = \binom{m}{2}$ and the rate is exactly $m/2$. Their conjecture is precisely the assertion $c_m = m/2$, that no block beats K_m. Exhaustive computation over all 2-connected graphs on at most nine vertices, of which there are 194066 at the top size alone, confirms that none does, for every $m \le 8$: the rate $m/2$ is attained and never exceeded. Since the rate of a bouquet of blocks is a weighted average of the rates of its blocks, a counterexample must contain a single 2-connected block that beats $m/2$ on its own, and by the computation such a block needs at least ten vertices.

That is not a defect of the computation, and it is worth saying exactly how far out of reach the truth is, because the reason is structural rather than a matter of processor time. The conjecture is false: Leonard disproved it at $m = 5$ with an explicit graph on 57 vertices [9], and Mader proved that for every $m \ge 6$ the difference $k_m(n) - mn/2$ is unbounded [5], so $c_m > m/2$ there. What replaces it is a lower bound of Sørensen and Thomassen [8]: for each fixed m,

$$c_m \;\ge\; \frac{m(m-1)-2}{2m-3} \;=\; \frac{m}{2} + \frac{1}{4} + O(1/m),$$

so the conjectured rate is wrong, but only by a constant.

Their witness is built by a recursion worth describing in the language of this section, since it says precisely why the blocks above cannot find it. Start from $G^0 = K_m$ minus an edge, and form G^j from two fresh copies of G^0 together with G^{j-1}, identifying one vertex of each with one vertex of the next, the three identifications arranged in a *cycle*. Each step adds $2m-3$ vertices and $m(m-1)-2$ edges, which is the rate above. The cyclic arrangement is the whole point: three graphs glued in a cycle at three vertices have no cut vertex at all, so G^j is 2-connected and is a *single block*. Theorem A.11 therefore sees it as one indivisible object of size $m + (2m-3)j$ and gets no purchase on it. What the graph does have is 2-cuts, one pair of identification vertices separating each copy from the rest, so the decomposition that would see it is the triconnected one, along 2-cuts rather than cut vertices, which is the same Tutte/SPQR machinery Lemma A.53 already uses on incidence graphs. That is the concrete next step this reformulation points at.

It also explains the sizes. The recursion beats the rate $m/2$ only once $j > 2/(m-4)$, so the smallest member of the family that outruns a bouquet of K_m's has 26 vertices at $m = 5$, 24 at $m = 6$ and 18 at $m = 7$, against the nine that exhaustive enumeration reaches. The construction was rebuilt and checked here at $m = 5, 6, 7$ and $j \le 3$: the vertex and edge counts match the formulas exactly, every member is 2-connected, and every member has $\kappa^{\max} = m - 1$, so each is feasible and none contains the forbidden m internally disjoint paths.

Remark A.12 (A degeneracy question, and what it would give). A general density bound is $k_m(n) < 2(m-1)n$: Mader's density theorem [17] would otherwise produce an m-connected subgraph. Feasibility says much more than that. It gives, for instance, that any two adjacent vertices have at most $m - 2$ common neighbours, since each common neighbour is a route of length two beside the edge.

The natural strengthening is that a feasible graph always has a vertex of degree at most $m - 1$. Since feasibility passes to induced subgraphs, that would make every feasible graph $(m-1)$-degenerate and give

$$k_m(n) \;\le\; (m-1)n - \binom{m}{2},$$

halving the constant in the only general upper bound known. It holds for $m = 3$, where it is classical: a graph of minimum degree at least three contains a subdivision of K_4, and two branch vertices of that subdivision are joined by three internally disjoint paths. Exhaustive search over all graphs with minimum degree at least m finds no feasible one for $m \le 6$ and $n \le 10$. It is stated here as a question rather than a conjecture, since the sizes reached are well below those at which the Bollobás–Erdős conjecture itself first fails.

A.3 The directed arc value at $m = 2$

Three short lemmas feed the induction. The first shows a hub forces a linear bound for every m, and is the proof behind the hub branch of Conjecture 1.14. The three lemmas below and the proof of Theorem 1.10 that they support are the author's own, and its finite base cases $2 \le n \le 7$ were verified by the author's program, through the exhaustive search of Chapter 2 whose completed range is recorded in Table A.1.

Lemma A.13 (Two-hop lemma)**.** *Let D be a simple digraph with $\lambda_D^{\max} \le m - 1$ containing a vertex r with $d^+(r) = n - 1$. Then every $v \ne r$ has at most $m - 2$ in-arcs from $V \setminus \{r, v\}$.*

Proof. If $u_1, \dots, u_{m-1}$ were distinct in-neighbours of v other than r, that is, distinct vertices each sending an arc to v, the paths $r \to v$ and $r \to u_i \to v$ for $i = 1, \dots, m-1$ would be m arc-disjoint routes, since the u_i are distinct and $d^+(r) = n - 1$ supplies every starting arc. That contradicts $\lambda_D(r, v) \le m - 1$. □

Theorem A.14 (Hub upper bound)**.** *If a simple digraph D with $\lambda_D^{\max} \le m-1$ has a vertex r with $d^+(r) = n-1$, then $|A(D)| \le m(n-1)$.*

Proof. Split the arcs into those leaving r (exactly $n-1$), those entering r (at most $n-1$), and the internal arcs avoiding r. By Lemma A.13 each of the $n-1$ other vertices receives at most $m-2$ internal arcs, so there are at most $(m-2)(n-1)$ internal arcs, and the total is at most $m(n-1)$. □

Lemma A.15 (No transitive triangle)**.** *A simple digraph with $\lambda^{\max} \le 1$ contains no three vertices x, y, z with all of $(x, y), (y, z), (x, z)$ present, since $x \to z$ and $x \to y \to z$ would be two arc-disjoint routes.*

Lemma A.16 (Deletion monotonicity)**.** *Write $D - v_0$ for the digraph obtained from D by deleting the vertex v_0 together with all arcs incident to it. For any digraph D and vertex v_0,*

$$\lambda^{\max}(D - v_0) \le \lambda^{\max}(D) \qquad \text{and} \qquad \kappa^{\max}(D - v_0) \le \kappa^{\max}(D),$$

since arc-disjoint paths in $D - v_0$ are still arc-disjoint paths in D, and internally vertex-disjoint paths in $D - v_0$ are still internally vertex-disjoint paths in D. The two statements are proved by the same one-line observation because deleting a vertex only removes routes, never creates one, and it cannot make two surviving routes share anything they did not share before.

In words: deleting a vertex can never manufacture independent routes. Every route surviving in $D - v_0$ was already a route in D, so no pair comes out of the deletion better connected than it went in. This single guarantee is what every vertex-deletion induction in this section rests on, and it holds for both separations alike.

Proof of Theorem 1.10. Write $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$. A direct computation shows $2(n-1) \ge \lfloor n^2/4 \rfloor$ exactly for $n \le 7$, with equality at $n = 7$, so $M(n) = 2(n-1)$ for $n \le 7$ and $M(n) = \lfloor n^2/4 \rfloor$ for $n \ge 7$. The lower bound $\ell_2^{\mathrm{dir}}(n) \ge M(n)$ is witnessed by the directed hub construction (Construction 1.9 at $m = 2$, the bidirected star) and the one-directional bipartite construction (Construction 1.8).

For the upper bound we show by strong induction on n that $\lambda_D^{\max} \le 1$ forces $|A(D)| \le M(n)$.

Base cases $2 \le n \le 7$. All six are settled uniformly by the pruned exhaustive search of Chapter 2, which walks every simple digraph with $\lambda^{\max} \le 1$ on up to seven vertices and reports the maxima $2, 4, 6, 8, 10, 12$, matching $M(n)$ at each size. The transcript is recorded in Table A.1, and the cut-counting certifier independently confirms the values within its reach through the $m = 2$ reduction of Corollary A.39. The two smallest are also immediate by hand, as a check on the machine rather than as a separate argument. At $n = 2$ there are at most 2 arcs. At $n = 3$ a fifth arc would leave at most one arc of the complete bidirected triangle missing, so three of the remaining arcs form a transitive triangle, which Lemma A.15 forbids. The point of the theorem is precisely that $n = 4, 5, 6, 7$ are *not* immediate by hand, which is what makes handing the case-check to the program the right move.

Inductive step $n \geq 8$. Let D be a simple digraph on n vertices with $\lambda_D^{\max} \leq 1$ and put $a := |A(D)|$. Choose v_0 minimising the total degree $d(v) = d^+(v) + d^-(v)$. Since $\sum_v d(v) = 2a$, averaging gives $d(v_0) \leq 2a/n$. By Lemma A.16 the digraph $D - v_0$ on $n-1 \geq 7$ vertices satisfies the induction hypothesis, so $a - d(v_0) \leq M(n-1) = \left\lfloor \frac{(n-1)^2}{4} \right\rfloor$. Combining,

$$a \;\leq\; \frac{2a}{n} + \left\lfloor \frac{(n-1)^2}{4} \right\rfloor \qquad \Longrightarrow \qquad a \;\leq\; \frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor .$$

It remains to check that this right-hand side never exceeds $\lfloor n^2/4 \rfloor$, and the two parities behave differently enough to treat them in turn.

For even $n = 2k$ with $k \geq 4$, first simplify $\lfloor (n-1)^2/4 \rfloor$. Expanding $(2k-1)^2 = 4k^2 - 4k + 1$ and dividing by 4 gives $k^2 - k + \frac{1}{4}$, whose floor is $k(k-1)$. The bound then reads

$$a \;\leq\; \frac{2k}{2k-2}\, k(k-1) \;=\; \frac{k}{k-1}\, k(k-1) \;=\; k^2,$$

and since $\lfloor n^2/4 \rfloor = \lfloor (2k)^2/4 \rfloor = k^2$ as well, the bound lands exactly on the target with no slack left over.

For odd $n = 2k+1$ with $k \geq 4$, the same simplification gives $\lfloor (n-1)^2/4 \rfloor = \lfloor (2k)^2/4 \rfloor = k^2$, so the bound is $a \leq \frac{2k+1}{2k-1}\, k^2$. To compare this fraction with the target, split off its integer part by dividing $(2k+1)k^2$ by $2k-1$. The remainder is exactly k, because $(2k-1)(k^2+k) = 2k^3 + k^2 - k$ and $(2k+1)k^2 = 2k^3 + k^2$ differ by k, so

$$\frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor \;=\; \frac{(2k+1)k^2}{2k-1} \;=\; k(k+1) + \frac{k}{2k-1}.$$

The leftover fraction $\frac{k}{2k-1}$ lies strictly between 0 and 1 for every $k \geq 2$ (the induction here is already past the base cases, so $k \geq 4$), so the bound puts a at most an integer $k(k+1)$ plus a positive amount below one. Since a is itself an integer it cannot reach the next integer up, so $a \leq k(k+1)$. The target $\lfloor n^2/4 \rfloor = \lfloor (2k+1)^2/4 \rfloor = k^2 + k = k(k+1)$ agrees, so for odd n it is integrality, not the raw arithmetic, that closes the last fraction of a unit. In both parities $a \leq \lfloor n^2/4 \rfloor \leq M(n)$, completing the induction. $\square$

Two remarks on the shape of this proof. First, the minimum-degree deletion closes the step on its own, and no separate hub case is needed, because the averaging bound holds whether or not D contains a hub, although Theorem A.14 shows directly that a hub digraph never exceeds the linear branch. Second, for even n the bound is exactly tight against the one-directional bipartite construction, which is $\frac{n}{2}$-regular in total degree, so every step of the chain is achieved by the conjectured extremiser, and for odd n the slack $\frac{k}{2k-1} < 1$ is absorbed by integrality. The proof is long not because it is deep but because the two regimes of $M(n)$ must be carried through the arithmetic separately, and that bookkeeping is precisely what Chapter 2 hands to the machine at the base cases.

Remark A.17 (Which way Whitney runs). It is worth pausing on the direction of Whitney's inequality before using it, because the tempting reading is the wrong one. From $\kappa_D(u,v) \le \lambda_D(u,v)$ for every pair it follows that $\kappa_D^{\max} \le \lambda_D^{\max}$, hence that

$$\{D : \lambda_D^{\max} \le m-1\} \subseteq \{D : \kappa_D^{\max} \le m-1\}.$$

Every digraph feasible for the *arc* problem is feasible for the *vertex* problem, and not the other way round. The vertex-feasible family is the larger of the two, so the inequality between the extremal numbers it hands over for free is

$$k_m^{\mathrm{dir}}(n) \ \ge\ \ell_m^{\mathrm{dir}}(n),$$

and an upper bound for the arc problem says nothing at all about the vertex problem. The gap is not vacuous: two routes that share an interior vertex but no arc are counted by λ and not by κ, so there really are digraphs in the difference of the two families. What Whitney gives is therefore exactly one thing, that an arc construction is an honest vertex *lower* bound, and the vertex upper bound has to be earned separately. The proof below earns it, and Chapter 3 keeps the same discipline for $m \ge 3$, where the vertex problem stays open even though the arc conjecture is stated.

Proof of Theorem 1.11. Write $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ as before.

Lower bound. The directed hub and the one-directional bipartite construction both have $\lambda^{\max} = 1$, so by Remark A.17 both have $\kappa^{\max} \le 1$ and are feasible for the vertex problem. Hence $k_2^{\mathrm{dir}}(n) \ge M(n)$.

Upper bound. We show by strong induction on n that $\kappa_D^{\max} \le 1$ forces $|A(D)| \le M(n)$. The induction is the one that proved Theorem 1.10, and every step of it survives the change of separation, because the only two properties of the ceiling that argument uses are that it is inherited by vertex deletion and that it holds at the base sizes.

The base cases $2 \le n \le 7$ are settled by the same pruned exhaustive search of Chapter 2, run with the vertex feasibility test in place of the arc one. It walks every simple digraph with $\kappa^{\max} \le 1$ on up to seven vertices and reports the maxima $2, 4, 6, 8, 10, 12$, matching $M(n)$ at each size. Table A.1 records this run beside the arc one.

For the inductive step $n \ge 8$, let D be a simple digraph on n vertices with $\kappa_D^{\max} \le 1$ and put $a := |A(D)|$. Choose v_0 of minimum total degree, so $d(v_0) \le 2a/n$ by averaging. By Lemma A.16, which holds for κ exactly as it does for λ, the digraph $D - v_0$ on $n-1 \ge 7$ vertices again satisfies $\kappa^{\max} \le 1$, so the induction hypothesis gives $a - d(v_0) \le M(n-1) = \lfloor (n-1)^2/4 \rfloor$. From here the two parity computations are verbatim those of Theorem 1.10: they use only the two displayed inequalities and the integrality of a, no property of the separation. They yield $a \le \lfloor n^2/4 \rfloor \le M(n)$, completing the induction.

Hence $k_2^{\mathrm{dir}}(n) = M(n) = \ell_2^{\mathrm{dir}}(n)$. The two problems have the same answer at $m = 2$, but this is a computed coincidence of the two searches meeting at every base size, not a formal consequence of Whitney's inequality. □

Remark A.18 (The directed vertex problem at $m \geq 3$). The proof above transfers the $m = 2$ induction but nothing more, and for $m \geq 3$ the vertex problem is genuinely separate: the arc conjecture, and the decomposition of Conjecture A.22 that would prove it, both concern the smaller family and leave $k_m^{\mathrm{dir}}(n)$ untouched. What can be said is computational. Running the exhaustive search of Chapter 2 under both feasibility tests at $m = 3$ gives

n	2	3	4	5	6
$\ell_3^{\mathrm{dir}}(n)$	2	6	9	12	15
$k_3^{\mathrm{dir}}(n)$	2	6	9	12	15

with every entry proved complete, so the two separations agree at every size the exhaustion reaches, and from $n = 3$ onwards both match the conjectured $\max(m(n-1), \lfloor (n+m-2)^2/4 \rfloor)$. The table stops at $n = 6$ because that is where the exhaustion stops: at $n = 7$ neither sweep closes inside an hour and a half, and each returns the construction's 18 as a bare lower bound. At $n = 2$ the conjecture's formula gives 3 where only 2 arcs exist at all, the usual degeneracy of the formula below $n = m$. Agreement over five sizes is evidence and not a proof, and the natural next question is structural rather than numerical: to separate the two problems one needs a digraph where some pair carries strictly more arc-disjoint routes than internally disjoint ones *and* where that slack pays for extra arcs. The two demands pull against each other, which is perhaps why no small example does both.

A.4 Structural propositions on the directed frontier

These are the proved ingredients of the reduction discussed in Chapter 3. The section ends with what the reduction still assumes, stated as Conjecture A.22, and with the one quadratic bound that assumes nothing, Proposition A.23.

Proposition A.19 (Mutually unreachable out-neighbourhoods). *Let D be a simple digraph with $\lambda_D^{\max} \leq 1$. For any vertex u and distinct out-neighbours v, w of u, there is no directed path from v to w in $D - u$.*

Proof. If P were a directed v–w path avoiding u, then $u \to w$ and $u \to v \xrightarrow{P} w$ would be two arc-disjoint u-to-w routes: the first uses only the arc (u, w), the second starts with (u, v) and continues along P, none of whose arcs touch u. This contradicts $\lambda_D(u, w) \leq 1$. The restriction to $D - u$ is not cosmetic, and Figure A.2 draws both the argument and the reason for the restriction: in the digraph with arcs (u, v), (v, u), (u, w) every local connectivity is 1, yet $v \to u \to w$ reaches w from v through u. □

(a) If a directed $v \rightsquigarrow w$ path (red) existed in $D - u$, then $u \to w$ and $u \to v \rightsquigarrow w$ would be two arc-disjoint u–w routes, against $\lambda(u, w) \leq 1$.

(b) The counterexample $\{(u, v), (v, u), (u, w)\}$: $\lambda^{\max} = 1$, yet w *is* reachable from v, but only *through* u.

Figure A.2. Why Proposition A.19 must exclude paths through u. **(a)** For out-neighbours v, w of u in a digraph with $\lambda^{\max} \leq 1$, no $v \rightsquigarrow w$ path can avoid u: such a path would combine with the lone arc $u \to w$ into two arc-disjoint routes. **(b)** The restriction to $D - u$ is essential. In the three-vertex digraph $\{(u, v), (v, u), (u, w)\}$ we have $\lambda^{\max} = 1$, yet v reaches w via $v \to u \to w$, a route that reuses the vertex u, which the proposition allows.

Proposition A.20 (Disjoint second out-neighbourhoods)**.** *Let D be a simple digraph with $\lambda_D^{\max} \leq 1$, let $u \in V$, and let $v_1 \neq v_2$ be out-neighbours of u. Then $(N^+(v_1) \setminus \{u\}) \cap (N^+(v_2) \setminus \{u\}) = \emptyset$.*

Proof. A common out-neighbour $w \neq u$ would give the two arc-disjoint routes $u \to v_1 \to w$ and $u \to v_2 \to w$, contradicting $\lambda_D(u, w) \leq 1$. □

Proposition A.21 (Two-hop budget on bipartite partitions)**.** *Let D be a simple digraph with $\lambda_D^{\max} \leq m - 1$. If V admits a partition $A \cup B$ with $A \neq \emptyset$ and every arc from A to B present, then every $b \in B$ has in-degree at most $m - 2$ from inside B, and consequently the number of arcs inside B is at most $(m - 2)|B|$.*

In words: once every arc from A to B is present, a vertex of B can afford very few in-arcs from inside B. Each such arc creates a fresh two-step detour on top of the direct arc that every vertex of A already has, and the ceiling allows only $m - 2$ of them.

Proof. Fix $a \in A$ and $b \in B$, and let $b'_1, \ldots, b'_d$ be the in-neighbours of b inside B, where $d = \deg_B^-(b)$. The direct arc $a \to b$ together with the two-step detours $a \to b'_i \to b$ are $1 + d$ arc-disjoint a-to-b routes, all present because $A \to B$ is complete. Feasibility forces $1 + d \leq m - 1$, so $d \leq m - 2$, and summing over B bounds the internal arcs by $(m - 2)|B|$. □

We now combine Proposition A.21 with the best choice of partition. Suppose every arc of D either runs from A to B or stays inside B: the partition $A \to B$ is complete, no arc lies inside A, and crucially *no arc runs from B back to A*. Write $b = |B|$, so that $|A| = n - b$. The arcs then come in exactly two kinds. The arcs crossing the partition number $|A|\,|B| = (n - b)\,b$, one for each ordered A–B pair. The arcs inside B number at most $(m - 2)\,b$: since no route between two vertices of B ever needs to leave B (there is no arc back to A to leave by), Proposition A.21 applies to B on its own and caps the in-degree of each $b \in B$ from inside B at $m - 2$. There are no other arcs, so, adding the two kinds,

$$|A(D)| \;\le\; (n-b)\,b \;+\; (m-2)\,b \;=\; b\,\bigl(n+m-2-b\bigr).$$

The right-hand side is a downward parabola in b. Writing $s = n+m-2$, it is $b(s-b) = -b^2+sb$, maximised at $b = s/2$, so over integers the best partition takes $b = \lceil s/2 \rceil$ (equivalently $\lfloor s/2 \rfloor$), and the maximum value is

$$\max_b \; b\,(s-b) \;=\; \left\lfloor \frac{s^2}{4} \right\rfloor \;=\; \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor .$$

This is the quadratic branch of Conjecture 1.14, reached uniformly in m. It is worth being exact about how much was assumed, because the count is often summarised as resting on the absence of backward arcs alone, and that summary is too generous. Four things were granted at once: that a partition $V = A \cup B$ exists at all, that every arc from A to B is present, that no arc lies inside A, and that no arc runs from B back to A. Only the last of the four is the backward-arc condition. The first three describe the shape of the extremiser, and nothing proved so far forces an arbitrary extremal digraph to have that shape. The honest statement of the gap is therefore a structural one:

Conjecture A.22 (The missing decomposition)**.** *Let n be large enough that the quadratic branch of Conjecture 1.14 strictly exceeds the linear one, that is, $\lfloor (n+m-2)^2/4 \rfloor > m(n-1)$. Then every extremal digraph D on n vertices with $\lambda_D^{\max} \le m-1$ admits a partition $V(D) = A \cup B$ such that every arc from A to B is present, no arc lies inside A, and no arc runs from B to A.*

The hypothesis on n has to be there, and it is worth seeing why, because the obvious phrasing is too narrow. On the linear branch the extremisers are *not* just the hub. At $m = 2$, every bidirected spanning tree is feasible and has exactly $2(n-1)$ arcs, since between two vertices the only route follows the unique tree path, so the whole extremal set on that branch is the set of bidirected trees. An exhaustive classification confirms it: at $n = 5$ and $n = 6$ there are exactly 3 and 6 extremal digraphs up to isomorphism, which are precisely the numbers of trees on 5 and 6 vertices, and none of them admits the decomposition. Excluding "the hub" would leave all the other trees behind, so the clean hypothesis is on the size rather than on the shape.

Given Conjecture A.22 the count above closes Conjecture 1.14 for $m \ge 3$. The backward-arc clause is the part where the natural delete-and-compensate exchange fails to be monotone, which is why it is the clause usually named, but the partition itself is not free either, and Chapter 3 states the position that way. Testing the conjecture by exhaustion is harder than it looks: the crossover sits at $n = 8$ for $m = 2$ and at $n = 9$ for $m = 3$, so every size an exhaustive sweep can reach at $m = 3$ still lies on the linear branch, where the statement says nothing. That is the practical reason this conjecture has resisted a computational probe rather than a proof.

What can be proved without any of the four is the leading constant, and the argument is short enough to give in full. It also says something the conjecture does not: that every feasible digraph is within a lower-order number of arcs of being one-directional bipartite.

Proposition A.23 (Unconditional quadratic bound and stability)**.** *Let D be a simple digraph on $n \geq 2$ vertices with $\lambda_D^{\max} \leq m-1$. Then*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; \sqrt{m}\, n^{3/2}.$$

Moreover D can be made one-directional bipartite, meaning every remaining arc runs from one part to the other and none returns, by deleting at most $\sqrt{m}\, n^{3/2}$ arcs. Consequently, for each fixed m,

$$\ell_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; O_m\!\left(n^{3/2}\right),$$

so the leading constant $\frac{1}{4}$ of Conjecture 1.14 holds unconditionally, with no hypothesis on the structure of the extremiser.

Proof. Step 1, a budget on two-step routes. Fix an ordered pair (u,v) with $u \neq v$, and let $p_2(u,v)$ count the vertices x with both $u \to x$ and $x \to v$ present. Such an x is automatically distinct from u and from v, since D has no loops, so each one gives a genuine two-step route $u \to x \to v$. Two of these routes with different midpoints $x \neq x'$ share no arc, since the first uses (u,x) and (x,v) and the second uses (u,x') and (x',v), and neither uses the arc (u,v), which has no midpoint at all. So the direct arc, when present, together with all the two-step routes, is a family of pairwise arc-disjoint u-to-v routes, and feasibility caps its size:

$$\mathbf{1}_{(u,v)\in A(D)} + p_2(u,v) \;\leq\; \lambda_D(u,v) \;\leq\; m-1.$$

Summing over all $n(n-1)$ ordered pairs gives $|A(D)| + \sum_{u\neq v} p_2(u,v) \leq (m-1)\,n(n-1)$.

Step 2, from pairs to degrees. Write Δ for the number of digons of D, meaning the pairs of vertices carrying an arc in each direction. Three small facts turn Step 1 into a statement about degrees alone.

(2a) Walks of length two, counted by their midpoint.

$$\sum_{u,\,v\in V} p_2(u,v) \;=\; \sum_{x\in V} d^-(x)\, d^+(x),$$

where the left-hand sum runs over *all* ordered pairs, the diagonal $u=v$ included. Fixing a vertex x, a walk $u \to x \to v$ is nothing more than an independent choice of an in-neighbour u of x and an out-neighbour v of x, so x is the midpoint of exactly $d^-(x)\, d^+(x)$ of them.

(2b) What the diagonal counts. A diagonal term $p_2(u,u)$ counts the closed walks $u \to x \to u$, and each digon contributes two such walks, one based at each of its endpoints. Hence

$$\sum_{u\in V} p_2(u,u) \;=\; 2\Delta.$$

(2c) Digons are few. Distinct digons use disjoint pairs of arcs, so $2\Delta \le |A(D)|$.

Splitting the all-pairs sum of (2a) along its diagonal and feeding in Step 1 now gives the inequality the rest of the argument runs on:

$$\begin{aligned}
\sum_{x\in V} d^+(x)\, d^-(x) \;&=\; \sum_{u\neq v} p_2(u,v) \;+\; 2\Delta && \text{by (2a) and (2b),}\\
&\le\; \big[(m-1)\, n(n-1) - |A(D)|\big] \;+\; 2\Delta && \text{by Step 1,}\\
&\le\; \big[(m-1)\, n(n-1) - |A(D)|\big] \;+\; |A(D)| && \text{by (2c),}\\
&=\; (m-1)\, n(n-1).
\end{aligned}$$

The point of the last two lines is that the digon term is absorbed exactly by the slack Step 1 left behind, so nothing is given away in passing from pairs to degrees. It is worth reading the conclusion in words: a vertex with a large in-degree *and* a large out-degree is expensive, because it manufactures two-step routes for many pairs at once, and the budget for those routes is only quadratic in n.

Step 3, the smaller half-degree is small on average. Since $\min(a,b) \le \sqrt{ab}$ for non-negative a, b, and by the Cauchy-Schwarz inequality in the form $\sum_x \sqrt{a_x} \le \sqrt{n \sum_x a_x}$,

$$\begin{aligned}
\sum_{x\in V} \min\big(d^+(x), d^-(x)\big) \;&\le\; \sum_{x\in V} \sqrt{d^+(x)\, d^-(x)} \;\le\; \sqrt{n \sum_{x\in V} d^+(x)\, d^-(x)}\\
&\le\; \sqrt{n\cdot(m-1)\, n(n-1)} \;=\; n\sqrt{(m-1)(n-1)} \;\le\; \sqrt{m}\, n^{3/2}.
\end{aligned}$$

Step 4, deleting the smaller half of every vertex. Call x *source-like* if $d^+(x) \ge d^-(x)$ and *sink-like* otherwise, and write S and T for the two classes, so $V = S \cup T$. Delete every in-arc of a source-like vertex and every out-arc of a sink-like vertex. A source-like x loses $d^-(x) = \min(d^+(x), d^-(x))$ arcs and a sink-like x loses $d^+(x) = \min(d^+(x), d^-(x))$ arcs, so the number of deleted arcs is at most $\sum_x \min(d^+(x), d^-(x))$, which Step 3 bounds by $\sqrt{m}\, n^{3/2}$. An arc (a, b) survives only if a is not sink-like and b is not source-like, that is, only if $a \in S$ and $b \in T$. So what remains runs entirely from S to T, which is the one-directional bipartite shape, and it has at most $|S|\,|T| \le \lfloor n^2/4 \rfloor$ arcs. Adding the two counts gives the bound.

The asymptotic statement follows because the augmented bipartite construction (Construction 1.13) already gives $\ell_m^{\mathrm{dir}}(n) \ge \lfloor (n+m-2)^2/4 \rfloor = n^2/4 + O_m(n)$, and $n^{3/2}$ dominates n. $\square$

The error term $n^{3/2}$ is larger than the $(m-2)n/2$ that Conjecture 1.14 predicts, so the proposition as it stands does not separate the conjectured formula from its near neighbours. It can be sharpened to the right order, and the extra idea is small enough to be worth giving, because it explains where the $\sqrt{n}$ was being wasted.

Step 3 is lossy exactly when every vertex carries a middling degree. The Cauchy-Schwarz step is tight only if $d^+(x)d^-(x)$ is about $(m-1)n$ at every vertex, which forces every degree down to about $\sqrt{mn}$. But a digraph with $\Theta(n^2)$ arcs has average degree $\Theta(n)$, and at a vertex of large degree the constraint $d^+(x)d^-(x) \le (m-1)n$ bites much harder: with $d^+ + d^-$ of order n, the smaller of the two must be of order m rather than $\sqrt{mn}$. The dense case and the tight case of Step 3 are therefore incompatible, and the way to exploit that is a dichotomy on the minimum degree, with vertex deletion to handle the sparse side.

Lemma A.24 (Small side of a high-degree vertex)**.** *For any vertex x of a digraph with total degree $d(x) = d^+(x) + d^-(x) > 0$,*

$$\min\bigl(d^+(x), d^-(x)\bigr) \;\le\; \frac{2\, d^+(x)\, d^-(x)}{d(x)}.$$

Proof. Write $\mu = \min(d^+(x), d^-(x))$ and $M = \max(d^+(x), d^-(x))$, so $\mu + M = d(x)$ and $\mu M = d^+(x)d^-(x)$. Since $\mu \le M$ we have $M \ge d(x)/2$, hence $d^+(x)d^-(x) = \mu M \ge \mu\, d(x)/2$, which rearranges to the claim. □

Theorem A.25 (The quadratic bound with a linear error)**.** *For all $m \ge 2$ and $n \ge 2$, every simple digraph D on n vertices with $\lambda_D^{\max} \le m - 1$ satisfies*

$$|A(D)| \;\le\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4(m-1)(n-1),$$

so for each fixed m,

$$\ell_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; \Theta_m(n) \qquad (m \ge 3),$$

and, for $m \ge 3$, both the leading constant $\frac{1}{4}$ and the order *of the second-order term of Conjecture 1.14 hold unconditionally. At $m = 2$, Theorem 1.10 instead gives $\ell_2^{\mathrm{dir}}(n) = n^2/4 + O(1)$ as $n \to \infty$.*

In words: a feasible one-way simple graph is never more than a linear amount denser than the balanced bipartite wall, which holds $\lfloor n^2/4 \rfloor$ arcs. The wall's quadratic term is therefore the truth of the matter, and the only thing still in question is how large the linear correction is. Nothing here assumes anything whatever about the shape of D.

Proof. Induction on n. For $n = 2$ a simple digraph has at most 2 arcs and the right-hand side is $1 + 4(m-1) \ge 5$, so the base holds. Let $n \ge 3$ and split on the minimum total degree.

Case 1: some vertex v has $d(v) < n/2$. Since $d(v)$ is an integer, $d(v) \le \lfloor n/2 \rfloor$. By Lemma A.16 the digraph $D - v$ on $n - 1$ vertices is still feasible, so the induction hypothesis applies to it and

$$|A(D)| \;\le\; \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 4(m-1)(n-2) + \left\lfloor \frac{n}{2} \right\rfloor \;=\; \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-2),$$

using Lemma A.30 for the last step, and this is below the claimed bound.

Case 2: every vertex has $d(x) \geq n/2$. By Lemma A.24 and Step 2 of Proposition A.23,

$$\sum_{x\in V} \min\bigl(d^+(x), d^-(x)\bigr) \;\leq\; \sum_{x\in V} \frac{2\,d^+(x)\,d^-(x)}{d(x)} \;\leq\; \frac{4}{n}\sum_{x\in V} d^+(x)\,d^-(x) \;\leq\; \frac{4}{n}\,(m-1)\,n(n-1),$$

which is $4(m-1)(n-1)$. Step 4 of Proposition A.23 deletes exactly that many arcs and leaves at most $\lfloor n^2/4 \rfloor$, so the bound holds in this case too.

For the two-sided asymptotic statement, Construction 1.13 gives

$$\ell_m^{\mathrm{dir}}(n) \;\geq\; \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor \;=\; \frac{n^2}{4} + \frac{(m-2)n}{2} + O_m(1),$$

so the second-order term is linear in n from both sides. □

What is left open is therefore a constant factor in one coefficient, not a shape. Conjecture 1.14 says the second-order coefficient is exactly $(m-2)/2$, and the theorem pins it between that and $4(m-1)$, a factor of $8(m-1)/(m-2)$: sixteen at $m = 3$, and tending to eight as m grows. Closing that gap is where the structure of Conjecture A.22 lives, and Remark A.26 rules out the most obvious way to try.

Remark A.26 (Case 2's coefficient needs a new inequality). One might hope to cut Case 2's constant using the observation that in the conjectured extremiser, one whole side has $\min(d^+, d^-) = 0$, so summing Lemma A.24's worst case at every vertex looks wasteful. That observation alone does not cut the coefficient. The value four is the best universal coefficient obtainable from the two facts used in Case 2, the budget $\sum_x d^+(x)\,d^-(x) \leq (m-1)n(n-1)$ of Step 2 in Proposition A.23 and the floor $d(x) \geq n/2$.

Think of the budget as money and of $t_x = \min(d^+(x), d^-(x))$ as what each vertex buys with it. At a vertex with $d(x) = s \geq n/2$, one also has $s \geq 2t$. Hence the cheapest possible cost for a given t is the two-branch function

$$c(t) = \min_{s \geq \max(n/2, 2t)} t(s-t) = \begin{cases} t(n/2 - t), & 0 \leq t \leq n/4, \\ t^2, & t \geq n/4. \end{cases}$$

On the first branch the value per unit cost is $t/c(t) = 1/(n/2 - t)$, which increases to $4/n$, whereas on the second it is $1/t$, which decreases from $4/n$. Thus $t \leq (4/n)c(t)$ for every t, with equality in the continuous relaxation at $t = n/4$. Summing and using the shared budget $Q = (m-1)n(n-1)$ gives

$$\sum_{x\in V} t_x \leq \frac{4}{n}\sum_{x\in V} d^+(x)d^-(x) \leq \frac{4Q}{n} = 4(m-1)(n-1).$$

Conversely, fix m and take n large. Assign $t = n/4$ to as many abstract vertices as the budget permits, assign one further $t \in [0, n/4]$ so that its continuous cost absorbs the remainder, and assign zero to the rest. This uses at most $16(m-1) + 1 \leq n$ vertices. Since the remainder is within $O_m(n)$ of the full cost $n^2/16$, the final t is $n/4 - O_m(\sqrt{n})$. The resulting total is $4(m-1)n - O_m(\sqrt{n})$, matching the coefficient four asymptotically. Therefore no smaller universal coefficient follows from these two inequalities alone.

The crude per-vertex bound therefore loses nothing within this relaxation, and the reason is that its two sources of slack close at that very point. Lemma A.24 is an equality only when $d^+(x) = d^-(x)$, and the floor substitution $d(x) \to n/2$ is an equality only when $d(x) = n/2$. Both hold at once exactly when $d^+(x) = d^-(x) = n/4$, which is the profile the relaxed optimum concentrates on. Cutting the constant therefore needs a second and independent inequality, one that constrains interference among the handful of expensive near-balanced vertices themselves rather than a sharper reading of the one already in hand.

A.4.1 What that proof actually used

Neither the proposition nor the theorem above ever used feasibility twice. Each used it once, in Step 1, to cap a single explicit family of routes, and everything after that is arithmetic on in-degrees and out-degrees. Promoting that one cap from a deduction to a hypothesis costs nothing here and buys two further results below, one of them for a problem Chapter 3 lists as untouched by the arc case.

Definition A.27 (Two-step budget)**.** Let D be a simple digraph and let $C \geq 1$. Call D *C-budgeted* if every ordered pair (u, v) of distinct vertices satisfies

$$\mathbf{1}_{(u,v)\in A(D)} + p_2(u, v) \;\leq\; C,$$

where $p_2(u, v)$ counts, as in Proposition A.23, the vertices $x \notin \{u, v\}$ with both (u, x) and (x, v) present.

Lemma A.28 (The counting core)**.** *Every C-budgeted simple digraph D on $n \geq 2$ vertices satisfies*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4C(n-1).$$

Proof. Read the proofs of Proposition A.23 and Theorem A.25 again with C written wherever $m-1$ stood. Step 1 of the proposition is now the hypothesis rather than a deduction, and it was the only place feasibility ever entered, since Steps 2, 3 and 4 manipulate in-degrees and out-degrees and nothing else. The theorem's induction needs one further point, that the hypothesis is inherited by induced subdigraphs, which Lemma A.16 supplied there by way of feasibility. Here it is immediate: deleting a vertex only deletes arcs and removes candidate midpoints, so neither term of the budget can rise. With those two observations the induction runs verbatim and delivers the stated bound. □

The first application is the directed *vertex* problem. Theorem 1.11 settles it at $m = 2$, and for $m \geq 3$ this thesis has been careful to record that it does not follow from the arc case: Whitney's

$\kappa \le \lambda$ makes the vertex-feasible family the larger of the two, so an arc upper bound restricts nothing there. That remains true of the *bound*. What does carry across is the route-counting, because the family Step 1 exhibits happens to be disjoint in the stronger sense as well, and once that is noticed the argument never needs the arc value at all.

Theorem A.29 (Directed vertex problem, leading constant and order). *For all $m \ge 2$ and $n \ge 2$, every simple digraph D on n vertices with $\kappa_D^{\max} \le m-1$ satisfies*

$$|A(D)| \;\le\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4(m-1)(n-1),$$

and consequently, for each fixed m,

$$k_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; \Theta_m(n) \qquad (m \ge 3).$$

Proof. Fix an ordered pair (u, v) of distinct vertices. The direct arc (u, v), when it is present, is a route with no interior vertex at all, and a two-step route $u \to x \to v$ has interior exactly $\{x\}$, one vertex, distinct from both u and v. Distinct midpoints give disjoint interiors, and the direct arc's interior is disjoint from all of them for want of any element, so these $\mathbf{1}_{(u,v)\in A(D)} + p_2(u, v)$ routes are pairwise internally vertex-disjoint and not merely arc-disjoint. Feasibility caps the family at $\kappa_D(u, v) \le m-1$, so D is $(m-1)$-budgeted and Lemma A.28 applies.

For the lower bound, the augmented bipartite digraph of Construction 1.13 has $\lambda^{\max} \le m-1$, hence $\kappa^{\max} \le m-1$ by Whitney, so for $n \ge m$ it witnesses $k_m^{\mathrm{dir}}(n) \ge \lfloor (n+m-2)^2/4 \rfloor$. For fixed $m \ge 3$ this is $n^2/4 + \Theta_m(n)$, and at $m = 2$ Theorem 1.11 gives the sharper $n^2/4 + O(1)$ asymptotic. □

The vertex problem at $m \ge 3$ is therefore open in the same narrow sense as the arc problem rather than in the wide one. Its leading constant is $\frac{1}{4}$ unconditionally, the term after it is linear in n, and since $\ell_m^{\mathrm{dir}}(n) \le k_m^{\mathrm{dir}}(n)$ with both now inside $n^2/4 + \Theta_m(n)$, the two values agree to first order. What is left is the coefficient of the linear term, the same quantity Conjecture A.22 is needed for on the arc side, together with the conjecture that the two values coincide exactly. It is worth being clear about what did and did not transfer. The upper bound of Theorem A.25 did not: it is proved over the smaller family and says nothing here. The route family behind it did, and that is a statement about a construction rather than about a bound, which is why Whitney's inequality does not block it.

A.5 The directed multigraph problem, solved in full

The body proves $L_m^{\mathrm{dir}}(n) = 2(n-1)(m-1)$ for $n \le 6$ by an optimisation that scales m out of the problem entirely (Theorem 2.2), and the section just below, on the simple digraph at $m = 2$, closes *every* n by repeatedly deleting one vertex of least degree and averaging. It is natural to hope that the same trick settles the multigraph case at every m too, and for a long while it very nearly does. Dividing every multiplicity by $m-1$, as in the scaling step of Chapter 2, turns a feasible multigraph into a weighting with values in $[0,1]$ whose pairwise maximum flows are all at most one. Deleting a vertex of least weighted degree from that weighting then drives the same averaging argument through cleanly on the linear branch, and on the quadratic branch as well whenever n is even. What breaks it is a single fractional remainder at odd n: a leftover of size $\frac{k}{2k-1}$ that an integer count could round away but a genuinely fractional weight cannot. Finding a specific vertex that is provably light enough to absorb that remainder turns out to resist every direct attempt.

This section abandons that attempt and proves the full value a different way. Instead of deleting one vertex at a time, it deletes one whole *sparse subgraph* at a time, chosen so that the deletion is guaranteed to lower every pair's connectivity by at least one in a single stroke. That change removes the fractional step altogether, along with the restriction to a particular m or a particular parity of n.

Lemma A.30 (Arithmetic identity)**.** *For all $n \geq 2$, $\left\lfloor \frac{(n-1)^2}{4} \right\rfloor + \lfloor n/2 \rfloor = \left\lfloor \frac{n^2}{4} \right\rfloor$.*

Proof. For $n = 2k$ the left side is $k(k-1) + k = k^2$, and for $n = 2k+1$ it is $k^2 + k = k(k+1)$. Both match the right side. □

Theorem A.31 (Directed multigraph value, every n and m)**.** *For all $n \geq 2$ and $m \geq 2$,*

$$L_m^{\mathrm{dir}}(n) \;=\; (m-1)\,M(n), \qquad M(n) := \max\!\Big(2(n-1),\ \left\lfloor \frac{n^2}{4} \right\rfloor\Big).$$

A.5.1 The lower bound: thickening

The lower bound is short, and it is worth isolating the one idea that makes it short: multiplying every arc's multiplicity by a fixed integer multiplies every cut's capacity, and hence every pairwise connectivity, by that same integer.

Lemma A.32 (Thickening)**.** *Let D_0 be a directed multigraph and $k \geq 1$ an integer. Write $k \cdot D_0$ for the multigraph obtained by multiplying every multiplicity $\mu(u,v)$ by k. Then $\lambda_{k\cdot D_0}(u,v) = k \cdot \lambda_{D_0}(u,v)$ for every ordered pair $u \neq v$, and $|A(k \cdot D_0)| = k\,|A(D_0)|$.*

Proof. By Theorem A.1, $\lambda_{D_0}(u,v)$ equals the minimum capacity, over all u–v cuts, of the total multiplicity crossing that cut. Multiplying every multiplicity by k multiplies the capacity of *every* cut by k at once, so it multiplies the minimum over all cuts by k as well, which is precisely $\lambda_{k\cdot D_0}(u,v) = k \cdot \lambda_{D_0}(u,v)$. The arc count scales by k because every one of its summands, each multiplicity, does. □

Construction A.33 (Thickened tree and thickened bipartite digraph)**.** For $m \geq 2$, take any spanning tree T of K_n and give every edge multiplicity $m-1$ in each direction. Between two vertices, T offers only its one tree path, so this is $m-1$ copies of the $m = 2$ bidirected tree of Theorem 1.10, feasible at $\lambda^{\max} = 1$ there, and Lemma A.32 with $k = m-1$ gives $\lambda^{\max} = m-1$ here and an arc count of $2(n-1)(m-1)$. Alternatively, take the one-directional complete bipartite digraph of Construction 1.8 (balanced parts A, B with $|A| = \lfloor n/2 \rfloor$, $|B| = \lceil n/2 \rceil$, feasible at $\lambda^{\max} = 1$) and give every arc multiplicity $m-1$: the same lemma gives $\lambda^{\max} = m-1$ and $(m-1)\left\lfloor n^2/4 \right\rfloor$ arcs.

Taking whichever of the two constructions is larger gives $L_m^{\mathrm{dir}}(n) \geq (m-1)M(n)$ for every n and m, which is the lower bound half of Theorem A.31. The rest of this section proves the matching upper bound.

A.5.2 The upper bound: a reachability skeleton

It helps to first ask what a single vertex deletion actually buys at $m = 2$: removing a vertex can only remove routes passing through it, so no pair's connectivity ever rises, and the whole $m = 2$ induction rests on that one guarantee together with an averaging bound on how much one well-chosen vertex carries. The multigraph problem at a general m wants those same two ingredients: a deletion that is guaranteed to lower every connectivity, paired with a bound on how much it removes. There, though, the natural analogue of "one vertex" breaks down. Once multiplicities are free to sit anywhere between 0 and $m-1$, no single vertex is guaranteed to carry a controllable share of the arcs, which is exactly the fractional remainder above. The fix is to stop insisting on a vertex and delete a whole *subgraph* instead: one general enough to always exist, sparse enough to bound, and specific enough that removing it is guaranteed to lower every pairwise connectivity by at least one.

Definition A.34 (Reachability skeleton)**.** Let D be a directed multigraph. A spanning subdigraph $R \subseteq D$, using at most one copy of each arc it uses (no parallel copies, even if D has them), is a *reachability skeleton* of D if, for every pair of vertices $u \neq v$, u reaches v by some directed path in R exactly when u reaches v by some directed path in D.

A reachability skeleton keeps every yes-or-no fact about who can reach whom and discards everything else, in particular every parallel copy of every arc. It is a caricature of D: thin, but faithful on the one question connectivity actually depends on. The next two lemmas turn that faithfulness into the whole induction. The first says a reachability skeleton always exists and is never too big. The second says deleting it always lowers every connectivity by at least one.

Lemma A.35 (A sparse reachability skeleton always exists)**.** *Every loopless directed multigraph D on $n \geq 2$ vertices has a reachability skeleton R with $|A(R)| \leq M(n)$.*

Building R takes three steps: handle the traffic *inside* each strongly connected piece of D, handle the traffic *between* pieces, then count. A *strongly connected component* (an SCC) is a maximal set of vertices any two of which reach each other, and every directed multigraph partitions uniquely into its SCCs, since mutual reachability is an equivalence relation. Figure A.3 carries one small running example through all three steps, and it is worth glancing at now, before the general argument, since every object named below appears in it concretely.

Step 1: inside each component. Fix one SCC of order $n_i \geq 1$ and pick any vertex r inside it as a root. Since r reaches every vertex of the component, a standard fact about digraphs gives a spanning *out-arborescence*: a tree of $n_i - 1$ arcs of D, every one pointing away from r, along which r alone reaches every other vertex of the component (grow it one hop at a time, always available since the component is strongly connected). Symmetrically, because every vertex of the component reaches r, there is a spanning *in-arborescence* of $n_i - 1$ arcs, every one pointing toward r, along which every vertex reaches r. Taking both trees together, any two vertices u, v of the component are joined by the route $u \to r \to v$, first the in-arborescence, then the out-arborescence, so the $2(n_i - 1)$ arcs of the two trees alone reconstruct *every* reachability relation inside this one component.

Step 2: between components. Contract each SCC to a single point and, between two contracted points, keep one arc whenever D has at least one arc from the first component to the second (dropping any further parallel arcs between the same two components at this stage). Call the result Q_0, a digraph on q points, one per component. Two different SCCs can never reach each other in both directions, since that would make them one strongly connected component after all, so Q_0 has no directed cycle: it is a DAG (a directed acyclic graph). Reachability between two different components of D is, by this very construction, exactly reachability between the two corresponding points of Q_0. Thin Q_0 down to an *inclusion-minimal* spanning subdigraph $Q \subseteq Q_0$ that still has the same reachability relation, deleting one arc at a time for as long as some arc is redundant, which must stop since Q_0 has only finitely many arcs.

The reason to thin Q_0 down to Q is that Q's *underlying undirected graph*, meaning Q with the arrows erased, has no triangle, and this deserves seeing directly. Suppose three points x, y, z of Q were pairwise joined by some arc of Q. Since $Q \subseteq Q_0$ and Q_0 has no cycle, Q has none either, so between any two of x, y, z the arc runs in only one direction, never both. Three points pairwise joined with a consistent direction on each pair admit only one pattern, up to relabelling: a linear order, say $x \to y$, $y \to z$, and $x \to z$. But then $x \to z$ is redundant, since x already reaches z via y, so deleting it would leave the reachability relation of Q unchanged, contradicting that Q is inclusion-minimal. So no three points of Q can be pairwise joined at all: that is triangle-freeness of the underlying graph.

A triangle-free graph on q vertices has at most $\lfloor q^2/4 \rfloor$ edges, Mantel's theorem [18], the $r = 2$ case of the more general theorem of Turán [19]. Applied to Q,

$$|A(Q)| \;\leq\; \left\lfloor \frac{q^2}{4} \right\rfloor .$$

Step 3: assembling R and counting. Build R from the two arborescences of Step 1 inside every SCC, together with one witness arc of D for every arc of Q (some arc of D realises it, by the very definition of Q_0). By Step 1, every pair inside one component reaches each other in R exactly as in D. By Step 2, and because within-component reachability is already secured, every pair in two different components reaches each other in R exactly as in D too: to travel from u to

v across the components Q's path visits, reach each witness arc's tail inside its own component (Step 1), cross the witness arc, and repeat. Since $R \subseteq D$ throughout, R can never certify a reachability D does not already have, so R is a genuine reachability skeleton. Writing $n_1, \ldots, n_q$ for the SCC orders, so $\sum_i n_i = n$, its arc count is at most the $2(n_i - 1)$ arborescence arcs of each component plus one witness per arc of Q. The first count is an upper bound rather than an equality, since the in- and out-arborescence of one component may share an arc, which only helps. So

$$|A(R)| \;\le\; \sum_{i=1}^{q} 2(n_i - 1) \;+\; |A(Q)| \;\le\; 2(n-q) \;+\; \left\lfloor \frac{q^2}{4} \right\rfloor \;=:\; f(q).$$

It remains to bound $f(q)$ over the range $1 \le q \le n$ that the number of components can actually take. Its increments are

$$f(q+1) - f(q) \;=\; -2 \;+\; \left\lfloor \frac{q+1}{2} \right\rfloor,$$

using the identity $\lfloor (q+1)^2/4 \rfloor - \lfloor q^2/4 \rfloor \;=\; \lfloor (q+1)/2 \rfloor$, checked on the two parities of q (at $q = 2j$ both sides are j, at $q = 2j+1$ both are $j+1$). Since $\lfloor (q+1)/2 \rfloor$ never decreases as q grows, neither do the increments of f, which is to say f is convex. A convex function on an interval attains its largest value at one of the two ends, because a value at an interior point is a weighted average of the two ends or less. Hence

$$f(q) \;\le\; \max\bigl(f(1),\, f(n)\bigr) \qquad \text{for } 1 \le q \le n.$$

At $q = 1$ the whole digraph is one component, so $f(1) = 2(n-1) + 0$. At $q = n$ every component is a single vertex, so $f(n) = 0 + \lfloor n^2/4 \rfloor$. Hence $f(q) \le \max\bigl(2(n-1), \lfloor n^2/4 \rfloor\bigr) = M(n)$ for every q in range, which is the lemma.

Lemma A.36 (Deleting a skeleton lowers every connectivity by one)**.** *Let D be a directed multigraph with $\lambda^{\max}(D) \le k$ for some $k \ge 1$, and let R be a reachability skeleton of D. Write $D - R$ for the multigraph obtained by deleting, from each multiplicity $\mu(u,v)$, the one copy (if any) that R uses on that pair. Then $\lambda^{\max}(D - R) \le k - 1$.*

In words: subtracting one copy of each skeleton arc costs every pair at least one route (a single pair can lose more, if some of its own routes happened to run through the deleted arcs too). The skeleton preserves reachability, so however many arc-disjoint routes survive the deletion, one further route can always be run entirely inside the arcs that were deleted, and that extra route is what the original ceiling has to pay for.

Proof. Fix an ordered pair $u \ne v$ and suppose $D - R$ carries $t \ge 1$ pairwise arc-disjoint u–v paths. These use arcs of $D - R$ and hence of D, since $D - R \subseteq D$, so u reaches v in D. Because R is a reachability skeleton, u then also reaches v in R: some directed path P from u to

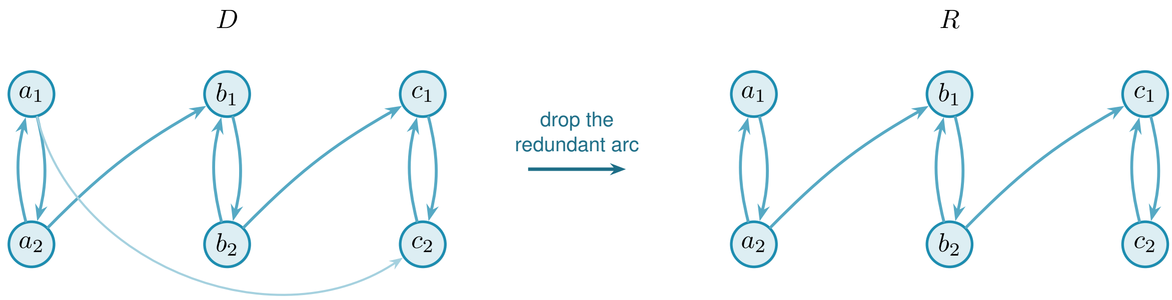


Figure A.3. A small instance of Lemma A.35. In D (left) three bidirected pairs $A = \{a_1, a_2\}$, $B = \{b_1, b_2\}$, $C = \{c_1, c_2\}$ are the strongly connected components, joined by $a_2 \to b_1$, $b_2 \to c_1$, and $a_1 \to c_2$. Contracting each pair to a point gives a condensation Q_0 on three points, carrying all three arcs $A \to B$, $B \to C$ and $A \to C$. That is exactly the transitively-closed triangle the proof rules out of the thinned Q. The arc $A \to C$ is redundant, since A already reaches C through B, so the transitive reduction Q keeps only $A \to B$ and $B \to C$. The underlying graph of Q is then a path on three points, triangle-free as claimed. The reachability skeleton R (right) reflects exactly this: every SCC keeps both of its internal arcs (an in- and an out-arborescence on two vertices is just the pair itself), while of the two component-crossing arcs of D only the two witnesses of Q's two arcs survive. The direct arc $a_1 \to c_2$ is dropped, yet a_1 still reaches c_2 in R: first $a_1 \to a_2$, then $a_2 \to b_1$, then $b_1 \to b_2$, then $b_2 \to c_1$, then $c_1 \to c_2$.

v uses only arcs of R. Every arc of R has been subtracted out of $D - R$, so P shares no arc with any of the t paths already found inside $D - R$. Together, P and those t paths are $t + 1$ pairwise arc-disjoint u–v paths in D, so $t + 1 \leq \lambda_D(u, v) \leq k$, that is, $t \leq k - 1$. Pairs with $t = 0$ trivially satisfy $t \leq k - 1$ once $k \geq 1$, so this holds for every pair, giving $\lambda^{\max}(D - R) \leq k - 1$. □

Proof of Theorem A.31. The lower bound is Construction A.33. For the upper bound, fix n and prove by induction on $k = m - 1 \geq 0$ that every directed multigraph D on n vertices with $\lambda^{\max}(D) \leq k$ has $|A(D)| \leq k\, M(n)$. At $k = 0$, every pair has $\lambda_D(u, v) \leq 0$, so no arc (u, v) can even be present, since a single arc is already one route, and $|A(D)| = 0 = 0 \cdot M(n)$.

For $k \geq 1$, take D with $\lambda^{\max}(D) \leq k$ and let R be a reachability skeleton, which exists with $|A(R)| \leq M(n)$ by Lemma A.35. By Lemma A.36, $D - R$ has $\lambda^{\max}(D - R) \leq k - 1$, and it is again a directed multigraph on the same n vertices, so the induction hypothesis gives $|A(D - R)| \leq (k - 1)M(n)$. Since R uses at most one copy of each arc it touches, the arcs of D split exactly into those of R and those left in $D - R$, so

$$|A(D)| \;=\; |A(R)| \;+\; |A(D - R)| \;\leq\; M(n) \;+\; (k - 1)M(n) \;=\; k\, M(n).$$

Taking $k = m - 1$ gives $L_m^{\mathrm{dir}}(n) \leq (m - 1)M(n)$, matching the lower bound. □

Nothing above depended on m or on the parity of n: the same three-step construction of R and the same one-line deletion bound run identically at every level. In particular, the induction closes exactly the finite computation the rest of this appendix once needed help to settle.

Corollary A.37. $L_3^{\mathrm{dir}}(7) = 24.$

Proof. $M(7) = \max(12, 12) = 12$, so Theorem A.31 at $m = 3$, $n = 7$ gives $L_3^{\mathrm{dir}}(7) = 2 \cdot 12 = 24$. Concretely, the induction peels two reachability skeletons off any 7-vertex witness in turn, each of at most $M(7) = 12$ arcs by Lemma A.35, first lowering $\lambda^{\max}$ from 2 to 1 and then from 1 to 0: $24 = 12 + 12$ spends the whole budget twice over, with nothing left for a third pass. This is a proof by hand of the one statement Chapters 2 and 3 once recorded as an outstanding computation. That computation was the mixed-integer program `prove_integral_arc_bound(7, 3, 25)`, whose `INFEASIBLE` verdict would have settled the same fact, and which never returned one: it reached its time limit on the bundled CBC solver, and a fourteen-hour uncapped enumeration was stopped without a verdict. It was never run on a stronger solver, and it no longer needs to be. The honest summary is that the machine route to this fact was abandoned unfinished and the hand proof above replaced it, rather than the two confirming each other. □

Corollary A.38 (The fractional relaxation is integral)**.** *For every $n \geq 2$, the m-free optimum $M^*(n)$ of Chapter 2 equals $M(n)$, and it is attained by a $\{0, 1\}$ weight matrix. Consequently the scaling identity*

$$L_m^{\mathrm{dir}}(n) \;=\; (m-1)\, M^*(n)$$

holds for every n and every $m \geq 2$, with no hypothesis on the optimum.

In words: allowing fractional weights buys nothing. The relaxed optimum is already attained by an honest multigraph with weights 0 and 1, so the scaling identity that carries one computed number to every m at once needs no integrality hypothesis attached to it.

Proof. For $M^*(n) \geq M(n)$, both constructions of Construction A.33 at multiplicity one are $\{0, 1\}$ matrices with all pairwise maximum flows equal to one, of total weight $2(n-1)$ and $\lfloor n^2/4 \rfloor$ respectively.

For the reverse, note first that the optimum is attained at a *rational* matrix. A weight matrix is feasible exactly when, for each ordered pair, some cut separating that pair has capacity at most one, so the feasible region is the union, over the finitely many ways of choosing one cut per pair, of the bounded rational polytopes cut out by those choices together with $0 \leq w \leq 1$. A linear objective over a bounded rational polytope attains its maximum at a rational vertex, and $M^*(n)$ is the largest of finitely many such maxima.

So let w be a feasible rational optimum and let q be a common denominator of its entries. Then $q\,w$ has integer entries, and multiplying every weight by q multiplies every cut capacity by q and hence every maximum flow by q, exactly as in Lemma A.32. So $q\,w$ is a directed multigraph with $\lambda^{\max} \leq q$, that is, feasible at $m - 1 = q$, and Theorem A.31 bounds its arc count by $q\,M(n)$. Dividing by q gives $M^*(n) = \sum_{u \neq v} w(u, v) \leq M(n)$.

The identity follows because Theorem A.31 gives $L_m^{\mathrm{dir}}(n) = (m-1)M(n)$. □

This settles the integrality question Chapter 2 raises when it introduces $M^*(n)$, namely whether the heaviest fractional weight matrix is always a $\{0, 1\}$ one. It is, and the reason is worth noting because it is backwards from the usual direction of such arguments. Nothing here inspects the fractional problem at all. An integral theorem, proved for every m at once, is strong enough to be applied to the scaled-up matrix $q\,w$, and the fractional statement falls out of the integral one rather than the other way round. That is only possible because Theorem A.31 holds at every m, however large, so no denominator is out of reach.

Corollary A.39 (The $m = 2$ case collapses to the simple digraph)**.** *At $m = 2$ (so $k = 1$), Theorem A.31 reads $L_2^{\mathrm{dir}}(n) = M(n)$, matching Theorem 1.10 exactly, and this is not a coincidence. A directed multigraph with $\lambda^{\max} \leq 1$ can carry no parallel arcs at all, since two parallel copies of one arc are already two arc-disjoint routes between their endpoints, so at $m = 2$ the multigraph problem and the simple digraph problem are literally the same problem. Theorem A.31 recovers Theorem 1.10 as its $k = 1$ case, rather than needing it as a separate input.*

Theorem A.31 settles the value at every n and m, but a value is not a full description of the extremal digraphs, and on the linear branch a companion fact about *which* multigraphs attain it is worth recording, because it comes almost for free and it is genuinely more than the value alone gives. Call a directed multigraph *everywhere-saturated* at level m when every ordered pair of distinct vertices has local connectivity exactly $m-1$. The doubled bidirected spanning trees at multiplicity $m-1$ form an extremal family on the linear branch and are everywhere-saturated: the $m-1$ routes between two vertices run along the doubled tree path, and cutting one doubled edge of that path is a cut of $m-1$ arcs.

Lemma A.40 (Saturated attachment)**.** *Let $m \geq 2$ and let D_0 be a directed multigraph on at least two vertices with $\lambda_{D_0}(x, y) = m-1$ for every ordered pair of distinct vertices. Let D be obtained from D_0 by adding one new vertex v together with any multiset of arcs incident to v. If $\lambda_D^{\max} \leq m-1$, then $d(v) \leq 2(m-1)$. If $d(v) = 2(m-1)$, then every arc at v joins v to one single vertex u, with $\mu(v, u) = \mu(u, v) = m-1$, and D is again everywhere-saturated.*

Proof. No mixed pair. Suppose v has an in-arc from u and an out-arc to w with $u \neq w$. By Menger and saturation D_0 carries $m-1$ pairwise arc-disjoint u-to-w routes, and the route $u \to v \to w$ uses only arcs incident to v, so it is arc-disjoint from all of them and $\lambda_D(u, w) \geq m$, a contradiction. Hence v is a pure source, a pure sink, or every arc at v touches one single vertex u.

A pure source has $d(v) \leq m-1$. Let $t = \sum_x \mu(v, x)$ and fix u with $\mu(v, u) \geq 1$. Any cut $(S, \bar{S})$ with $v \in S$ and $u \in \bar{S}$ has capacity $\sum_{x \in \bar{S}} \mu(v, x) + e_{D_0}(S \setminus \{v\}, \bar{S})$. There are two kinds. If $S = \{v\}$ the capacity is exactly t. If S also contains a vertex x of D_0, then $S \setminus \{v\}$ and $\bar{S}$ separate x from u inside D_0, so the second term is at least $\lambda_{D_0}(x, u) = m-1$, while the first is at least $\mu(v, u)$. Every cut therefore has capacity at least $\min\bigl(t,\ \mu(v, u) + m - 1\bigr)$, and by max-flow min-cut so does $\lambda_D(v, u)$, which feasibility caps at $m-1$. Since $\mu(v, u) \geq 1$, the second argument of that minimum exceeds $m-1$, so the minimum is t and $t \leq m-1$.

A pure sink has $d(v) \leq m-1$. Reverse every arc: the reverse of an everywhere-saturated multigraph is everywhere-saturated, a pure sink becomes a pure source, and the previous step applies.

A single partner. If every arc at v touches one vertex u, then parallel arcs are already that many arc-disjoint one-step routes, so $\mu(v,u) \le m-1$ and $\mu(u,v) \le m-1$, giving $d(v) \le 2(m-1)$.

Equality. Since $2(m-1) > m-1$, a degree of $2(m-1)$ rules out the pure cases, so v has a single partner at full multiplicity both ways. The result is feasible and again everywhere-saturated: a route through v must enter and leave through u, so it adds nothing to any flow between old vertices, and $\lambda_D(x,v) = \min\bigl(\lambda_{D_0}(x,u), \mu(u,v)\bigr) = m-1$ for every old x, symmetrically for $\lambda_D(v,x)$. □

Remark A.41 (The linear branch grows one leaf at a time)*.* The equality case of Lemma A.40 attaches a new leaf to an arbitrary vertex at full multiplicity, and that is the only attachment reaching $2(m-1)$. Iterating from a single doubled edge, itself everywhere-saturated on two vertices, generates all doubled bidirected spanning trees. Thus this extremal family is closed under maximal attachment and grows one leaf at a time. The lemma does not assert that every linear-branch extremiser admits such a leaf-deletion sequence. The program confirms the lemma exhaustively at $m = 3$: over every doubled tree on five and six vertices and every one of the 3^{2n} possible attachment patterns, the densest feasible attachment has degree exactly $4 = 2(m-1)$, and the degree-4 patterns are precisely the single-partner attachments at full multiplicity, one per choice of partner.

Remark A.42 (The degree-capped 24-arc extremisers at $n = 7$)*.* Lemma A.40 proves rigidity for maximal one-vertex attachments to an everywhere-saturated graph, but it does not classify every extremiser on the linear branch. At $n = 7$, $m = 3$, the two branches tie at 24 arcs, and every doubled bidirected tree is already an extremiser, giving eleven tree isomorphism classes before the bipartite examples are counted. The separate machine run discussed here answered only a degree-capped question relevant to extension arguments. The tool was the sound generation enumerator of Chapter 2, in the call `enumerate_extremal_directed_multigraphs_via_generation(7, 3, 24, max_degree=8)`, which prunes only with proved bounds and is therefore complete within the imposed maximum-total-degree cap. It ran for about seven hours across ten processor cores and returned exactly three isomorphism classes satisfying that cap. Two of them are the bipartite shapes $2 \cdot B_{3,4}$ and $2 \cdot B_{4,3}$. The third is the doubled bidirected path P_7, the one non-star tree whose maximum degree (2) is compatible with an 8-regular graph one vertex up. Each of the three was re-verified by the exact checker. This is a restricted classification, not a classification of all 24-arc extremisers. The reachability-skeleton induction proves the value 24 without it, and the unrestricted extremal family at this tie size is substantially larger.

A.5.3 Extremal classification on the quadratic branch

The classification above is one finite data point, and the linear branch is settled by Lemma A.40, so what is missing is the quadratic branch at a general n. The reachability-skeleton proof looks unpromising for this, since it is designed never to need to know which multigraph it is taking apart. But that indifference is only in the direction of the *bound*. Run backwards, from the assumption that the bound is met exactly, the same construction is unexpectedly rigid: it pins down the number of strongly connected components, then the shape of the skeleton, and finally the multigraph.

Two consequences of equality do the work, and both come from reading the count of Lemma A.35 as an equation rather than an inequality.

Lemma A.43 (What equality forces)**.** *Let $n \geq 8$, so that $\lfloor n^2/4 \rfloor > 2(n-1)$ and $M(n) = \lfloor n^2/4 \rfloor$, and let D be a directed multigraph on n vertices with $\lambda_D^{\max} \leq m-1$ and $|A(D)| = (m-1)\lfloor n^2/4 \rfloor$. Then*

(1) D is acyclic, and

(2) D has a reachability skeleton R whose underlying undirected graph is exactly the balanced complete bipartite graph $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$, so R is an acyclic orientation of it.

Proof. The proof of Theorem A.31 peels reachability skeletons $R_1, \ldots, R_{m-1}$, each deletion dropping every pairwise connectivity by at least one, so that after $m-1$ peels no arc survives and $|A(D)| = \sum_i |A(R_i)|$. Each $|A(R_i)| \leq M(n)$, so equality in the hypothesis forces $|A(R_i)| = M(n)$ for every i, and in particular for $R := R_1$.

Now read the count of Lemma A.35 backwards. It gives $|A(R)| \leq f(q) := 2(n-q) + \lfloor q^2/4 \rfloor$, where q is the number of strongly connected components of D, so $f(q) \geq \lfloor n^2/4 \rfloor = f(n)$. The increments $f(q+1) - f(q) = -2 + \lfloor (q+1)/2 \rfloor$ are non-decreasing, so f is convex on $q \in \{1, \ldots, n\}$. Write $d_i := f(i+1) - f(i)$. Suppose toward a contradiction that $q < n$. If $q = 1$ then $f(q) = f(1) = 2(n-1)$, and the hypothesis $n \geq 8$ gives $\lfloor n^2/4 \rfloor > 2(n-1)$, so $f(q) < f(n)$, contradicting $f(q) \geq f(n)$. So $1 < q < n$. Since $f(1) < f(n) \leq f(q)$, the sum $d_1 + \cdots + d_{q-1} = f(q) - f(1)$ is positive, and since the d_i are non-decreasing this forces the last term d_{q-1} to be positive too (otherwise every term up to it would be at most $d_{q-1} \leq 0$ and the sum could not be positive). Non-decreasing again gives $d_q, \ldots, d_{n-1} \geq d_{q-1} > 0$, so $f(n) - f(q) = d_q + \cdots + d_{n-1} > 0$, that is $f(n) > f(q)$, contradicting $f(q) \geq f(n)$. Both cases are impossible, so $q = n$: every strongly connected component is a single vertex, which is (1).

With $q = n$ the two arborescences of Step 1 of that proof are empty, so R consists of the witness arcs of Q alone and $|A(Q)| = \lfloor n^2/4 \rfloor$. The underlying undirected graph of Q is triangle-free, proved there from the inclusion-minimality of Q. A triangle-free graph on n vertices with $\lfloor n^2/4 \rfloor$ edges meets Mantel's bound [18] with equality, and the extremal graph for Mantel's theorem is unique: it is $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$. That is (2), and R is acyclic because D is. ☐

The next lemma is where the skeleton's minimality, which had only been used to make it small, is used to make it *shallow*. It costs nothing and it is what makes the theorem work.

Lemma A.44 (A minimal bipartite skeleton has no long path)**.** *Let R be as in Lemma A.43, with parts A and B. Then R contains no directed path with three arcs.*

Proof. Suppose $u \to v \to w \to x$ were such a path. A path in a bipartite graph alternates sides, so u and x lie on opposite sides, and since R's underlying graph is the *complete* bipartite graph, R contains an arc between them. It cannot be $x \to u$, which would close a directed cycle against the path, and D is acyclic. So $u \to x$ lies in R. But then deleting $u \to x$ from R changes no reachability, since u still reaches x along the path, contradicting the inclusion-minimality of Q. $\square$

Theorem A.45 (Extremal classification on the quadratic branch)**.** *Let $m \geq 2$ and $n \geq \max(8, 2m)$, so that $M(n) = \lfloor n^2/4 \rfloor$ and n is on the quadratic branch. If D is a directed multigraph on n vertices with $\lambda_D^{\max} \leq m-1$ and $|A(D)| = L_m^{\text{dir}}(n) = (m-1)\lfloor n^2/4 \rfloor$, then*

$$D \in \left\{(m-1)\cdot B_{\lceil n/2 \rceil,\, \lfloor n/2 \rfloor},\ (m-1)\cdot B_{\lfloor n/2 \rfloor,\, \lceil n/2 \rceil}\right\},$$

where $B_{a,b}$ has all arcs from its part of size a to its part of size b, and every displayed arc has multiplicity exactly $m-1$. Thus there is one isomorphism class when n is even and two when n is odd, interchanged by reversing every arc.

In words: past $n = \max(8, 2m)$ the underlying bipartite shape is forced. Split the vertices as evenly as possible, choose either part as the source, run every arc from it to the other part, and repeat each arc $m-1$ times. When n is odd the two choices give two non-isomorphic digraphs, because the source has different size, and they are reversals of one another.

Proof. Take R, A and B from Lemma A.43, and write $\alpha = |A| \geq \beta = |B| = \lfloor n/2 \rfloor$.

Step 1: R has no directed path with two arcs. Suppose $a \to b \to a'$ with $a, a' \in A$ and $b \in B$. If a had an incoming arc $b_0 \to a$ in R, then $b_0 \to a \to b \to a'$ would be a path with three arcs, which Lemma A.44 forbids. So a has no incoming arc in R, and since R's underlying graph is complete bipartite, $a \to b''$ for *every* $b'' \in B$. Symmetrically a' has no outgoing arc, so $b'' \to a'$ for every $b'' \in B$. The β routes $a \to b'' \to a'$, one through each $b'' \in B$, use pairwise distinct arcs, and each of their arcs is present in D because $R \subseteq D$. Hence $\lambda_D(a, a') \geq \beta = \lfloor n/2 \rfloor \geq m$, against feasibility. The case of a path $b \to a \to b'$ with middle vertex in A is identical with the sides exchanged and gives $\lambda_D(b, b') \geq \alpha \geq \beta \geq m$, equally impossible. This is the one step that uses $n \geq 2m$.

Step 2: R is one-directional. By Step 1 no vertex of R has both an incoming and an outgoing arc, so every vertex is a source or a sink. Fix $a \in A$. It is adjacent in R to every vertex of B, so it is not isolated. If a is a source then every $b \in B$ has an incoming arc, hence is a sink, hence has no outgoing arc, so every arc at every b runs into it: all arcs of R run from A to B. If instead a is a sink the same argument gives all arcs from B to A. Thus R is either $B_{\alpha,\beta}$ or $B_{\beta,\alpha}$, according to which part is the source. The remaining steps apply identically to either orientation, so from now on use A for the source part and B for the sink part without imposing an order on their sizes.

Step 3: D has no arc outside R. R has the same reachability as D, and in a one-directional bipartite digraph the only ordered pairs joined by a directed path are the pairs $(a, b) \in A \times B$, each by the single arc $a \to b$. So if D had an arc (u, x), then u reaches x in R, forcing $(u, x) \in A \times B$, and R already contains that arc. Hence D and R use exactly the same $\alpha\beta = \lfloor n^2/4 \rfloor$ ordered pairs.

Step 4: every multiplicity is $m-1$. A pair joined by $\mu(a,b)$ parallel arcs has $\lambda_D(a,b) \geq \mu(a,b)$, so feasibility gives $\mu(a,b) \leq m-1$ on each of the $\lfloor n^2/4 \rfloor$ pairs. Their total is $|A(D)| = (m-1)\lfloor n^2/4 \rfloor$, so every one of them attains the ceiling. Hence D is the multiplicity-$(m-1)$ one-directional complete bipartite digraph on these two parts, and their product $\lfloor n^2/4 \rfloor$ forces their sizes to be $\lceil n/2 \rceil$ and $\lfloor n/2 \rfloor$. Either size may be the source size, giving exactly the two displayed possibilities (which coincide up to isomorphism when n is even). □

Three remarks. First, on where the hypothesis $n \geq 2m$ is used and what it costs. It enters once, in Step 1, where a two-arc path in the skeleton is converted into $\lfloor n/2 \rfloor$ arc-disjoint routes between one pair, and feasibility kills that only when $\lfloor n/2 \rfloor$ reaches m. For $n < 2m$ the theorem is not proved, and it is not known to fail either. What can be said is that nothing in the argument suggests a real boundary there: the step is a lower bound on one pair's connectivity, and losing it removes a contradiction rather than supplying a counterexample. Second, a degree fact falls out of the same equality and is worth recording on its own, since it is what makes a direct search for extremisers finite in practice. Deleting a vertex leaves at most $(m-1)M(n-1)$ arcs, so every vertex of an extremiser carries degree at least $(m-1)\lfloor n/2 \rfloor$. For even n the degree sum $2(m-1)\lfloor n^2/4 \rfloor = n \cdot (m-1)(n/2)$ leaves no slack at all, so every vertex has degree exactly $(m-1)n/2$: an extremiser at even n is regular. Third, the shape of the argument is worth noticing. Theorem A.31 is indifferent to which multigraph it dismantles, and Chapter 3 records that indifference as the reason the value says nothing about uniqueness. That is true of the proof read forwards. Read backwards from equality, the same skeleton is a rigid object, and the two facts that had been mere bookkeeping, that f is maximised at one end and that a minimal skeleton is triangle-free, become a determination of q and an application of the equality case of Mantel's theorem.

A.6 The hypergraph bounds

The body's hypergraph claims rest on two theorems: a Menger theorem for Berge routes, which also certifies the helper-point checker of Chapter 2, and a hypergraph Gomory–Hu theorem from the recent literature.

Theorem A.46 (Menger for hypergraphs)**.** *For a hypergraph $\mathcal{H}$ and vertices u, v, the maximum number of hyperedge-disjoint Berge u–v paths equals the minimum size of a set C of hyperedges whose removal separates u from v.*

Proof. Build the helper-point network of Figure 2.3b: for each hyperedge e a helper arc $e^- \to e^+$ of capacity one, and arcs $x \to e^-$ and $e^+ \to x$ of unbounded capacity for each member $x \in e$. The proof matches the number of disjoint Berge paths to the size of a minimum cut by translating between Berge-path language and flow language in each direction: a family of hyperedge-disjoint

Berge paths becomes a family of arc-disjoint flow paths, and conversely a flow becomes a family of Berge paths, as the next two paragraphs spell out in turn.

Berge paths become flow. A Berge path $u, e_1, v_1, \ldots, e_k, v$ becomes the directed walk $u \to e_1^- \to e_1^+ \to v_1 \to \cdots \to v$, crossing one helper arc per hyperedge it uses. The e_i along a single Berge path are distinct, and two hyperedge-disjoint Berge paths share no hyperedge at all, so they use disjoint sets of helper arcs and translate to *arc-disjoint* flow paths, paths that share no arc, each carrying one unit of flow.

Flow becomes Berge paths. Conversely, because every capacity is an integer, a maximum flow may be taken integral, and an integral flow of value k splits into k arc-disjoint unit paths. Reading a helper crossing $\cdots x \to e^- \to e^+ \to y \cdots$ back as the single hyperedge step x, e, y, that is, *suppressing* the two helper nodes, turns each unit path into a Berge path, and no two of them can share a hyperedge, since that would send two units across some capacity-one helper arc.

Cuts. Finally, the member arcs are uncapped, so any cut of finite capacity uses helper arcs only, and a set of helper arcs separates u from v in the network exactly when the corresponding hyperedges separate them in $\mathcal{H}$. The max-flow min-cut theorem equates the largest number of arc-disjoint flow paths with the smallest helper-arc cut, which is precisely the claim. $\square$

Theorem A.47 (Hypergraph Gomory–Hu)**.** *Every r-uniform hypergraph $\mathcal{H}$ on vertex set V has a weighted tree T^* on V, its* hypergraph Gomory–Hu tree*, such that $\lambda_{\mathcal{H}}(u, v)$ equals the minimum weight on the u–v tree path, and each tree edge t induces a vertex partition $(S_t, V \setminus S_t)$ with $|\delta_{\mathcal{H}}(S_t)| = c^*(t)$, where $\delta_{\mathcal{H}}(S)$ is the set of hyperedges incident to both sides, that is, having at least one vertex in S and at least one outside it.*

This is the exact hypergraph echo of Theorem A.2, the same tree-of-cuts summary of all pairwise connectivities, with the single change that the capacity of a cut now counts the hyperedges straddling it, $|\delta_{\mathcal{H}}(S)|$, in place of the edges. Gomory and Hu's construction [16] needs nothing about the cut function beyond it being symmetric and submodular, and the hyperedge-cut function $|\delta_{\mathcal{H}}(\cdot)|$ has both properties exactly as the ordinary edge-cut function does, so the identical uncrossing construction builds this tree too. We use only the two properties in the statement, the path-minimum reading of $\lambda_{\mathcal{H}}$ and the identity $|\delta_{\mathcal{H}}(S_t)| = c^*(t)$.

Proof of Proposition 1.15. Upper bound. Let $\mathcal{H}$ be r-uniform with $\lambda_{\mathcal{H}}^{\max} \leq m - 1$, and let T^* be its hypergraph Gomory–Hu tree (Theorem A.47), a weighted tree on the same n vertices with weights c^*. The plan is to charge each hyperedge to tree edges and count the charges two ways, exactly as in the Mader proof, with the one change that a hyperedge spans r vertices rather than two.

Step 1: each hyperedge crosses at least $r-1$ tree cuts. Fix a hyperedge e, a set of r vertices, and let $\mathrm{ST}(e)$ be its *smallest connected subtree*, the minimal subtree of T^* that contains all r of them (its Steiner tree). A tree spanning at least r vertices has at least $r-1$ edges, so $\mathrm{ST}(e)$

has at least $r-1$ of them. Each such edge t is a genuine cut for e: deleting t from T^* splits the vertices into the two sides $(S_t, V \setminus S_t)$, and because t lies on $\mathrm{ST}(e)$ the hyperedge e has at least one vertex on each side. So e touches both sides, that is, $e \in \delta_{\mathcal{H}}(S_t)$. Therefore

$$\big|\{\, t \in E(T^*) : e \in \delta_{\mathcal{H}}(S_t) \,\}\big| \;\geq\; r-1 \qquad \text{for every hyperedge } e.$$

Step 2: count the incidences two ways. Sum that inequality over all hyperedges, then exchange the order of summation. This is legitimate because both of the middle sums below simply count the same pairs (e,t) with $e \in \delta_{\mathcal{H}}(S_t)$, once grouped by the hyperedge and once by the tree edge:

$$(r-1)\,|E(\mathcal{H})| \;\leq\; \sum_e \big|\{\, t : e \in \delta_{\mathcal{H}}(S_t) \,\}\big| \;=\; \sum_t \big|\{\, e : e \in \delta_{\mathcal{H}}(S_t) \,\}\big| \;=\; \sum_t |\delta_{\mathcal{H}}(S_t)|.$$

Step 3: read off the bound. By the Gomory–Hu property each tree edge has $|\delta_{\mathcal{H}}(S_t)| = c^*(t)$. Moreover $c^*(t)$ is itself a local connectivity $\lambda_{\mathcal{H}}$ of t's two endpoints: since those endpoints are adjacent in T^*, the u–v tree path between them is the single edge t, so the path-minimum reading of Theorem A.47 gives $\lambda_{\mathcal{H}}$ of that pair exactly $c^*(t)$, hence at most $m-1$ by feasibility. Summing over the $n-1$ tree edges, $\sum_t |\delta_{\mathcal{H}}(S_t)| = \sum_t c^*(t) \leq (m-1)(n-1)$. Chaining Steps 2 and 3,

$$(r-1)\,|E(\mathcal{H})| \;\leq\; (m-1)(n-1),$$

and dividing by $r-1$, then rounding down because $|E(\mathcal{H})|$ is an integer, gives $|E(\mathcal{H})| \leq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$. *Lower bound.* When $(r-1) \mid (n-1)$, take a hub h and split the other $n-1$ vertices into groups $G_1, G_2, \ldots$ of size $r-1$ each. Form the star hypertree whose hyperedges are $\{h\} \cup G_i$, one per group, and give each hyperedge multiplicity $m-1$. Every non-hub vertex lies in exactly $m-1$ hyperedges, which form a cut isolating it, so by Theorem A.46 every pair has Berge connectivity at most $m-1$, while the count is $\frac{(m-1)(n-1)}{r-1}$. For general n under the hypothesis $m-1 \leq \binom{n-2}{r-2}$, Theorem A.48 below attains the floor exactly, without even needing repeated hyperedges. $\square$

The multiplicities in the lower bound can in fact be avoided: for $r \geq 3$, *simple* r-uniform hypergraphs already attain the same bound, in sharp contrast to the graph case $r = 2$, where simplicity costs the factor that separates Mader's $\lfloor m(n-1)/2 \rfloor$ from the multigraph value $(m-1)(n-1)$.

Theorem A.48 (Simplicity is free for $r \geq 3$)**.** *Let $r \geq 3$, $n \geq r$, and $2 \leq m$ with $m - 1 \leq \binom{n-2}{r-2}$. Then the maximum number of hyperedges in a* simple *r-uniform hypergraph on n vertices with Berge edge-connectivity at most $m-1$ is exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$.*

Proof. The upper bound is Proposition 1.15, since a simple hypergraph is a multihypergraph. For the lower bound, fix a hub h and let $V' = V \setminus \{h\}$. By Lemma A.51 applied with rank $r-1$ and degree ceiling $d = m-1$ on the $n-1$ vertices of V', there is an $(r-1)$-uniform simple hypergraph

$\mathcal{G}^*$ on V' with maximum degree at most $m-1$ and exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. Adding h to each hyperedge produces distinct r-sets, so the result $\mathcal{H}$ is simple, and every non-hub vertex w lies in $\deg_{\mathcal{G}^*}(w) \le m-1$ hyperedges, which form a cut isolating w. By Theorem A.46, $\lambda_{\mathcal{H}}^{\max} \le m-1$. $\square$

Remark A.49 (The hypothesis is not decorative). Outside this range the displayed value can fail for simple hypergraphs, even though it stays a valid upper bound (Proposition 1.15) and, when $(r-1) \mid (n-1)$, stays exactly attained by a multihypergraph. The smallest case already shows it. At $n = r = m = 3$ there is only one 3-subset of a 3-element vertex set, so a simple hypergraph carries at most one hyperedge. The hypothesis needs $m - 1 \le \binom{n-2}{r-2} = \binom{1}{1} = 1$, and $m - 1 = 2$ misses it, so the formula's value $\lfloor 2 \cdot 2/2 \rfloor = 2$ is not attained by any simple hypergraph there. It is attained by the multihypergraph star, two parallel copies of that one triple. Theorem 1.17 carries the identical hypothesis for the vertex separation, for the same reason. Table 3.1's two hypergraph rows and the twelve-panel program's hyper enumeration, which counts simple hypergraphs only, both carry this distinction, stated where each formula is used.

The sparse hypergraph the construction needs is a corollary of a classical partition theorem, which we state before using it.

Theorem A.50 (Baranyai [20])**.** *If r divides n, the $\binom{n}{r}$ distinct r-subsets of an n-element set can be partitioned into $\binom{n-1}{r-1}$ classes, each a* perfect matching*, meaning a family of n/r disjoint r-sets that together cover every vertex exactly once. More generally, for any prescribed class sizes $a_1 + \cdots + a_t = \binom{n}{r}$, and with no divisibility assumed, the r-subsets can be partitioned into classes of exactly those sizes so that in the class of size a_i every vertex lies in either $\lfloor ra_i/n \rfloor$ or $\lceil ra_i/n \rceil$ of the sets, as evenly balanced as the size permits.*

The first statement is the memorable one. When r divides n the complete r-uniform hypergraph, the one holding every r-set, comes apart into $\binom{n-1}{r-1}$ perfect matchings, the exact hypergraph analogue of splitting a regular bipartite graph into perfect matchings by König's theorem. The proof is an induction that keeps the partial partition as balanced as possible at every step, an integer-flow argument in the spirit of Hall's marriage theorem rather than an explicit construction, and its full form is in Baranyai [20]. We need only the general equitable form, and only its conclusion about degrees.

Lemma A.51 (Sparse uniform hypergraphs exist)**.** *Let $r \ge 2$, $n \ge r$, and $0 \le d \le \binom{n-1}{r-1}$. There exists an r-uniform simple hypergraph on n vertices with maximum degree at most d and exactly $\left\lfloor \frac{dn}{r} \right\rfloor$ hyperedges.*

Proof. Put $e := \left\lfloor \frac{dn}{r} \right\rfloor \le \frac{n}{r}\binom{n-1}{r-1} = \binom{n}{r}$. Apply the equitable form of Theorem A.50 with two classes, of sizes e and $\binom{n}{r} - e$. Within each class every vertex is covered either $\lfloor re/n \rfloor$ or $\lceil re/n \rceil$ times, so the class of size e is a simple r-uniform hypergraph with $e = \left\lfloor \frac{dn}{r} \right\rfloor$ hyperedges and maximum degree at most $\lceil re/n \rceil \le \lceil d \rceil = d$, the last step because $re/n \le r(dn/r)/n = d$ and d is an integer. $\square$

A.7 The hypergraph vertex problem

The whole vertex problem translates into a graph problem through one observation, the one the rest of this section rests on and which Figure A.4 draws in full. Write $I(\mathcal{H})$ for the bipartite *incidence graph* of a hypergraph $\mathcal{H}$: one node per vertex, one node per hyperedge, an edge when the vertex lies in the hyperedge. A Berge path with distinct hyperedges is exactly a path of $I(\mathcal{H})$ between two vertex-class nodes. The interior of such a path consists of both kinds of node, the hyperedges it rides and the vertices where it changes hyperedge, so two paths of $I(\mathcal{H})$ are internally disjoint in the ordinary graph sense exactly when the two Berge routes share neither a hyperedge nor an intermediate vertex. That is precisely the separation fixed in Chapter 1, and it is the reason both halves were required there. Hence

$$\kappa_{\mathcal{H}}(u,v) \;=\; \kappa_{I(\mathcal{H})}(u,v),$$

the maximum number of internally disjoint u–v paths in $I(\mathcal{H})$, and the hypergraph vertex problem asks for the maximum number of degree-r nodes on one side of a bipartite graph whose other side holds n nodes, subject to no two of those n nodes being joined by m internally disjoint paths. Figure A.4a traces a single Berge path through both pictures, and Figure A.4b shows the star hypertree of Theorem 1.16, whose incidence graph is a tree.

Theorem 1.16, stated in Chapter 1, follows at once from the incidence translation.

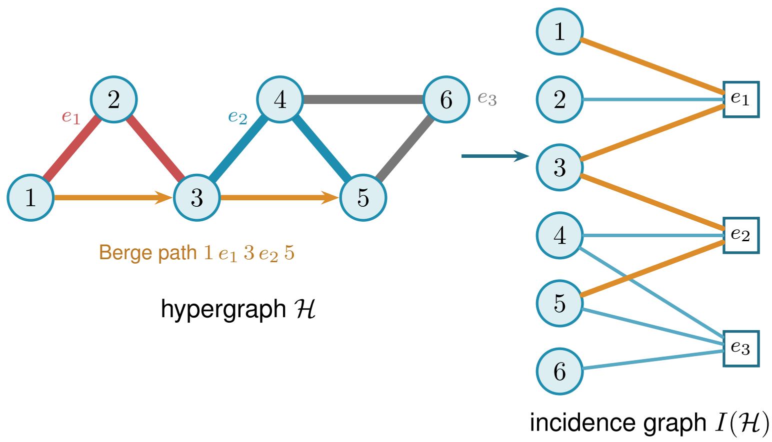
Proof of Theorem 1.16. We claim $\kappa_{\mathcal{H}}^{\max} \le 1$ holds if and only if $I(\mathcal{H})$ is a forest. If $I(\mathcal{H})$ contains a cycle $v_1, e_1, v_2, e_2, \ldots, v_\ell, e_\ell, v_1$ (it alternates classes, so $\ell \ge 2$, all v_i and e_i distinct), then v_1, e_1, v_2 and $v_1, e_\ell, v_\ell, e_{\ell-1}, \ldots, e_2, v_2$ are two Berge v_1–v_2 paths with disjoint hyperedge sets and disjoint interior vertex sets, so $\kappa(v_1,v_2) \ge 2$. (At $\ell = 2$ this is the repeated pair: two hyperedges sharing two vertices.) Conversely, two such paths between u and v concatenate to a cycle of $I(\mathcal{H})$. This proves the claim. A forest on $n+q$ nodes has at most $n+q-1$ edges, while $I(\mathcal{H})$ has exactly rq edges, so $rq \le n+q-1$, i.e. $q \le (n-1)/(r-1)$, and q is an integer. The star hypertree has $\lfloor (n-1)/(r-1) \rfloor$ hyperedges and a forest as incidence graph (one tree together with any unused isolated vertices) because each block vertex lies in one hyperedge and every cycle would need two hyperedges sharing two vertices, so the floor is attained. □

Proposition A.52 (General lower bound)**.** *For $r \ge 3$, $m \ge 2$ and $m-1 \le \binom{n-2}{r-2}$,*

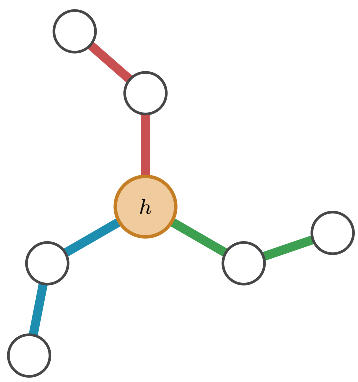
$$k_m^{(r)}(n) \;\ge\; \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor,$$

attained by a simple hypergraph.

Proof. The simple construction of Theorem A.48 has Berge edge-connectivity at most $m-1$, and Whitney's inequality $\kappa \le \lambda$ holds pairwise (an internally vertex-disjoint family is in particular hyperedge-disjoint under our definition), so the same hypergraph is feasible for the vertex problem. □



(a) A 3-uniform hypergraph $\mathcal{H}$ (left) and its bipartite incidence graph $I(\mathcal{H})$ (right): circles for vertices, hollow squares for hyperedges, an edge when the vertex lies in the hyperedge. The Berge path $1\, e_1\, 3\, e_2\, 5$ is traced in orange in both pictures. In $\mathcal{H}$ it rides the red line e_1 and the blue line e_2, changing at the shared vertex 3, while the grey line e_3 lies off it. In $I(\mathcal{H})$ it is the ordinary path $1 - e_1 - 3 - e_2 - 5$ alternating sides.



(b) A star hypertree: every hyperedge $\{h\} \cup G_i$ shares only the hub h (orange), and the other $r - 1 = 2$ vertices of each block sit in a single hyperedge. Its incidence graph is a tree, so $\kappa^{\max} \leq 1$.

Figure A.4. The translation the vertex problem rests on. Internally vertex-disjoint Berge paths in $\mathcal{H}$ correspond exactly to internally disjoint paths in $I(\mathcal{H})$, so $\kappa_{\mathcal{H}}(u, v) = \kappa_{I(\mathcal{H})}(u, v)$. **(top)** A hypergraph and its incidence graph with one Berge path drawn in both. **(bottom)** The star hypertree of Theorem 1.16, whose incidence graph is a tree.

At $m = 3$ the upper-bound side closes completely, and equality follows in the construction range stated below. The engine is a rank bound for graphs in which one side of a partition must stay below local 3-connectivity. The hypergraph theorem then falls out of the incidence translation. Throughout, $\operatorname{rank}(G) = |E(G)| - |V(G)| + 1$ denotes the cycle rank of a connected multigraph, that is, the number of independent cycles it carries. A tree has cycle rank 0, and each edge added beyond a spanning tree closes exactly one new cycle and raises the rank by one, so bounding the rank is the same as bounding how many independent cycles the graph can hold.

Primer: global connectivity and the pieces of a graph

The lemma below speaks of a graph being 2-connected or 3-connected and of taking it apart at small separators. These are the global cousins of the local $\kappa(u, v)$ from Chapter 1, so we fix the words once. A graph on more than k vertices is *k-connected* if it stays connected after the removal of any $k - 1$ vertices. By Menger this is the same as asking that every pair of vertices be joined by at least k internally vertex-disjoint paths, so k-connected means $\kappa(u, v) \geq k$ for all pairs at once, the uniform version of the local measure. Thus 2-connected means no single vertex disconnects the graph, and 3-connected means no two vertices do. Three named obstructions go with this. A *cut vertex* is a vertex whose removal disconnects the graph, so a connected graph on at least three vertices is 2-connected exactly when it has no cut vertex. A *separating pair* is a set of two vertices whose removal disconnects the graph, the obstruction a 2-connected graph must lack in order to be 3-connected. Finally a *block* is a maximal piece with no cut vertex of its own, hence either a single bridge edge or a maximal 2-connected subgraph. Two blocks overlap in at most one vertex, always a cut vertex, and the blocks and cut vertices hang together in a tree, the *block-cut tree*, with one node per block and one per cut vertex. Step 1 of the proof is precisely an induction over that tree.

Lemma A.53 (Incidence rank lemma)**.** *Let G be a connected multigraph with vertex set $X \cup Z$ such that (i) Z is independent, (ii) no edge incident to Z is parallel to another edge, (iii) every $z \in Z$ has degree at least 2, and (iv) $\kappa_G(x, x') \leq 2$ for all distinct $x, x' \in X$. Then $\operatorname{rank}(G) \leq |X| - 1$.*

In words: this is the engine behind the $m = 3$ hypergraph result, stated for a bipartite graph rather than for a hypergraph. Read X as the vertices of a hypergraph and Z as its hyperedges, so that G is the incidence graph and $\operatorname{rank}(G)$, the number of independent cycles it carries, is the quantity the hyperedge count is eventually read off from. The four hypotheses say in turn that the picture really is an incidence graph (i), that no hyperedge repeats a vertex (ii), that no hyperedge dangles as a leaf (iii), and that the hypergraph is feasible at $m = 3$ (iv). The conclusion caps the independent cycles by the number of vertices.

Proof. Induction on $|V| + |E|$. The proof peels G down to a core that cannot exist. Step 1 handles a cut vertex by arguing block by block. Steps 2 and 3 remove a degree-two Z-vertex or X-vertex without changing the bound. What survives is 2-connected with minimum degree at least three, and Step 4 shows that such a graph already violates the connectivity ceiling (iv), so it never arises.

Base $|V| \leq 2$. A single vertex has degree 0, too little for the degree-at-least-2 floor (iii) imposes on a Z-vertex, so it cannot lie in Z and must lie in X, giving $0 \leq |X| - 1$. On two vertices the same reasoning applies to either one: a vertex in Z could reach only the other vertex, and (ii) forbids that single edge from being parallel to another, so it would have degree at most 1, again short of (iii). Hence (i)–(iii) force both vertices into X. Parallel edges between them are internally disjoint paths (each has empty interior, so no two can share an interior vertex), so (iv) caps the multiplicity at 2 and $\operatorname{rank} \leq 1 = |X| - 1$.

Step 1: a cut vertex. Suppose G has a cut vertex, so it is not yet 2-connected. Break it into its *blocks* $B_1, \ldots, B_c$ with $c \geq 2$, a block being a maximal piece with no cut vertex of its own, that is, either a single bridge edge or a maximal 2-connected subgraph. Two blocks meet in at most one vertex, and any shared vertex is a cut vertex of G. No cycle can pass through two different blocks: since two blocks share at most one vertex, a cycle straddling both would have to visit that shared vertex twice, which a simple cycle cannot do. So the cycle rank adds up over blocks, $\operatorname{rank}(G) = \sum_i \operatorname{rank}(B_i)$, and it is enough to bound each $\operatorname{rank}(B_i)$ by $|X \cap B_i| - 1$ and sum.

Each block inherits the hypotheses. A bridge block is one edge, so its rank is 0, and by (i) at least one endpoint lies in X, giving $0 \leq |X \cap B| - 1$. Every other block is 2-connected, hence has minimum degree at least 2, so (i) to (iv) hold inside it and the induction hypothesis gives $\operatorname{rank}(B_i) \leq |X \cap B_i| - 1$.

Adding these requires care, because a cut vertex is shared by several blocks and would be counted several times. Write $b(v)$ for the number of blocks containing v, so $b(v) = 1$ except at cut vertices. Counting each X-vertex once for every block it lies in,

$$\sum_i |X \cap B_i| = |X| + \sum_{x \in X \text{ cut}} (b(x) - 1).$$

The number of blocks satisfies the block–cut tree identity $c = 1 + \sum_{v \text{ cut}} (b(v) - 1)$. This follows from counting the edges of the block-cut tree two ways. That tree has c block-nodes and one node per cut vertex, so being a tree its edge count is one less than its total number of nodes, $c + \#\{\text{cut vertices}\} - 1$. The same edge count also equals $\sum_{v \text{ cut}} b(v)$, since each cut vertex v is joined in the tree to exactly the $b(v)$ blocks that contain it. Equating the two expressions for the edge count and solving for c gives the identity. Subtracting this count of blocks from the previous display and separating the cut vertices into their X- and Z-parts leaves

$$\operatorname{rank}(G) = \sum_i \operatorname{rank}(B_i) \leq \sum_i (|X \cap B_i| - 1) = |X| - 1 - \sum_{z \in Z \text{ cut}} (b(z) - 1) \leq |X| - 1,$$

where the last inequality holds because each $b(z) - 1 \geq 0$, so the Z-cut sum only pushes the bound further below $|X| - 1$.

Step 2: a Z-vertex of degree two. From now on G is 2-connected, every cut vertex having been handled. Suppose some $z_0 \in Z$ has degree exactly 2. Both its neighbours lie in X by (i), and they are distinct, since two edges to the same vertex would be parallel at a Z-vertex, which

(ii) forbids. *Suppress* z_0 by deleting it and joining its two neighbours x, x' with one new edge. This loses two edges and one vertex and gains one edge, so $|E|$ and $|V|$ each drop by one and $\mathrm{rank} = |E| - |V| + 1$ is unchanged. The new edge runs X to X, so $|X|$ is unchanged and (i), (ii), (iii) still hold for the remaining Z-vertices. Replacing the path x, z_0, x' by a direct edge alters no route among the surviving vertices, so every $\kappa(x_1, x_2)$ is preserved and (iv) holds. The graph is strictly smaller, so the induction hypothesis applies and returns the bound for G.

Step 3: an X-vertex of degree two. Now G is 2-connected and, Step 2 being used up, every Z-vertex has degree at least 3. Suppose some $x_0 \in X$ has degree 2, and delete it. Deleting one vertex from a 2-connected graph leaves it connected. Stripping a degree-2 vertex removes two edges and one vertex, so rank drops by exactly one. Each neighbour of x_0 loses one edge, and a neighbour in Z only falls from degree at least 3 to degree at least 2, so (iii) survives, while deleting a vertex cannot raise a connectivity, so (iv) survives. Here $|X|$ drops by one, and $|X| \geq 2$ to begin with, for if $|X| \leq 1$ then by (i) and (ii) every z would send all its edges to one X-vertex without repeats, hence have degree at most 1, against (iii). The smaller graph satisfies the hypotheses, so induction gives $\mathrm{rank}(G) - 1 \leq (|X| - 1) - 1$, that is $\mathrm{rank}(G) \leq |X| - 1$.

Step 4: the remaining case is impossible. Now G is 2-connected with $|V| \geq 3$ and every degree at least 3. First, G is simple: a parallel class lies inside X by (ii), and if $\mu(x, x') \geq 2$ then a third internally disjoint route exists: walk from x to any third vertex w inside $G - x'$, then from w to x' inside $G - x$, both walks available because deleting a single vertex from a 2-connected graph leaves it connected, exactly as already used in Step 3. The concatenation visits x only first and x' only last, so it contains an x–x' path with at least one interior vertex, giving $\kappa(x, x') \geq 3$, against (iv). If G is 3-connected, Menger puts every pair at $\kappa \geq 3$, so two vertices of X would violate the ceiling $\kappa(x, x') \leq 2$ that (iv) imposes on every pair drawn from X, forcing $|X| \leq 1$, and then every z has degree at most 1 by (i), absurd.

Otherwise G is 2-connected but not 3-connected. Tutte's triconnected decomposition [21] expresses it as a tree of cycle, bond, and 3-connected pieces joined along virtual edges. For a leaf piece L its unique virtual edge $\{a, b\}$ represents a real a–b path through the rest of G, internally outside L; vertices of L away from that edge keep their degree in G. These are the only properties of the decomposition used below.

Now G itself is none of the three piece types, since it is not a single cycle (its degrees are at least three), not a bond (it is simple), and not 3-connected (the case just handled), so the tree has at least two leaves. Let L be one, with unique virtual edge $\{a, b\}$. If L is a cycle, a vertex of L off $\{a, b\}$ has degree 2 in G, absurd. If L is a bond, it holds at least two real parallel edges, absurd in a simple graph. If L is 3-connected, then any two of its vertices are joined by three internally disjoint paths inside L, at most one through the virtual edge, which the far-side path replaces: so $\kappa_G(u, v) \geq 3$ for *all* $u, v \in V(L)$, forcing $|X \cap V(L)| \leq 1$ by (iv). But $|V(L)| \geq 4$ leaves a Z-vertex $z^* \notin \{a, b\}$. Its neighbours lie in $X \cap V(L)$ by (i), at most one vertex, reached by at most one edge since G is simple, giving degree at most 1, which is absurd. □

The rank bound now yields Theorem 1.17, stated in Chapter 1. In words: at the route limit

$m = 3$ a hyperedge network can hold no more hyperedges under the stricter separation than the displayed edge bound. When $r \geq 3$ and $2 \leq \binom{n-2}{r-2}$, that count is reached without repeating a hyperedge, so the two separations agree in this range exactly as they did at $m = 2$.

Proof of Theorem 1.17. Apply Lemma A.53 to each connected component of the incidence graph $I(\mathcal{H})$, with X the vertex class and Z the hyperedge class: Z is independent, incidences are simple even when hyperedges repeat, hyperedge-nodes have degree $r \geq 2$, components without an X-node are impossible, and $\kappa_I(u, v) = \kappa_{\mathcal{H}}(u, v) \leq 2$ for vertex pairs. Sum $\operatorname{rank} \leq |X_c| - 1$ over the C components. The incidence graph $I(\mathcal{H})$ has rq edges (each of the q hyperedges contributes r incidences) and $n+q$ nodes, so $\sum_c \operatorname{rank} = rq - (n+q) + C$, while $\sum_c (|X_c| - 1) = n - C$. The inequality reads $rq - (n+q) + C \leq n - C$, hence $q(r-1) \leq 2(n-1) + 2(1-C) \leq 2(n-1)$, the last step using $C \geq 1$. The lower bound is Proposition A.52 at $m = 3$. □

Remark A.54 (Edges as gates at $r = 2$, and the boundary of the method). At $r = 2$ the theorem speaks of multigraphs under the hyperedge-as-gate convention, in which parallel edges are distinct one-step routes, giving the multigraph value $2(n-1)$ of Theorem 1.3, complementing the body's convention for graphs, where parallel adjacencies count once. The proof of Lemma A.53 leans on the triconnected decomposition exactly where $\kappa \leq 2$ lives. For $m = 4$, the analogous question asks for a rank bound relative to $\kappa \leq 3$, where no comparably clean decomposition exists. Thus the proof is deliberately confined to $m \leq 3$; no general $m = 4$ formula is claimed.

A.8 The directed hypergraph

The remaining two of the twelve variants orient the Berge edge. They are grouped here rather than with the undirected hypergraph bounds above because the questions they raise are different ones: not which decomposition controls the rank of an incidence graph, but how much a single edge with one tail and many heads can carry, and whether the way it is oriented changes the answer. The first result is a pair of bounds a factor of four apart, the last closes that factor for the model the program measures.

Proposition A.55 (Directed hypergraphs: first bounds)**.** *Model a directed r-uniform hyperedge as a tail vertex together with $r-1$ head vertices, a Berge route entering at the tail and leaving at a head. If every pair of vertices has at most $m-1$ arc-disjoint directed Berge routes, then*

$$|E(\mathcal{H})| \;\leq\; \frac{(m-1)\,n(n-1)}{r-1},$$

and, writing $A(n, r, m)$ for the set of tail-class sizes α with $1 \leq \alpha \leq n-1$ and $m - 1 \leq \binom{n-\alpha-1}{r-2}$, there are feasible constructions with $\max_{\alpha \in A(n,r,m)} \alpha \left\lfloor \frac{(m-1)(n-\alpha)}{r-1} \right\rfloor$ hyperedges. The balanced choice $\alpha = \lfloor n/2 \rfloor$ lies in $A(n, r, m)$ as soon as $m - 1 \leq \binom{\lceil n/2 \rceil - 1}{r-2}$ and already gives $\approx (m-1)n^2/(4(r-1))$, so the value is quadratic in n.

Proof. Upper bound: a hyperedge with tail u serves the $r-1$ ordered pairs (u, h), h a head. If some ordered pair (u, v) were served by m distinct hyperedges, those would be m arc-disjoint one-step routes, so each ordered pair is served at most $m-1$ times. Summing that cap over all $n(n-1)$ ordered pairs counts every hyperedge once for each of its $r-1$ tail-head pairs, so the sum is $(r-1)|E|$ on one side and at most $(m-1)n(n-1)$ on the other, giving $(r-1)|E| \leq (m-1)n(n-1)$. Lower bound: split V into tails A, $|A| = \alpha$, and heads B. By Lemma A.51 there is a simple $(r-1)$-uniform hypergraph $\mathcal{G}^*$ on B with maximum degree $m-1$ and $\lfloor (m-1)|B|/(r-1) \rfloor$ edges. Give every tail $a \in A$ the hyperedges (a, H), $H \in \mathcal{G}^*$. No vertex of B is a tail, so every route is a single step and $\lambda(a, b) = \deg_{\mathcal{G}^*}(b) \leq m-1$, while pairs inside A or starting in B have no routes. □

The forward model is one of three ways to orient a Berge edge. In general a directed r-uniform hyperedge is a split of its r vertices into a non-empty *tail set* T and a non-empty *head set* H, a route entering at a tail and leaving at a head. The *forward* model is $|T| = 1$, the *backward* model is $|H| = 1$, and the *general* model allows any split (genuinely mixed splits, both parts at least two, exist from $r \geq 4$). The next two results place all three. The first is that backward needs no separate study, and the second is that the general model, despite admitting more hyperedges on a single vertex set, obeys the very same first-order bound as forward.

Lemma A.56 (Backward is the reverse of forward)**.** *Reversing every hyperarc, $(T, H) \mapsto (H, T)$, is an edge-count-preserving bijection between the forward directed r-uniform hypergraphs and the backward ones on the same vertices, and the reverse $\mathcal{H}^R$ satisfies $\lambda_{\mathcal{H}}(u, v) = \lambda_{\mathcal{H}^R}(v, u)$ and $\kappa_{\mathcal{H}}(u, v) = \kappa_{\mathcal{H}^R}(v, u)$. Hence $\lambda^{\max}$ and $\kappa^{\max}$ are invariant under reversal, and the backward edge and vertex extremal numbers equal the forward ones, for every r, m and n.*

In words: the backward model is the forward model seen in a mirror. Turning every hyperarc around swaps the tails with the heads and reverses every route, which leaves the number of independent routes between a pair untouched. The two models have identical extremal numbers, so nothing below needs to treat backward as a separate case.

Proof. Reversal sends a backward hyperarc ($|T| = r-1$, $|H| = 1$) to a forward one ($|T| = 1$) and is its own inverse, so it is an edge-count-preserving bijection of the two families. Let $u = x_0, e_1, x_1, \ldots, e_k, x_k = v$ be a directed Berge route in $\mathcal{H}$, where x_{i-1} is a tail and x_i a head of e_i. The reversed edge $e_i^R = (H_i, T_i)$ has x_i as a tail and x_{i-1} as a head, so $v = x_k, e_k^R, x_{k-1}, \ldots, e_1^R, x_0 = u$ is a directed Berge route from v to u in $\mathcal{H}^R$ on the same edges and the same internal vertices. This is a bijection between u–v routes in $\mathcal{H}$ and v–u routes in $\mathcal{H}^R$ that preserves which edges and which internal vertices any two routes share, so it carries an edge-disjoint (respectively internally vertex-disjoint) family to one of equal size. Therefore $\lambda_{\mathcal{H}}(u, v) = \lambda_{\mathcal{H}^R}(v, u)$ and $\kappa_{\mathcal{H}}(u, v) = \kappa_{\mathcal{H}^R}(v, u)$, and maximising over ordered pairs gives $\lambda^{\max}(\mathcal{H}) = \lambda^{\max}(\mathcal{H}^R)$ and the same for κ. Since reversal is a bijection preserving feasibility and edge count, the two models share every extremal number. □

Proposition A.57 (The general model obeys the forward bound)**.** *Let a directed r-uniform hyperedge be any split into a non-empty tail set T and head set H. If every ordered pair has at most $m-1$ arc-disjoint, or at most $m-1$ internally vertex-disjoint, directed Berge routes, then*

$$|E(\mathcal{H})| \;\le\; \frac{(m-1)\,n(n-1)}{r-1},$$

the same bound as the forward model of Proposition A.55. The forward construction of that proposition is itself a set of general hyperedges, so the general value obeys the same upper bound and inherits the same quadratic lower bound, and in particular is $\Theta(n^2)$.

Proof. Fix an ordered pair (u,v). A hyperedge (T,H) with $u \in T$ and $v \in H$ carries the one-step route $u \to v$ that enters at the tail u and leaves at the head v. Distinct such hyperedges give routes that share no edge and no internal vertex, so if k hyperedges had u in their tail set and v in their head set then $\lambda(u,v) \ge k$ and $\kappa(u,v) \ge k$, so feasibility forces $k \le m-1$. Each hyperedge (T,H) is counted by exactly the $|T|\,|H|$ ordered pairs (u,v) with $u \in T$ and $v \in H$, so summing the per-pair bound over all pairs gives $\sum_{(T,H)} |T|\,|H| \le (m-1)\,n(n-1)$. As $|T|+|H| = r$ with both parts at least one, $|T|\,|H| \ge 1 \cdot (r-1) = r-1$, whence $(r-1)|E| \le (m-1)n(n-1)$. The forward family of Proposition A.55 consists of general hyperedges, so it witnesses the same $\approx (m-1)n^2/(4(r-1))$ lower bound. □

The two models are therefore trapped between the same pair of bounds, which differ by a factor of 4. That is enough to fix the *order* of both values and not enough to fix either leading constant, so it does not follow that the forward and general models agree asymptotically, only that any disagreement between them lives inside that factor. The rest of this section closes that gap for the forward model, which leaves the comparison between the two models as the question that survives.

It is worth saying which end of that gap is the likely culprit, because the two bounds are not equally crude and the analogy with the simple digraph case is informative. The upper bound charges each hyperedge to the ordered tail-head pairs it occupies and then allows all $n(n-1)$ ordered pairs to be used to capacity. The construction uses only the $\alpha(n-\alpha) \le n^2/4$ pairs that run from the tail class to the head class, and leaves the pairs inside each class, and those running backwards, entirely idle. The factor of four is exactly that difference, n^2 against $n^2/4$, and it is the same difference that appears in the simple directed problem, where Construction 1.13 also uses only a one-directional bipartite pattern. There, Theorem A.25 now proves that the bipartite pattern is right to first order, that no arrangement recovers the idle pairs. If that phenomenon is a feature of directedness rather than of graphs specifically, then it is the upper bound here that should come down by a factor of four, not the construction that should climb, and that is what happens.

The obstacle to running the argument of Theorem A.25 here is real and worth naming before it is disposed of, because the way around it is not to remove it. There, two-step routes through distinct midpoints were automatically arc-disjoint. Here a single hyperedge carries $r-1$ heads at once, so a pair of two-step routes with different midpoints may enter through the very same hyperedge, and the family of two-step routes is not edge-disjoint at all. The resolution is to accept the loss. A packing can always be extracted from that family at a cost of a factor $r-1$, and that factor turns out to be free, because of where it lands. The route count does not bound the hyperedges directly. It bounds the arcs of an auxiliary digraph, and there the factor $r-1$ enters only the *linear* error term of Lemma A.28, never the $n^2/4$ leading term. Meanwhile a second, independent factor of $r-1$ is gained when the auxiliary digraph is exchanged back for hyperedges, and that one does divide the leading term. The two factors are not the same factor, and only one of them is paid where it matters.

Definition A.58 (One-step shadow)**.** Let $\mathcal{H}$ be a forward directed r-uniform hypergraph on V. Its *one-step shadow* is the simple digraph $R(\mathcal{H})$ on V with an arc (u,v) exactly when some hyperedge of $\mathcal{H}$ has tail u and v among its heads. Write $\operatorname{cod}(u,v)$ for the number of hyperedges realising that arc.

Theorem A.59 (The directed hypergraph constant)**.** *Let $r \geq 2$ and $m \geq 2$. Every forward directed r-uniform multihypergraph $\mathcal{H}$ on $n \geq 2$ vertices with $\kappa_{\mathcal{H}}^{\max} \leq m-1$ satisfies*

$$|E(\mathcal{H})| \;\leq\; \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4(m-1)^2(n-1).$$

The same bound therefore holds under $\lambda_{\mathcal{H}}^{\max} \leq m-1$, which is the stronger hypothesis by $\kappa \leq \lambda$. Hence for fixed r and m both the edge- and the vertex-disjoint maxima are $\big(1+o(1)\big)(m-1)n^2/\big(4(r-1)\big)$, and the bipartite construction of Proposition A.55 is asymptotically optimal for both.

In words: the bipartite construction of Proposition A.55 is asymptotically the best there is. Its count already matches the leading term $\frac{m-1}{r-1}\left\lfloor n^2/4 \right\rfloor$, so the factor of four that used to separate the known bounds is gone and the remaining uncertainty sits entirely in a term linear in n. The bound covers both separations at once.

Proof. Write $R = R(\mathcal{H})$ for the one-step shadow and note it has no loops, since a hyperedge's tail is not one of its heads.

Step 1: hyperedges against shadow arcs. Suppose $\operatorname{cod}(u,v) = k$. Those k hyperedges each give a one-step Berge route from u to v, and the routes use one hyperedge apiece and different ones, so they are pairwise hyperedge-disjoint, while having no interior vertex at all they are internally vertex-disjoint as well. Hence $k \leq \kappa_{\mathcal{H}}(u,v) \leq m-1$. Every hyperedge is counted by exactly its $r-1$ tail-head pairs, so summing over the arcs of R,

$$(r-1)\,|E(\mathcal{H})| \;=\; \sum_{(u,v)\in A(R)} \operatorname{cod}(u,v) \;\leq\; (m-1)\,|A(R)|.$$

This is the count of Proposition A.55 stopped one step earlier. That proof then bounded $|A(R)|$ by the trivial $n(n-1)$, which is exactly where its factor of four was lost, and the rest of this argument replaces that step.

Step 2: the shadow is budgeted. Fix an ordered pair (u,v) of distinct vertices, and let $T \subseteq V$ collect v itself when $(u,v) \in A(R)$ together with every midpoint counted by $p_2(u,v)$, so $|T| = \mathbf{1}_{(u,v)\in A(R)} + p_2(u,v)$. Every $t \in T$ is a head of at least one hyperedge whose tail is u. Form the bipartite graph joining each $t \in T$ to each such hyperedge containing it, and let M be a maximum matching in it.

Then $|M| \geq |T|/(r-1)$. To see it, take any $t \in T$ left unmatched by M. Every hyperedge containing t must already be matched, since otherwise M could be extended along that pair. So every element of T, matched or not, lies in some hyperedge used by M, and each hyperedge holds at most $r-1$ elements of T, giving $|T| \leq |M|\,(r-1)$.

Now read M as a family of routes. For a matched pair (v,e) take the one-step route $u \to v$ through e, and for a matched pair (x,e) with x a midpoint take $u \to x \to v$, entering through e and leaving through any hyperedge whose tail is x and which has v as a head, one of which exists because $(x,v) \in A(R)$. These routes are pairwise hyperedge-disjoint: the entering hyperedges are distinct because M is a matching, the leaving hyperedges have pairwise distinct tails and so are distinct from each other, and no leaving hyperedge is an entering one, since the entering ones have tail u and the leaving ones have tail a midpoint, which is not u. They are also internally vertex-disjoint, since the one-step route has empty interior and each two-step route has for interior the single vertex it is matched to, and M matches distinct targets. Hence

$$\frac{\mathbf{1}_{(u,v)\in A(R)} + p_2(u,v)}{r-1} \;\leq\; |M| \;\leq\; \kappa_{\mathcal{H}}(u,v) \;\leq\; m-1,$$

so R is $(r-1)(m-1)$-budgeted in the sense of Definition A.27.

Step 3: assembling. Lemma A.28 with $C = (r-1)(m-1)$ gives $|A(R)| \leq \lfloor n^2/4 \rfloor + 4(r-1)(m-1)(n-1)$, and substituting that into Step 1,

$$|E(\mathcal{H})| \;\leq\; \frac{m-1}{r-1}\left[\left\lfloor \frac{n^2}{4} \right\rfloor + 4(r-1)(m-1)(n-1)\right] \;=\; \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)^2(n-1).$$

The lower bound is Proposition A.55, whose bipartite family has

$$\max_{\alpha}\; \alpha \left\lfloor \frac{(m-1)(n-\alpha)}{r-1} \right\rfloor \;=\; \frac{(m-1)n^2}{4(r-1)} - O_{m,r}(n)$$

hyperedges, where the balanced split $\alpha = \lfloor n/2 \rfloor$ already attains the right-hand side on its own, and which is feasible for both separations at once, since it is λ-feasible and $\kappa \leq \lambda$. (The integer maximum is occasionally taken a little off balance, since the floor can favour an unbalanced split by a unit or two, but never by enough to move the n^2 term.) Upper and lower bound agree in their n^2 term, which is the asymptotic claim. □

Two remarks on the statement. First, the two upper bounds are not comparable and both are worth keeping: the bound of Proposition A.55 is the smaller one until n is around $16(m-1)(r-1)/3$, since the linear term here carries a constant of $4(m-1)^2$, and the theorem is an asymptotic improvement rather than a uniform one. Second, at $r = 2$ the shadow is the digraph itself and cod is the multiplicity, so the theorem recovers the quadratic branch of Theorem A.31 up to its linear error term, which is the consistency check one wants, though it does not reprove that theorem, which is exact.

A third remark used to say that the general orientation model was out of reach, and it is worth recording why, because the obstacle was real and the way past it is short. The proof above uses the forward model twice, once for the entering hyperedges and once for the leaving ones, in both cases to know that a hyperedge has a single tail. That is what makes Step 2's matching legitimate: the leaving hyperedges have pairwise distinct tails and so are automatically distinct from each other, and none of them can be an entering hyperedge, since the entering ones have tail u and the leaving ones do not. Allow several tails and both statements fail at once. Two midpoints may leave through one shared hyperedge, and from $r \geq 4$ a single hyperedge with two tails and two heads may serve as one target's entrance and another's exit. So the conflicts run both ways and no matching on one side controls them.

The repair is to stop building a large family and instead take a maximum one and ask what stopped it. A maximum family cannot be extended, so every target it leaves unserved must be obstructed by a hyperedge the family has already spent, and a hyperedge has only r vertices in all to obstruct with. That counts the leftovers without ever needing the two sides to be independent, which is exactly what the general model denies.

Theorem A.60 (The general orientation model shares the constant)**.** *Let $r \geq 2$ and $m \geq 2$, and let a directed r-uniform hyperedge be any split (T, H) into a non-empty tail set and a non-empty head set, as in Proposition A.57. Every such multihypergraph $\mathcal{H}$ on $n \geq 2$ vertices with $\kappa_{\mathcal{H}}^{\max} \leq m-1$ satisfies*

$$|E(\mathcal{H})| \;\leq\; \frac{m-1}{r-1}\left\lfloor\frac{n^2}{4}\right\rfloor \;+\; \frac{4(2r+1)(m-1)^2}{r-1}\,(n-1).$$

The same bound therefore holds under $\lambda_{\mathcal{H}}^{\max} \leq m-1$. Hence for fixed r and m the general model has the same leading term $\big(1+o(1)\big)(m-1)n^2/\big(4(r-1)\big)$ as the forward model, under both separations, and the bipartite construction of Proposition A.55 is asymptotically optimal for it too.

In words: how a hyperedge is oriented does not move the leading term. Whether one vertex points at the other $r-1$ of them, or several do, the maximum carries the same $\frac{m-1}{r-1}\lfloor n^2/4\rfloor$ in front, and only the linear term notices the difference, through the factor $2r+1$. With Lemma A.56 this closes the orientation axis to first order.

Proof. Let R be the one-step shadow read in the general model, so $(u,v) \in A(R)$ exactly when some hyperedge has u among its tails and v among its heads, and $\operatorname{cod}(u,v)$ counts the hyperedges realising it. A hyperedge's tail set and head set are disjoint, so R has no loops.

Step 1: hyperedges against shadow arcs. The $\operatorname{cod}(u,v)$ hyperedges realising an arc give that many one-step routes, one hyperedge apiece and different ones, with no interior vertex, so they are pairwise hyperedge-disjoint and internally vertex-disjoint and $\operatorname{cod}(u,v) \le \kappa_{\mathcal{H}}(u,v) \le m-1$. A hyperedge (T,H) is counted by exactly the $|T|\,|H|$ ordered pairs it fills, and $|T|+|H|=r$ with both parts non-empty gives $|T|\,|H| \ge r-1$. Hence

$$(r-1)\,|E(\mathcal{H})| \;\le\; \sum_{(T,H)\in E(\mathcal{H})} |T|\,|H| \;=\; \sum_{(u,v)\in A(R)} \operatorname{cod}(u,v) \;\le\; (m-1)\,|A(R)|.$$

This is the count of Proposition A.57 stopped one step early, exactly as Step 1 above stopped Proposition A.55 one step early.

Step 2: the shadow is budgeted. Fix an ordered pair (u,v) of distinct vertices and let $\mathcal{T}$ collect v itself when $(u,v)\in A(R)$ together with every midpoint counted by $p_2(u,v)$, so $|\mathcal{T}| = \mathbf{1}_{(u,v)\in A(R)} + p_2(u,v)$. Call a hyperedge e an *entrance* for $t\in\mathcal{T}$ if u is a tail of e and t a head, and an *exit* for a midpoint t if t is a tail of e and v a head. Every $t\in\mathcal{T}$ has at least one entrance, and every midpoint at least one exit, by the definition of R. No hyperedge is both an entrance and an exit for the *same* t, since that would put t in its tail set and its head set at once.

Consider families of routes indexed by distinct targets: for the target v the one-step route through an entrance, and for a midpoint t the two-step route $u\to t\to v$ through an entrance and an exit. Such a family is admissible when its hyperedges are pairwise distinct. Let $\mathcal{F}$ be one of maximum size M, serving the target set S, and let U be the hyperedges it uses, so $|U|\le 2M$. The routes of $\mathcal{F}$ are pairwise hyperedge-disjoint by construction and internally vertex-disjoint because their interiors are the empty set and distinct singletons, so

$$M \;\le\; \kappa_{\mathcal{H}}(u,v) \;\le\; m-1.$$

Now take $t\in\mathcal{T}\setminus S$. If some entrance $e\notin U$ existed and, for a midpoint, some exit $f\notin U$ as well, then $e\ne f$ by the paragraph above and the route through them could be added to $\mathcal{F}$, serving a target not yet served and using no hyperedge already spent. That contradicts maximality. So every entrance of t lies in U, or t is a midpoint all of whose exits lie in U. Either way t lies in the tail set or the head set of some member of U, and therefore

$$\mathcal{T}\setminus S \;\subseteq\; \bigcup_{e\in U}(T_e\cup H_e), \qquad \text{so} \qquad |\mathcal{T}| \;\le\; M + \sum_{e\in U}\big(|T_e|+|H_e|\big) \;=\; M + r\,|U|.$$

The last equality is where the two sides are counted together rather than separately, and it is what keeps the constant at r rather than $2(r-1)$. With $|U|\le 2M$ this gives $|\mathcal{T}|\le(2r+1)M\le(2r+1)(m-1)$, so R is $(2r+1)(m-1)$-budgeted in the sense of Definition A.27.

Step 3: assembling. Lemma A.28 with $C=(2r+1)(m-1)$ gives $|A(R)|\le\lfloor n^2/4\rfloor + 4(2r+1)(m-1)(n-1)$, and substituting into Step 1 gives the stated bound. The hypothesis

$\lambda^{\max} \leq m-1$ implies $\kappa^{\max} \leq m-1$ by Whitney, so the bound holds there too. For the lower bound, the bipartite family of Proposition A.55 consists of general hyperedges, so it witnesses $\approx (m-1)n^2/(4(r-1))$ here as well, and upper and lower bound agree in their n^2 term. □

Two things about that proof are worth separating. The constant $2r+1$ is not optimised and is not claimed to be sharp: it is what the crude estimate $|U| \leq 2M$ gives, and a search over deliberately adversarial instances, in which every target shares both its entrance and its exit with another, never drives $|\mathcal{T}|/\kappa(u,v)$ above $r-1$, the forward model's own constant. Nothing rests on the difference, since the whole point of Lemma A.28 is that this constant lands in the linear term and never touches the $n^2/4$ that carries the leading constant. The second point is what the theorem does to the orientation axis of Section 3.3.1. Forward and backward were already known equal, exactly, by Lemma A.56. The general model was known only to sit between the same two bounds a factor of four apart, which left room for it to be genuinely denser to first order. It is not. All three orientation models have the same leading term, and the computed differences between them at $n \approx r$ are a small-size effect that the asymptotics cannot see.

A.9 The multigraph vertex problem under the other convention

Remark 2.1 collapses parallel copies in vertex mode, which makes the two multigraph vertex variants restatements of their simple counterparts and is why they carry no theorem of their own. This section takes the other reading seriously, because under it the question is genuinely new, and records what can be proved and what the machine finds. Throughout, q parallel copies of the edge uv count as q internally vertex-disjoint u-v routes, since a single edge is a path with no interior. Write $K_m^{\mathrm{multi}}(n)$ for the largest total multiplicity of a multigraph on n vertices with $\kappa^{\max} \leq m-1$ under this reading.

The first thing to notice is that the two kinds of route never interfere, which splits the measure cleanly in two.

Lemma A.61 (Splitting the vertex measure)**.** *Let G be a multigraph with multiplicity function μ and underlying simple graph G_0. For any two vertices $u \neq v$, write $\pi(u,v)$ for the largest number of pairwise internally disjoint u-v paths of length at least two in G_0. Then, under the convention above,*

$$\kappa_G(u,v) \;=\; \mu(u,v) \;+\; \pi(u,v).$$

Proof. A parallel copy of uv is a path with empty interior, so any two copies are internally disjoint from each other and from every longer route. A maximum family may therefore be assumed to contain all $\mu(u,v)$ copies, and what it can add to them is a family of pairwise internally disjoint routes of length at least two, whose interiors are disjoint sets of vertices. Since parallel copies are irrelevant to a route of length at least two, which is determined by its vertex sequence, the largest such family has size $\pi(u,v)$, computed in G_0. □

So feasibility says exactly that $\mu(u,v) \le m-1-\pi(u,v)$ on every edge, and that $\pi(u,v) \le m-1$ on every pair. At $m=2$ this is already decisive.

Theorem A.62 (The other convention at $m=2$)**.** *$K_2^{\mathrm{multi}}(n) = n-1$, attained exactly by the spanning trees.*

Proof. By Lemma A.61, $\kappa^{\max} \le 1$ forces $\mu(u,v)+\pi(u,v) \le 1$ on every edge. If G_0 contained a cycle, an edge uv of that cycle would have $\mu(u,v) \ge 1$ and $\pi(u,v) \ge 1$ from the rest of the cycle, a contradiction, so G_0 is a forest and every multiplicity is one. A forest has at most $n-1$ edges, and a spanning tree attains it. □

For $m \ge 3$ at least three constructions compete, and which one wins depends on how n compares with m, exactly as in every other variant of this thesis.

- *The thickened tree.* Take a spanning tree and give every edge multiplicity $m-1$. Tree edges have $\pi = 0$ and non-adjacent pairs have $\pi = 1$, so this is feasible for every $m \ge 2$, and it carries $(m-1)(n-1)$ edges, the multigraph *edge* value of Theorem 1.3.
- *The thickened complete graph.* Give every edge of K_n the same multiplicity q. Here $\pi(u,v) = n-2$, so feasibility asks $q \le m+1-n$, and the count is $\binom{n}{2}(m+1-n)$, positive only when $n \le m+1$.
- *The thickened theta.* Choose two *poles* and join each of the $n-2$ remaining vertices to both of them at multiplicity $m-2$, then join the poles to each other at multiplicity $m+1-n$ when that is positive. A pole and a middle vertex have $\pi = 1$, two middle vertices have $\pi = 2$, and the two poles have $\pi = n-2$, so this is feasible exactly when $n \le m+1$, and it carries $2(n-2)(m-2) + \max(0,\, m+1-n)$ edges.

The theta family is the one that makes this problem genuinely different from the edge problem, and it was found by the search rather than by hand. Whenever it is feasible, which is to say for $n \le m+1$, it matches the thickened tree exactly at $m=4$ and beats it for every $m \ge 5$, and it is an extremiser in every such case the exhaustion has settled. Comparing the three counts, the tree leads once $n \ge m+2$, where the other two die for lack of route budget.

An exhaustive include-or-exclude search, independently checked by a separate implementation on the cells where both completed, confirms these constructions at the small sizes used below. The structural statements, rather than the cell-by-cell output, are what matter here.

Two things stand out. First, the problem is *not* the edge problem in disguise: at $m=5, n=5$ the value is 19 against the edge value 16, so the separation axis genuinely buys something for multigraphs under this reading. Second, the small-n regime $n \le m+1$, where the complete graph is feasible for the simple problems too, is tempting to read as the *only* place a cycle pays for itself, with the thickened tree optimal once $n \ge m+2$. It is not: a single triangle grafted anywhere onto the thickened tree already beats it, for every $m \geq 5$ and every such n, which the next theorem makes precise.

By Lemma A.61, a feasible multigraph on top of a fixed connected simple graph G_0 maximises its total multiplicity by setting $\mu(u,v) = m - 1 - \pi(u,v)$ on every edge, giving total $(m-1)\,|E(G_0)| - \sum_{uv \in E(G_0)} \pi(u,v)$. Writing $|E(G_0)| = n - 1 + \operatorname{rank}(G_0)$, the thickened tree is beaten by any G_0 with

$$\sum_{uv \in E(G_0)} \pi(u,v) \;<\; (m-1)\operatorname{rank}(G_0).$$

The obvious guess is that cycles must always pay for themselves against this bound, each independent cycle costing $m-1$ units of route capacity. The next theorem shows a single triangle already fails to pay, for $m \geq 5$, and packing many of them compounds the failure into a gap that grows with n.

Theorem A.63 (A bouquet of thickened cliques, and how it beats the thickened tree)**.** *Fix $3 \leq r \leq m$ and set $q = m + 1 - r \geq 1$, the multiplicity of a thickened K_r that saturates the vertex ceiling on r vertices (the "thickened complete graph" construction above, restricted to a block of size r rather than all of n). Let $1 \leq k \leq \lfloor (n-1)/(r-1) \rfloor$. Take k copies of K_r that all share one common vertex and are otherwise disjoint, thicken every edge of every copy to multiplicity q, and attach any leftover vertices as a pendant path at multiplicity $m-1$. This multigraph is feasible, $\kappa^{\max} \leq m-1$, and its total multiplicity is exactly*

$$(m-1)(n-1) \;+\; k \cdot \operatorname{gain}(r,m), \qquad \operatorname{gain}(r,m) := \frac{(r-1)(r-2)(m-1-r)}{2}.$$

The count therefore exceeds the thickened tree's $(m-1)(n-1)$ exactly when $\operatorname{gain}(r,m) > 0$, that is when $r \leq m-2$, which is possible for some r in range precisely when $m \geq 5$.

Proof. Write G_0 for the underlying simple graph.

Feasibility. The shared vertex is a cut vertex of G_0, and so is every vertex along the pendant path, since the blocks meet only there. So a path of length ≥ 2 between two vertices of the same block cannot leave that block and come back: to do so it would have to pass through the shared vertex twice. Hence for an edge uv inside a block, every route counted by $\pi(u,v)$ stays inside its K_r, where deleting uv leaves u, v joined by exactly $r-2$ pairwise internally-disjoint two-step routes, one through each of the other block vertices, and by Menger no more, since those same $r-2$ vertices are a cut of that size. So $\pi(u,v) = r-2$ and, by Lemma A.61, $\kappa_G(u,v) = q + (r-2) = m-1$ exactly, never over. For u, v in different blocks, or nonadjacent with either endpoint on the pendant path, a single cut vertex separates them, so $\kappa_G(u,v) \leq 1$. If u, v are adjacent along the pendant path, then the edge lies on no cycle, and Lemma A.61 gives $\kappa_G(u,v) = \mu(u,v) + \pi(u,v) = (m-1) + 0 = m-1$.

Count. Since $q \geq 1$, every block edge is genuinely present, so G_0 is connected and $|E(G_0)| =$

$n-1+\operatorname{rank}(G_0)$. This is where the hypothesis $r \le m$ earns its keep: at $r = m+1$ the multiplicity q would drop to zero, the blocks would carry no edges at all, and G_0 would fall apart into isolated vertices. Only the k blocks contribute to $\operatorname{rank}(G_0)$ or to $\sum \pi$, since the pendant edges have $\pi = 0$ and lie in no cycle. Each K_r has rank $(r-1)(r-2)/2$ and contributes $\binom{r}{2}(r-2)$ to the sum, since each of its $\binom{r}{2}$ edges has $\pi = r-2$. Substituting into $(m-1)(n-1+\operatorname{rank}(G_0)) - \sum \pi$,

$$(m-1)\Big[n-1+k\tfrac{(r-1)(r-2)}{2}\Big] - k\tfrac{r(r-1)(r-2)}{2} \;=\; (m-1)(n-1) + k\,\operatorname{gain}(r,m),$$

using $(m-1)\frac{(r-1)(r-2)}{2} - \frac{r(r-1)(r-2)}{2} = \frac{(r-1)(r-2)\big[(m-1)-r\big]}{2} = \operatorname{gain}(r,m)$. □

At $r = 3$, $\operatorname{gain}(3,m) = m-4$: zero at $m = 4$, matching the exhaustive finding that $m \le 4$ has no counterexample, and strictly positive for every $m \ge 5$. So the guess that the thickened tree is optimal once $n \ge m+2$ is false for every $m \ge 5$: it held only within the reach of the exhaustive table, $m \le 4$. The smallest witness is small enough to check by hand. Take $m = 5$ and $n = 7$, the first size the guess covers. Put a triangle on $\{u_1, u_2, u_3\}$ at multiplicity 3 and hang a path of four further vertices off u_1 at multiplicity 4. Two triangle vertices have three parallel copies plus one route through the third vertex, so $\kappa = 4$, and every other pair is separated by a single cut vertex or joined only by its own parallel copies, so nothing exceeds $4 = m-1$. The total multiplicity is $3 \cdot 3 + 4 \cdot 4 = 25$, one more than the $(m-1)(n-1) = 24$ the guess allows. Packing k blocks instead of one multiplies the excess by k, and k may be taken as large as $\lfloor (n-1)/(r-1) \rfloor$, so the gap over the thickened tree grows linearly in n.

Sharing a vertex is what buys that count. Disjoint blocks joined by bridges would fit only $\lfloor n/r \rfloor$ of them, since each bridge spends a vertex on nothing but the join, and the difference is real rather than cosmetic: at $m = 10$, $r = 6$ and $n = 11$ the shared-vertex form carries 150 against the bridged form's 120.

Optimising r widens the gap further. Treated as continuous, the gain per vertex $\operatorname{gain}(r,m)/(r-1)$ peaks near $r \sim (m+1)/2$, where it is $\Theta(m^2)$. Packing blocks of that size therefore gives $K_m^{\text{multi}}(n) \ge \big((m-1) + \Theta(m^2)\big)(n-1-O(m))$. In particular the per-vertex rate is quadratic in m once $n/m \to \infty$, a different order from the tree's linear rate.

Even this is not the truth. At $m = 5$, $n = 7$ the construction above gives 27, a partial branch and bound found 28, and the exact block computation below gives 29. What is not open is the order, and to say that the bouquet is a lower bound of the right order there has to be a matching upper bound. There is one, and it comes from the underlying simple graph rather than cycle rank.

Proposition A.64 (An upper bound of the same order)**.** *For all $m \ge 2$ and $n \ge 2$,*

$$K_m^{\text{multi}}(n) \;\le\; (m-1)\,k_m(n) \;<\; 2(m-1)^2\, n.$$

Proof. Let G be feasible, with underlying simple graph G_0.

Every multiplicity is at most $m-1$. By Lemma A.61, $\kappa_G(u,v) = \mu(u,v) + \pi(u,v) \ge \mu(u,v)$, so feasibility gives $\mu(u,v) \le m-1$ on every pair and the total is at most $(m-1)|E(G_0)|$.

G_0 is itself feasible for the simple vertex problem. Take two vertices $u \ne v$. If they are adjacent in G_0 then $\mu(u,v) \ge 1$, and the routes of G_0 between them are the single edge together with the internally disjoint routes of length at least two, so $\kappa_{G_0}(u,v) = 1 + \pi(u,v) \le \mu(u,v) + \pi(u,v) \le m-1$. If they are not adjacent then $\kappa_{G_0}(u,v) = \pi(u,v) \le m-1$ directly. So $\kappa_{G_0}^{\max} \le m-1$ and $|E(G_0)| \le k_m(n)$ by definition of k_m, which gives the first inequality.

The simple problem is linearly bounded. Suppose G_0 had average degree at least $4(m-1)$. By Mader's density theorem [17], a graph of average degree at least $4k$ contains a $(k+1)$-connected subgraph, so with $k = m-1$ some subgraph $H \subseteq G_0$ is m-connected. An m-connected graph has more than m vertices and, by the global form of Menger's theorem, any two of them are joined by m internally disjoint paths inside H and hence inside G_0. That contradicts $\kappa_{G_0}^{\max} \le m-1$. So the average degree is below $4(m-1)$, that is $2|E(G_0)|/n < 4(m-1)$, giving $k_m(n) < 2(m-1)n$ and the second inequality. $\square$

Combining the two directions, first letting $n/m \to \infty$ so that optimised blocks can be packed with only $O(m)$ unused vertices, and then letting $m \to \infty$,

$$\Big(\tfrac{1}{8} + o_m(1)\Big) m^2 n \;\le\; K_m^{\mathrm{multi}}(n) \;<\; 2m^2 n,$$

where the $o_m(1)$ vanishes as m grows, so $K_m^{\mathrm{multi}}(n) = \Theta(m^2 n)$ in that large-n regime. This qualification is necessary: for example $K_m^{\mathrm{multi}}(2) = m-1$, so no estimate of order $m^2 n$ can hold uniformly when n is fixed. The bouquet of Theorem A.63 has the right joint order once many blocks fit, and what separates the two sides is a constant factor tending to 16. Theorem A.67 below improves the lower constant and brings that factor down to $6 + 4\sqrt{2} \approx 11.66$. The upper bound is deliberately crude, since it throws away the $-\sum \pi$ correction entirely and charges every edge the full $m-1$. The first inequality, $K_m^{\mathrm{multi}}(n) \le (m-1)k_m(n)$, is the more informative half. It says that this variant can never outrun the simple vertex problem by more than the multiplicity ceiling, which ties the least understood of the twelve to the one with the longest history. It is tight at $m=2$, where both sides read $n-1$.

A.9.1 The value is a block problem

The lower bounds above are constructions and the upper bound is a density count, and neither says what an extremiser looks like. This section says exactly what it looks like, in the sense that it reduces the whole question to one about 2-connected graphs, which is a genuine reduction and not a restatement: the number of vertices in play stops being n and becomes the size of a single block.

Two observations do it. The first is already in the section, one line further than it was taken.

Lemma A.65 (The objective in closed form)**.** *For a simple graph G_0 on n vertices with $\kappa_{G_0}^{\max} \le m-1$, the largest total multiplicity of a feasible multigraph with underlying simple graph exactly*

G_0 is

$$W_m(G_0) \;:=\; \sum_{uv\in E(G_0)} \bigl(m-\kappa_{G_0}(u,v)\bigr),$$

and every simple graph arising as the underlying graph of a feasible multigraph satisfies $\kappa_{G_0}^{\max} \le m-1$. Hence

$$K_m^{\mathrm{multi}}(n) \;=\; \max\bigl\{\, W_m(G_0) \;:\; G_0 \textit{ simple on } n \textit{ vertices},\; \kappa_{G_0}^{\max} \le m-1 \,\bigr\}.$$

Proof. By Lemma A.61 feasibility reads $\mu(u,v) \le m-1-\pi(u,v)$ on every edge, so the best choice is $\mu(u,v) = m-1-\pi(u,v)$, which is what the paragraph before Theorem A.63 records. On an edge $\kappa_{G_0}(u,v) = 1+\pi(u,v)$, since the routes of G_0 between adjacent u and v are the single edge together with the routes of length at least two. Substituting turns $m-1-\pi(u,v)$ into $m-\kappa_{G_0}(u,v)$ and the total into $W_m(G_0)$.

Two things have to be checked for that maximum to be attained by a multigraph over G_0 itself. Feasibility of G_0 is the middle step of Proposition A.64. And the chosen multiplicities are all at least one, so the underlying graph really is G_0 and no edge silently disappears: $\kappa_{G_0}(u,v) \le m-1$ on an edge gives $\pi(u,v) \le m-2$, hence $\mu(u,v) = m-1-\pi(u,v) \ge 1$. Conversely any feasible multigraph is one of the multigraphs considered over its own underlying graph, so nothing is lost by maximising over G_0 first. ☐

That the objective is a sum over edges of a *local* quantity is what makes the second observation bite. Recall that a *block* of a graph is a maximal connected subgraph with no cut vertex of its own, so every block is either 2-connected or a single edge, every edge lies in exactly one block, and two blocks meet in at most one vertex.

Theorem A.66 (The multigraph vertex problem is a knapsack over blocks)**.** *For a 2-connected graph B, or a single edge, write*

$$g_m(b) \;:=\; \max\bigl\{\, W_m(B) \;:\; B \textit{ 2-connected or an edge},\; |V(B)| = b,\; \kappa_B^{\max} \le m-1 \,\bigr\},$$

with $g_m(b) := -\infty$ when no such B exists. Then for all $m \ge 2$ and $n \ge 2$,

$$K_m^{\mathrm{multi}}(n) \;=\; \max\Bigl\{ \sum_i g_m(b_i) \;:\; b_i \ge 2,\; \sum_i (b_i-1) \le n-1 \Bigr\},$$

the maximum over all finite multisets of block sizes. In particular the value is determined by the finitely many numbers $g_m(2), \dots, g_m(n)$.

Proof. The objective is additive over blocks. Let G_0 be feasible and let u, v be adjacent, lying in the block B. Every u-v path of G_0 stays inside B: a path leaving B must leave through a cut vertex, and to return it would have to pass through that same cut vertex a second time, which no path does. So $\kappa_{G_0}(u,v) = \kappa_B(u,v)$, and since every edge lies in exactly one block,

$$W_m(G_0) \;=\; \sum_{\text{blocks } B} W_m(B).$$

Each block inherits feasibility, $\kappa_B^{\max} \le \kappa_{G_0}^{\max} \le m-1$ by the same identity, so $W_m(B) \le g_m(|V(B)|)$.

The sizes fit. Suppose G_0 has c components and blocks $B_1,\dots,B_t$. Within a single component, contracting the block tree makes $\sum(|V(B_i)|-1)$ equal to the number of vertices of that component minus one. Isolated components contribute zero to both sides, so summing gives $\sum_i(|V(B_i)|-1) = n-c \le n-1$. With Lemma A.65 this proves $\le$.

Every multiset is realised. Given blocks $B_1,\dots,B_t$ with $\sum(b_i-1) \le n-1$, take one common vertex and attach the B_i to it, pairwise disjoint otherwise, leaving any spare vertices isolated. The result uses $1+\sum(b_i-1) \le n$ vertices. Its blocks are exactly the B_i, so by the first paragraph its κ agrees with κ_{B_i} on pairs inside a block and equals at most 1 on pairs split between two blocks, which the shared vertex separates. So the graph is feasible and scores $\sum_i W_m(B_i)$, and choosing each B_i optimal gives $\sum_i g_m(b_i)$. This proves $\ge$. □

The reduction changes what has to be searched: enumerate 2-connected simple graphs, evaluate W_m, and solve the resulting knapsack. A sweep through eight vertices agrees with the independent exhaustive routine wherever both finish. It shows that the winning blocks are unbalanced and bipartite-like rather than complete, which motivates the family below. The retained transcript is `figures/multi_vertex_blocks_log.txt`.

Theorem A.67 (Thickened complete bipartite blocks)**.** *Let $1 \le s \le t \le m-1$. Thicken every edge of $K_{s,t}$ to multiplicity $m-s$. The result is feasible, $\kappa^{\max} \le m-1$, on $s+t$ vertices with total multiplicity $st(m-s)$. Hanging k such blocks off one shared vertex gives, for $n = k(s+t-1)+1$,*

$$K_m^{\text{multi}}(n) \;\ge\; \frac{st\,(m-s)}{s+t-1}\,(n-1).$$

Optimised over s and t this rate is $\big(3-2\sqrt{2}+o_m(1)\big)m^2$, attained at $t = m-1$ and $s \sim (\sqrt{2}-1)m$, against the $\big(\tfrac{1}{8}+o_m(1)\big)m^2$ of Theorem A.63. The improvement is by a factor $8(3-2\sqrt{2}) = 24-16\sqrt{2} \approx 1.373$.

Proof. Write S and T for the two sides, $|S| = s$ and $|T| = t$.

The connectivities of $K_{s,t}$. For $u \ne u'$ in S, the u-u' paths of length at least two are the t two-step routes through the vertices of T, and T is a cut of size t, so $\kappa(u,u') = \pi(u,u') = t$. Symmetrically $\kappa(v,v') = s$ for $v \ne v'$ in T. For adjacent $u \in S$ and $v \in T$, delete the edge uv: the sets $T \setminus \{v\}$ and $S \setminus \{u\}$ both separate u from v, so $\pi(u,v) \le \min(s-1,t-1) = s-1$, and the $s-1$ routes u, v_i, u_i, v through distinct $u_i \in S \setminus \{u\}$ and distinct $v_i \in T \setminus \{v\}$ attain it, which needs $s-1 \le t-1$. So $\pi(u,v) = s-1$ and $\kappa(u,v) = s$.

Feasibility and count. By Lemma A.61 the multigraph has $\kappa = (m-s)+(s-1) = m-1$ on an edge, and $\kappa = t \le m-1$ or $s \le m-1$ on a non-adjacent pair, so nothing exceeds the ceiling. The count is st edges at multiplicity $m-s$. Equivalently, by Lemma A.65, $W_m(K_{s,t}) = st(m-s)$.

The bouquet. If $s \ge 2$, then $K_{s,t}$ is 2-connected, so Theorem A.66 applies and k copies at one shared vertex are feasible with total $k\,st(m-s)$ on $n = k(s+t-1)+1$ vertices. If $s=1$, then $K_{1,t}$ is a tree (a single edge when $t=1$), and after thickening every edge to multiplicity $m-1$, sharing vertices among copies still produces a thickened tree and is feasible directly. Its count is again $k\,t(m-1) = k\,st(m-s)$. Thus the displayed rate holds throughout the theorem's full range $1 \le s \le t$.

The optimum. For fixed $s>1$, the factor $t/(s+t-1)$ is strictly increasing in t, and for $s=1$ it is constant. Thus some optimum has $t=m-1$, its largest permitted value. Put $s=\alpha m$. The rate becomes

$$\frac{s\,(m-1)(m-s)}{s+m-2} \;=\; \frac{\alpha(1-\alpha)}{1+\alpha}\,m^2 \;+\; O(m),$$

and $h(\alpha) = \alpha(1-\alpha)/(1+\alpha)$ has $h'(\alpha) = (1-2\alpha-\alpha^2)/(1+\alpha)^2$, vanishing at $\alpha = \sqrt{2}-1$, where $h = 3-2\sqrt{2}$. Since $3-2\sqrt{2} > \frac{1}{8}$, the family beats the clique bouquet, by the stated factor. $\square$

The optimum is lopsided: one side has size about $0.41m$ and the other reaches the ceiling $m-1$. Finite block sweeps also show that adding a few edges inside the small side can improve this clean family. Thus the answer is a block problem with dense bipartite-like blocks, and its asymptotic constant lies between $3-2\sqrt{2}$ and 2.

A.10 Computational audit

The authoritative code and audit instructions are maintained in the versioned repository at `https://github.com/louisvandenbruwaene/thesis`, and the submitted snapshot should record an immutable commit. From `program/`, `python erdos915_unified.py` runs the principal checks and `python -m unittest discover -s tests -v` runs the regression suite. `python make_figures.py` regenerates the computational figures.

The base cases of the directed induction (Theorem 1.10) run to $n=7$, and are settled by a pruned exhaustive search over all simple digraphs with $\lambda^{\max} \le 1$. That search is exact and complete at these sizes, so each completed row is an upper-bound proof at that n. The vertex problem needs its own base cases, since Remark A.17 shows they cannot be inherited from the arc ones, and the same driver supplies them with only the inner feasibility test swapped. Table A.1 records both runs. The two separations agree at every size, which is what closes Theorem 1.11, and that agreement is a computed coincidence of the two searches rather than a formal consequence of Whitney's inequality.

The historical cut-counting optimisation closes the directed multigraph checks through $n=6$ but emits no independently replayable matching bound, so no theorem rests on it. Its raw log

n	arc-disjoint	vertex-disjoint	$M(n)$	nodes	status
2	2	2	2	5	complete
3	4	4	4	22	complete
4	6	6	6	537	complete
5	8	8	8	17 823	complete
6	10	10	10	788 101	complete
7	12	12	12	45 122 129	complete

Table A.1. The directed $m = 2$ base cases under both separations: exhaustive branch and bound over all simple digraphs on n vertices with $\lambda^{\max} \leq 1$, and the same run with $\kappa^{\max} \leq 1$. Same driver, same prunes, only the inner feasibility test changes. Both columns agree with $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ at every size, and every run completed. The node counts coincide because the prunes do not depend on the separation. The wall times do not, the vertex run at $n = 7$ taking $3125\,$s against the arc run's $718\,$s. Full logs: `figures/basecase_search_log.txt` and `figures/basecase_search_vertex_log.txt`.

is `figures/certificate_log.txt`. Exact flow measures a supplied object, completed enumeration proves a finite optimum, and heuristic search supplies only a checked witness and lower bound. The file `program/README.md` records the dependencies, commands, entry points, and archival checklist.

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